End of line (EOL) test description
下线检测

Customer:	**General**
System:	**ABS 9.x** **ESP® 9.x** **ESP®plus 9.x,**
Brake split distribution:	**Diagonal (X); front/rear (II)**
Transmission:	**Rear wheel drive / front wheel drive / four wheel drive**
Number of HU channel:	**3, 4**
Test rig:	**2 axle brake dynamometer** **respectively** **Roller test bench with 4 independent and detachable rolls**
Diagnostic protocol:	**KWP2000, UDS (customer specific)**
Tester requests via:	**K-Line (opt.), CAN, FlexRay (customer specific)**
”D”	
Distribution list:	**Customer**

Department:	Date:	Signature:
CC/EVA4-CN	10.12.2009	Jiang Zhigang
CC/EVA-CN	10.12.2009	Benjamin.Przybyla

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Modifications:

No.	Date	Page	Description	Signature
00	10.08.2009	All	New document	sgd. A. Horvath
01	10.12.2009	All	Chinese version	Paul & Xiang

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Table of contents 目录

1 ATTACHMENTS AND ABBREVIATIONS 6

2 INTRODUCTION 导言 8

3 TEST SCOPE 测试范畴 9

4 TEST STEP DESCRIPTION 测试步骤描述 11

4.1 INITIAL START UP ACTIONS & STANDSTILL TESTS 初始起动动作和静态测试 16

4.1.1 Start of the customer specific diagnostic mode 开始客户指定诊断模式 16

4.1.2 ECU identification 识别 ECU 18

4.1.3 Check of the process bytes (optional) 检查过程字节（可选） 19

4.1.4 Coding (e.g. engine- and gear box variant, vehicle identification number, etc.) - optional 写入变量编码（例如发动机和变速器变量，车架号（VIN）等）– 可选 21

4.1.5 Lateral (AQS) and longitudinal (AXS) acceleration sensor offset value calibration - optional 侧向(AQS)与纵向(AXS)加速度传感器偏移量校准 - 可选 23

4.1.6 Calibration of the steering angle sensor (SAS) - optional 方向盘转角传感器（SAS）校准 - 可选 27

4.1.7 Check of the input / output signals 检查输入/输出信号 29

4.1.8 Static Test - optional 静态测试 – 可选 32

4.1.9 Pump motor slow-down time test (only ESP® systems) – optional 泵马达减速时间测试（仅限 ESP®系统）– 可选 35

4.1.10 Evaluation and erasing of the fault code memory 评估及清除故障码内存 42

4.2 DYNAMIC TESTS ON A 2-AXLE BRAKE DYNAMOMETER (LOW SPEED) 双轴制动测力计动态测试（低速） 45

4.2.1 Wheel speed sensor correlation, signal quality and WSS11i tests (optional) 轮速传感器关联性、信号质量和 WSS 11i 测试（可选） 45

4.2.1.1 Graphical presentation of the entire wheel speed sensor tests (test step 1-4) 轮速传感器整个测试示意图（测试步骤 1-4） 50

4.2.1.2 Evaluation diagrams of the WSS signal quality test WSS 信号质量测试评估图 51

4.2.1.3 Exemplary determination of the allowed WSS signal tolerance WSS 信号公差允许值定义示范 53

4.2.1.4 Evaluation proposal for the WSS test (signal quality regarding single pitch deviation, eccentricity, etc.): WSS 测试（测试信号质量包括孤峰偏离、偏心等）的评估建议 54

4.2.1.5 Evaluation of the additional information “rotation direction” of a intelligent WSS (e.g. Bosch WSS11i) – optional 智能 WSS（如博世 WSS11i）附加信息“旋转方向”的评估–可选 56

4.2.1.6 Evaluation of the additional information “mounting position” of a intelligent WSS (e.g. Bosch WSS11i) – optional 智能 WSS（如博世 WSS11i）附加信息“安装位置”的评估–可选 56

4.2.2 Base brake- and ABS test 基础制动测试和 ABS 测试 58

4.2.2.1 Test sequence of the ABS test ABS 测试序列 62

4.2.2.2 Diagram of the ABS test (example at one wheel) ABS 测试示意图（以单个车轮为例） 65

4.2.2.3 Diagram of the ABS test at all wheels 所有车轮的 ABS 测试示意图 67

4.2.2.4 Proposal of a evaluation logic for the optional leakage and ABS test 可选泄漏测试和 ABS 测试的评估逻辑建议 69

4.2.3 ESP® test ESP®测试 73

4.2.3.1 Schematically sequence exemplary at one wheel 单一车轮的测试序列示意图 75

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.2.3.2 Graphical presentation of the entire ESP® tests on a 2 axle brake dynamometer 基于双轴制动测力计的ESP®完整测试示意图 77
4.2.3.3 Test sequence for the ESP® test ESP®测试序列 78
4.2.3.4 Parameter (Base brake, ABS and ESP® test) 参数（基础制动测试、ABS 测试和 ESP®测试） 81
4.2.3.5 Proposal of a evaluation logic for the ESP® test ESP®测试的评估逻辑建议 83
4.3 ALTERNATIVE: DYNAMIC TEST ON A 4 ROLLER TEST BENCH WITH 4 INDEPENDENT ROLLS (HIGH SPEED) 替代方案：在 4 滚筒测试台上进行 4 轮独立旋转动态测试（高速） 85
4.3.1 Wheel speed sensor quality tests (during acceleration of the vehicle – rotation direction: forward) 轮速传感器质量测试（在车辆加速过程中 – 旋转方向：向前） 85
4.3.1.1 Graphical presentation of the entire wheel speed senor tests inclusive the additional WSSi information e.g. Bosch WSS11i during the acceleration phase 加速阶段进行的轮速传感器完整测试（包括 WSSi 附加信息，如博世 WSS11i）示意图 89
4.3.1.2 Evaluation diagrams to the WSS signal quality test (test step 1) during the acceleration phase 加速阶段进行的 WSS 信号质量测试（测试步骤 1）评估图 90
4.3.1.3 Test step 1: Exemplary definition of the maximal allowed difference speed 测试步骤 1：最大允许速度差异定义示范 93
4.3.1.4 Evaluation proposal for the WSS test (signal quality regarding single pitch deviation, eccentricity, etc.): WSS 测试（测试信号质量包括孤峰偏离、偏心等）评估建议： 93
4.3.2 Base brake test and dynamic ABS test (function test „brake pipe correlation, pressure increase and release“) 基础制动测试和动态 ABS 测试（“制动管路关联性，增压和减压”功能测试） 95
4.3.2.1 Diagram of the dynamic ABS test at all wheels 所有车轮的动态 ABS 测试示意图 98
4.3.2.2 Diagram of the dynamic ABS test (exemplary at one wheel) 动态 ABS 测试示意图（以单个车轮为例） 99
4.3.2.3 Test sequence for the dynamic ABS test 动态 ABS 测试序列 101
4.3.2.4 Proposal of the evaluation of the base brake and dynamic ABS test 基础制动测试和动态 ABS 测试的评估逻辑建议 104
4.3.3 Base brake and dynamic ESP® test (function test „brake pipe correlation, pressure increase and release“) 基础制动和动态 ESP®测试（功能测试“制动管路关联性，增压减压”） 106
4.3.3.1 Diagram base brake and dynamic ESP® test (independent of the brake split distribution) 基础制动测试和动态 ESP®测试（与制动回路分布类型无关） 111
4.3.3.2 Diagram to the dynamic ESP® test exemplary at one wheel 单个车轮的动态 ESP®测试示意图 113
4.3.3.3 Test sequence for the dynamic ESP® test (independent of the brake split distribution) 动态 ESP®测试序列（与制动回路分布类型无关） 117
4.3.3.4 Parameter (base brake, dynamic ABS and ESP® test) 参数（基础制动测试、动态 ABS 测试和动态 ESP®测试） 121
4.3.3.5 Proposal of the evaluation of the base brake and dynamic ESP® test 基础制动测试和动态 ESP®测试的评估逻辑建议 123
4.4 FINISHING ACTIONS & STANDSTILL TESTS 结束动作和静态测试 126
4.4.1 Evaluation of the fault code memory 评估故障码内存 126
4.4.2 Warning lamp test (change to the default diagnostic mode) 告警灯测试（切换为默认诊断模式） 127
4.5 EOL DRIVING TEST (ONLY ESP® SYSTEMS - OPTIONAL) 下线检测行驶测试（仅限 ESP®系统 – 可选） 129
5 BOSCH SPECIFIC EOL DIAGNOSTIC ROUTINES AT GENERATION 9 诊断程序说明 130
5.1 DIAGNOSTIC ROUTINE „STATIC TEST“ 130
5.1.1 Start parameter „Static Test“ 130
5.1.2 Evaluation parameter „Static Test“ 130

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

5.2 DIAGNOSTIC ROUTINE „PUMP MOTOR SLOW-DOWN TIME TEST“ 131
5.2.1 Start parameter „Pump motor slow-down time test“ 131
5.2.2 Evaluation parameter „Pump motor slow-down time test“ 131
5.3 DIAGNOSTIC ROUTINE „WHEEL SPEED SENSOR TEST“ 132
5.3.1 Start parameter „Wheel Speed Sensor Test“ 132
5.3.2 Evaluation parameter „Wheel Speed Sensor Test“ 133
5.4 DIAGNOSTIC ROUTINE „WHEEL SPEED SENSOR TEST – ACCELERATED/DECELERATED VEHICLE“ 134
5.4.1 Start parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“ 135
5.4.2 Evaluation parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“ 135
5.5 DIAGNOSTIC ROUTINE „ACTUATOR TEST“ 136
5.5.1 Start parameter „Actuator Test“ 136
5.5.2 Evaluation parameter „Actuator Test“ 137
5.6 DIAGNOSTIC ROUTINE „DYNAMIC TEST“ 138
5.6.1 Start parameter „Dynamic Test “ 138
5.6.2 Evaluation parameter „Dynamic Test“ 139
5.7 DIAGNOSTIC ROUTINE „EXTENDED DYNAMIC TEST“ – ONLY FOR ESP® SYSTEMS AVAILABLE 140
5.7.1 Start parameter „Extended Dynamic Test” 140
5.7.2 Evaluation parameter „Extended Dynamic Test“ 141

6 PUMP MOTOR AND VALVE PROTECTION MEASURES RESPECTIVELY NOTES TO THE PERCENTAGED ACTIVATION 阀保护措施 142

7 TEST RIG INFORMATION 测试设备信息 145

ATTACHMENTS 附件 147

7.1 HYDRAULIC CIRCUIT DIAGRAMS 液压图 147
7.1.1 ABS 9 – independent of the brake split distribution 147
7.1.2 ESP® 9- independent of the brake split distribution 147
7.2 ELECTRICAL CIRCUIT DIAGRAM ESP® 9I 电路图 148

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

1 Attachments and abbreviations

Attachments:

No.	BOSCH – No.:	Name
1	---	Hydraulic circuit diagram ABS 9
2	---	Hydraulic circuit diagram ESP® 9i
3	Y 265 B32 347/E	Electrical circuit diagram ESP® 9i

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Abbreviations:

ABS	antilock braking system
ABS-SILA	ABS warning light
AC	Accumulator chamber
AV	outlet valve
ALS (Ax)	longitudinal acceleration sensor
AQS (Ay)	lateral acceleration sensor
BLS	brake light switch
BS	Brake switch
DTC	diagnostic trouble code
EBD	electronic brake distribution
ECU	electronic control unit
ESP®	electronic stability program
ESP®-MFL	ESP multifunction (warning) light
EV	inlet valve
FA	front axle
FCM	fault code memory
FL	front left
FR	front right
HAS	handbrake switch (park brake switch)
HSV	high pressure switching valve (priming valve)
MCC	master cylinder circuit
PATA	passive taster (TCS/ ESP® passive taster)
PB	process byte
PS (DG, DS)	pressure sensor
RA	rear axle
RFP	recirculation pump
RL	rear left
RR	rear right
SAS (LWS)	steering angle sensor
SV	solenoid valve
TCS (ASR)	Traction control system
UDS	Unified diagnostic services
UMR	pump motor feedback
USV	pilot valve
UVR	Valve relay feedback
WSS	wheel speed sensor
YRS	yaw rate sensor

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

2 Introduction 导言

This document describes test steps which are recommended to ensure the correct function of the Bosch ABS / ESP® system and its interfaces after the vehicle assembly. The tests consist of static- and dynamic function tests.
为确保博世 ABS/ESP®系统能正常工作，本文描述了博世推荐的正确测试方案。包含静态测试与动态测试两部分。

The tests are based on the customer specific diagnosis. The dynamic tests are carried out on a brake dynamometer or selectively on a roller test bench with 4 independent and detachable rolls.

The appropriate version of the customer/vehicle specific ”customer specific descriptions of diagnosis” provides information on the handling and function of the diagnosis.
本测试是基于客户的诊断功能描述进行的，其中动态测试部分需要在转毂上进行。同时诊断功能描述会提供所有的诊断通信数据。

Requirements for the test facilities, which are provided by the customer 对测试设备的要求:

- Communication with the ABS/ESP® ECU based on the corresponding diagnostics 与博世电子控制单元通信
- Control of the test sequence 测试序列的控制
- Control of the dynamometer 测力计的控制
- Evaluation of the test data 测试数据的评估
- Issue test protocol 发布流程协议

Product of Bosch 博世提供的相关产品:

- Functional description of diagnosis (customer specific) 诊断功能描述（基于客户规范）
- Bosch specific EOL diagnostic routines 下线检测程序

The test steps and the test timing is optimized in order to meet a short cycle time and to avoid double checks of the system.
本测试步骤与时间已经经过优化，有效避免了双重检测。

Note 备注:

If the dynamic ABS / ESP® EOL tests are performed in another way as described in this end of line test description, the ABS / ESP® control functions could be activated or the system disabled due to error detection. Bosch cannot be liable for such of problems respectively critical driving situations caused by a not specified test procedure.
To avoid damages at the valve coils by overheating after a previous executed evac/fill or repair bleed process a defined time has to be waited before further valve activivations are executed at the EOL process.
对于不按照本文档进行下线检测而造成的 ABS/ESP®系统失效的情况，博世不承担责任。为了避免阀组过热而烧坏，在执行完抽真空加注或人工排气之后须等待阀组冷却后再进行下线检测。具体信息在博世提供的抽真空加注/人工排气温当中有描述。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

3 Test scope 测试范畴

All the Bosch ABS / ESP® components, such as ECU, hydraulic modulator, wheel speed sensors are checked individually in the Bosch assembly line prior to shipment to the customer's assembly plants.

博世提供的所有部件（如 ABS/ESP®模块，轮速传感器等）在出厂前都进行过严格的测试。

The entire system including harnesses, dashboard lights and power supply of the vehicle is assembled the first time at the customer plant and must be checked for proper connections and functions.

The ECU is capable to run a number of complexity self monitoring software features, such as continuity and plausibility checks and in case of errors to identify these faults and to store fault code numbers.

This feature is used at the end of line test by reading out the fault codes as error identification as well as information for repair works.

ABS/ESP®系统除了博世提供的部分，还包括线束、仪表板指示灯、车辆电瓶等。在车辆组装的时候需确保其他相关部件的性能完好，线束的连接正确。同时 ABS/ESP®的 ECU 会随时监控系统其他部件，当出现故障时 ECU 会自动记录并储存故障码。在下线检测时使用检测设备访问 ECU 并将故障报出，以指导维修工作。

However, some of the failures can not be detected by the ECU itself (such as cross connections of hydraulic pipes or wheel speed sensors and correct functioning of some input signals, i.e. brake light switch). In order to find these failures static or dynamic tests should be performed while the vehicle is at rest or during a dynamic test on a dynamometer.

然而有一些故障是不会被 ECU 监测到的（比如制动管路接反、轮速传感器安装不规范等），这就需要通过动态测试来检查了。

<u>The major items of this test steps is to check for 主要检查项目:</u>

- correct functioning of all input and output signals of the ECU 检查 ECU 的输入输出信号的正确性
- correct functioning of all wheel speed sensors (signal quality and correlation) 检查轮速传感器信号的准确性（信号质量与信号相关性）
- correct functioning of all valves and hydraulic components (functioning and correlation) 检查阀组等液压单元以及各轮缸制动力的准确性（液压单元的功能以及各轮缸的制动压力等）

The ABS / ESP® hydraulic modulator is shipped to the customer in a dry state (no brake fluid). Therefore it needs to be evacuated and filled on line by using the customer specific diagnostics for valve and pump motor activation's. To ensure a proper process, the process information (e.g. ok or nok) can be stored in the EEPROM of the ECU. The value of the process bytes are stored by the evac/fill or bleed equipment and can be checked at the end of line test to verify the previous processes. The delivery state of the process bytes contains a value which should be used for nok process.

在装车时，ABS / ESP®模块是干式的（内部没有制动液），这就需要在线上进行抽真空加注或人工排气，而加注或排气的结果作为过程字节会被储存在 ECU 的芯片中。在下线检测的开始会检查过程字节，以确保抽真空加注/人工排气准确完成。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

The entire test procedure is designed to cover all the important items of the system and to minimize the testing time.

整个测试流程包含了所有重要的项目，并在测试时间上做了优化处理。

Important notes to the ABS/ESP® EOL tests 下线检测的注意事项:

a) Variant coding 变量代码:
Preconditions for a faultless ABS/ESP® system are, that all required variant data are coded.
关于车辆的变量代码需完整的写入 ABS / ESP®系统中。

b) Calibrations 校准:
Preconditions for a faultless ABS/ESP® system are, that all integrated and connected sensors (e.g. longitudinal- (AXS), lateral- (AQS) acceleration and steering angle) are calibrated.
ABS / ESP®系统中所有的相关部件需连接完好，并被校准过。（如加速度传感器、方向盘转向叫传感器等）

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4 Test step description 测试步骤描述

Test step		Description
4.1		**Initial start up actions & standstill tests:** 初始起动动作和静态测试
	.1	**Start of the customer specific diagnostic mode** 开始客户指定诊断模式
	.2	**ECU identification** 识别 **ECU**
	.3	**Check of the process bytes (optional)** 检查过程字节（可选）
	.4	**Coding (e.g. engine and gear box variant, vehicle identification number (VIN), etc.) - optional** 写入变量编码（例如发动机和变速器变量，车架号（**VIN**）等）**–** 可选
	.5	**Calibration of the in the ABS/ESP® ECU integrated Ax/Ay sensors (optional)** **ABS/ESP® ECU** 集成的纵向/侧向加速度传感器校准（可选）
	.6	**Calibration of the steering angle sensor (only ESP® systems – optional)** 方向盘转角传感器校准（仅限 **ESP®**系统 **–** 可选）
	.7	**Check of the input and output signals** 检查输入输出信号
	.8	**Execution of the static test – optional** 执行静态测试 **–** 可选
	.9	**Execution of the pump motor slow-down time test (only ESP® systems) – optional** 执行泵马达减速时间测试（仅限 **ESP®**系统） **–** 可选
	.10	**Read and clear fault code memory** 读取及清除故障码内存
4.2		**Dynamic tests on a 2 axle brake dynamometer (low speed):** 双轴制动测力计动态测试（低速）：
	.1	**Wheel speed sensor tests** 轮速传感器测试 — **WSS correlation test** 轮速传感器关联性测试 **WSS signal quality test** 轮速传感器信号质量测试 **Rotation direction and mounting position tests (optional e.g. Bosch WSS 11i sensors used)** 旋转方向和安装位置测试（可选，例如使用博世 **WSS 11i** 轮速传感器时）

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

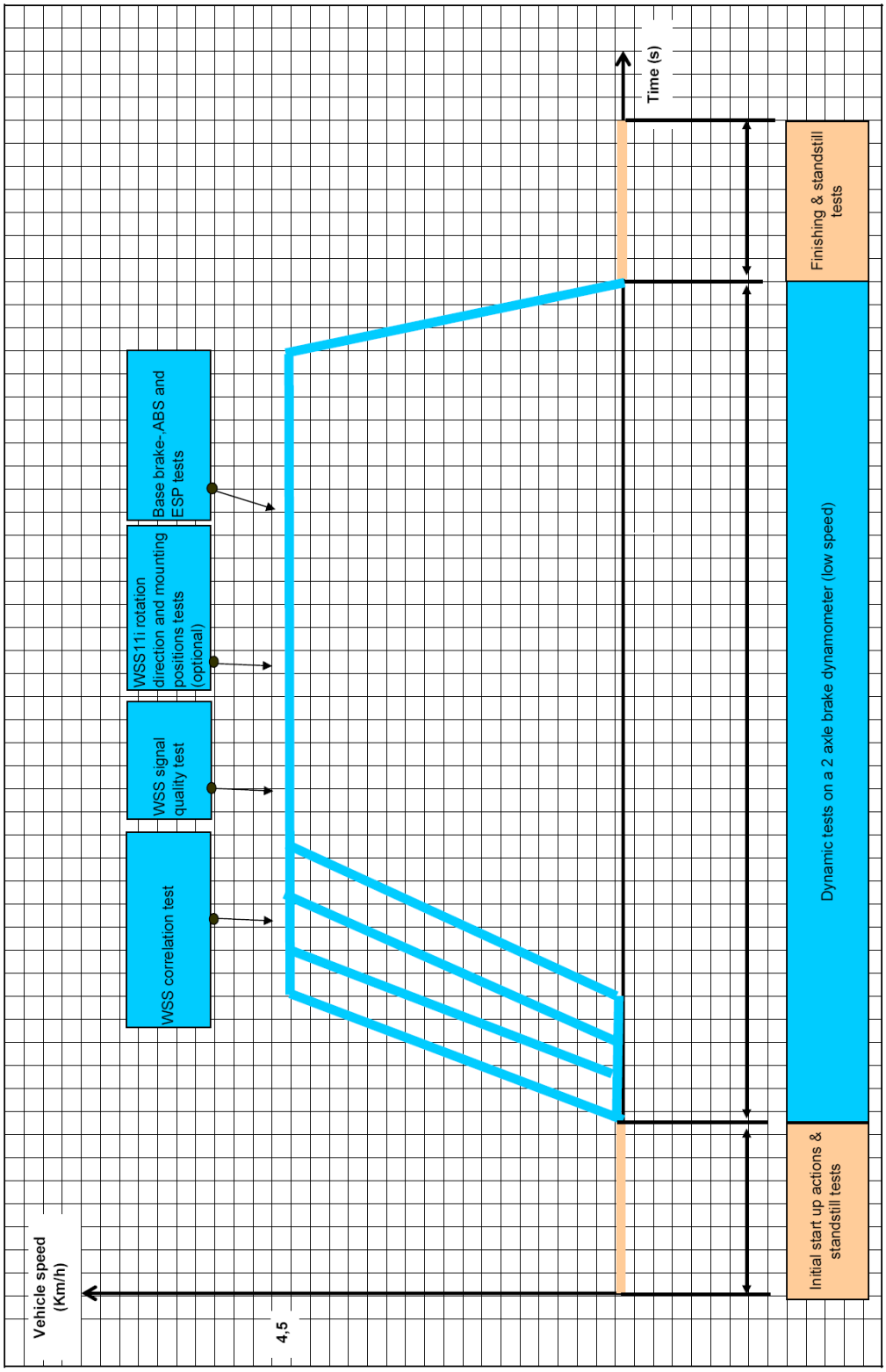
对本文有任何疑问，以英文为准。

	.2	Dynamic tests 动态测试	Base brake- and ABS test 基础制动测试和 ABS 测试
	.3		ESP® test ESP®测试
4.3		Alternate: Dynamic tests on a 4 roller test bench with 4 independent rolls (high speed): 替代方案： 在 4 滚筒测试台上进行 4 轮独立旋转动态测试（高速）	
	.1	Wheel speed sensor tests 轮速传感器测试	WSS signal quality test 轮速传感器信号质量测试 Rotation direction and mounting position tests (optional e.g. Bosch WSS 11i sensors used) 旋转方向和安装位置测试（可选，例如使用博世 WSS 11i 轮速传感器时）
	.2	Dynamic tests 动态测试	Base brake- and ABS test (incl. WSS correlation test) 基础制动测试和 ABS 测试（包括轮速传感器关联性测试）
	.3		ESP® test (incl. WSS correlation test) ESP®测试（包括轮速传感器关联性测试）
4.4		Finishing actions & standstill tests: 结束动作和静态测试	
	.1	Read fault code memory 读取故障码内存	
	.2	ABS/ESP® warning and safety lamp tests / change into normal mode (end of the EOL test procedure) ABS/ESP®告警安全灯测试/切换为常规模式（下线检测过程结束）	
4.5		Driving EOL test (only ESP® systems - optional) 下线检测行驶测试（仅限 ESP®系统 – 可选）	

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Graphical presentation of the complete EOL test procedure on a 2 axle brake dynamometer (low speed): 基于双轴制动测力计（低速）的下线检测完整流程示意图



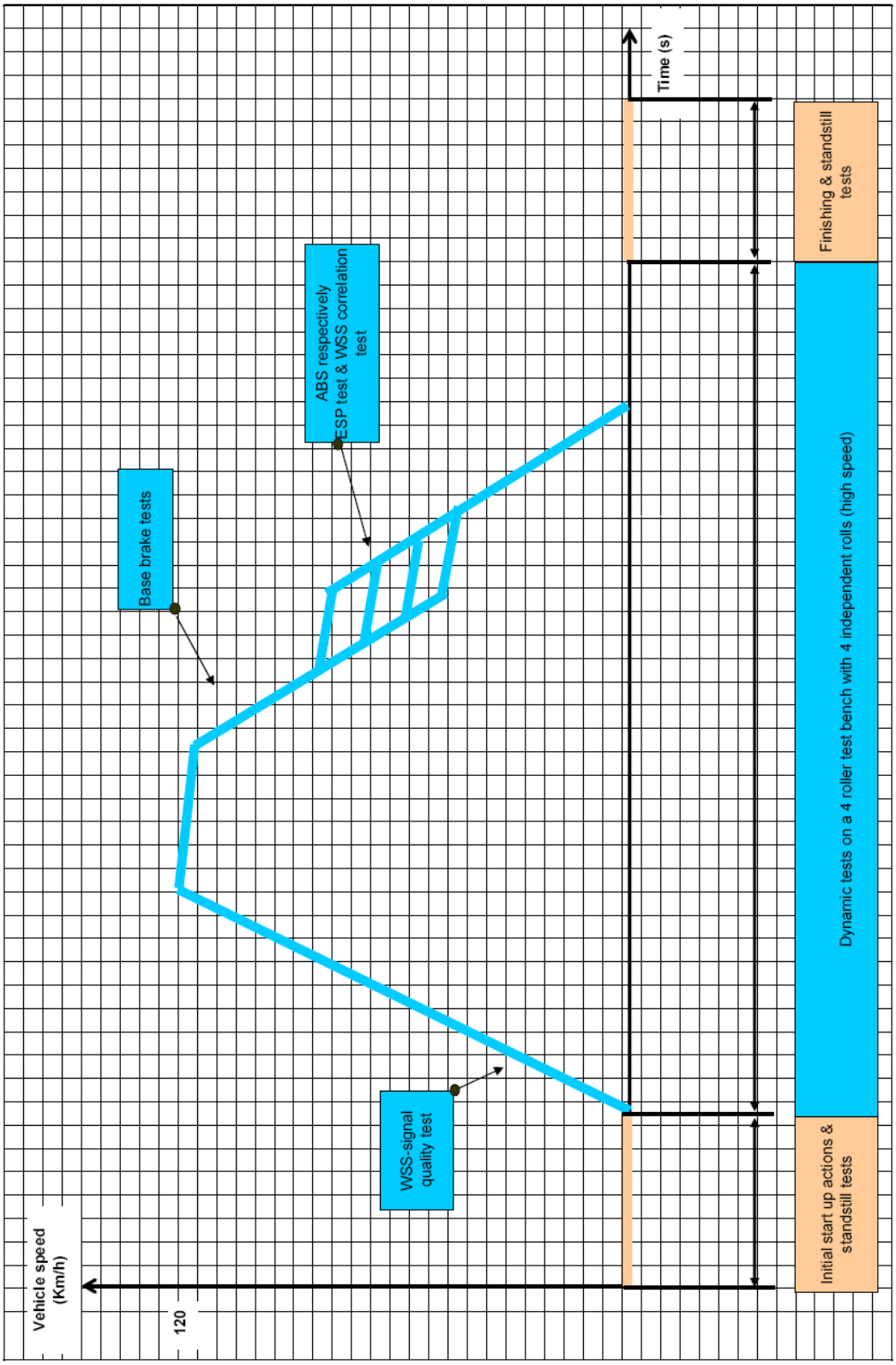
© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Graphical presentation of the complete EOL test procedure on a 4 roller test bench (high speed): 基于 4 滚筒测试台（高速）的下线检测完整流程示意图

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.1 Initial start up actions & standstill tests 初始起动动作和静态测试

4.1.1 Start of the customer specific diagnostic mode 开始客户指定诊断模式

Start of the customer specific diagnostic mode 开始客户指定诊断模式	
Operator: 操作者	Tester / ECU: 测试设备/ECU
→ Ignition off. 关闭点火开关。 → Connect diagnostic interface (OBD2 connector). 连接诊断接口（OBD2 接头）。 → Ignition on. 打开点火开关。 → Wait till the ABS/ESP® system check (initial tests after ignition on) is finished: **Time: t_{ABS/ESP_system_check}** 等待 ABS/ESP®系统自检（点火后初始化测试）结束： 时间：**t_{ABS/ESP_system_check}**	
	Start of customer specific diagnostic mode.开始客户指定诊断模式。

Remarks: 备注：

Before the EOL tests are executed the ignition should be switched off to be sure that the ABS/ESP® system is active and has not been switched off by a previous assembling process which set the ABS/ESP® system into an unwanted faulty mode.
在下线检测开始之前必须先关闭点火开关，以确保之前装配过程中的错误系统设置不会引起 ABS/ESP®系统关闭。

After every switching on of the ignition, the ABS/ESP® system check is started. To avoid to interrupt this system self tests (ECU self diagnosis), the customer specific diagnostic mode should be started after the defined **time t_{ABS/ESP_system_check}**. A premature change into the customer specific diagnostic mode might stop this system self test. In this case the static test should be executed.
在每次打开点火开关后将开始 ABS/ESP®系统自检。为了避免打断系统自检（ECU 自我诊断），必须在规定时间 **t_{ABS/ESP_system_check}** 后方能启动客户指定诊断模式。如果提早启动客户指定诊断模式，系统自检可能被打断，此时必须执行静态测试。

The communication between the tester and the ABS/ESP® ECU is based on the customer specific diagnostic interface.
测试设备与 ABS/ESP® ECU 之间的通讯是基于客户指定诊断接口实现的。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Before e.g. the pump motor slow-down time test (only available at ESP® systems - optional), the WSS tests, the dynamic ABS/ESP® test can be executed, the ABS/ESP® SG must be set into the customer specific diagnostic mode. Only in this special diagnostic mode such of EOL functions are available respectively executable.
在泵马达减速时间测试（仅限 ESP®系统-可选）、轮速传感器测试、动态 ABS/ESP®测试执行之前，ABS/ESP® 的 ECU 必须设置为客户指定诊断模式。只有在这种特殊诊断模式下，下线检测功能才可用。

In case of a not successful execution of the function „start of communication“, it should be repeated minimum 3 times before the process is interrupted.
如果“开始通讯”功能执行不成功，在中断整个流程之前必须将该指令至少再重复 3 次。

The customer specific diagnostic mode must be kept alive with sending a valid diagnostic service (e.g. tester present service) < P3 Timeout, otherwise this diagnostic mode will become invalid.
为了保持客户指定诊断模式可用，必须在 <P3 Timeout 的时间内发送有效的诊断服务（如“测试设备存在中”），否则将退出客户指定诊断模式。

Conditions for a change into the customer specific diagnostic mode (EOL mode):
切换为客户指定诊断模式（下线检测模式）的前提条件：

- No system control 系统无控制动作
- **Speed of the vehicle (km/h) ≤ V_1. 车速(km/h) ≤ V_1**

<u>Parameter:</u>

Kind of parameter:	Parameter name:	Unit:	Physical value:
Condition 条件	Vehicle speed 车速 V_1	km/h	tbd
Diagnostic 诊断	P3 Timeout 超时	Milliseconds (ms)	5000
Time 时间	t_{ABS/ESP_system_check}		5000

More details to the start of communication can be read in the customer specific documentations.
关于“开始通讯”更多信息请参考客户规范文件。

All with tbd signed parameters are customer specific.
所有数值为 tbd 的参数为客户指定。

Start of communication successful ⇒ Continue the EOL test procedure

Start of communication not successful ⇒ Abort, print out fault message and send vehicle to repair area

开始通讯成功 ⇒ 继续下线检测流程
开始通讯不成功 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.1.2 ECU identification 识别 ECU

ECU identification 识别 ECU	
Tester: 测试设备	ECU:
Read and evaluate the ECU identification data Selection of the right valve activation respectively EOL test sequence. 读取并评估 ECU 识别数据。选择正确的阀动作序列和下线检测序列。	Output of the requested ECU data. 输出所需的 ECU 数据

Notes: 注意：

The ECU identification data which has to be evaluated must be confirmed by the customer and its assembly plants.
待评估的 ECU 识别数据应经过客户及其装配工厂确认。

The ECU identification data must be compared with the in the tester software stored desired ECU data. This is required to select the right valve activation respectively EOL test sequence. The ECU identification data might be changed in case of an ABS/ESP® software update respectively modification, therefore it must be easily possible with editor menu to update the data base of the tester respectively EOL equipment.
ECU 识别数据必须和测试设备软件中存储的默认 ECU 数据进行对比，以确保所选择的阀动作序列和下线检测序列正确无误。当 ABS/ESP® 软件升级时，ECU 识别数据将发生变更，因此测试设备和下线检测设备中的数据库应当能用编辑菜单进行方便的升级。

More details to the chapter “ECU identification” can be read in the customer specific documentations.
关于“识别 ECU”更多信息请参考客户规范文件。

The evaluation of the ECU respectively vehicle identification has following reason:
对 ECU 和车辆识别数据进行评估的原因如下：

1) To ensure that the expected ABS/ESP® system is installed 确保安装的 ABS/ESP®系统与车辆相吻合
2) Selection of the required test sequence 选择正确的测试序列

Known system ⇒ **Continue**

Unknown system ⇒ **Abort, print out fault message and send vehicle to repair area**

已知系统 ⇒ 继续下线检测流程
未知系统 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.1.3 Check of the process bytes (optional) 检查过程字节（可选）

Check of the process bytes (optional) 检查过程字节（可选）	
Tester: 测试设备	ECU:
Requests and evaluates the process bytes. 请求并评估过程字节。	Reads the process bytes from the E^2PROM and responses the requested data. 读取 E^2PROM 中的过程字节并回复给测试设备。

Notes: 注意：

The process byte must be equal to the demanded data which are stored in the tester software data base. The evaluation of the process bytes is optional and customer specific.
过程字节应当与测试设备软件的数据库中储存的默认值相同。评估过程字节是由客户指定的可选步骤。

The process bytes contain the information if the previous vacuum and filling (online) or the previous repair bleed (offline) process was successfully executed.
过程字节包含了之前线上抽真空加注或线下人工排气过程是否成功完成的信息。

Precondition to evaluate these data is, that as well as the vacuum and filling and the repair bleed facility supports the process byte writing.
评估这些数据的前提条件是抽真空加注和人工排气设备支持写入过程字节。

The ABS/ESP® ECU is delivered with predefined value (customer specific). This value is written into the ABS/ESP® EEPROM by the Bosch plant. In case of none successful communication with the ECU at the customer assembly plant, the delivery value is still stored. These conditions must be considered as a not successful respectively faulty evacuation and filling or repair bleed process.
ABS/ESP® ECU 在发送至客户时已由博世工厂在 ABS/ESP® EEPROM 中预先写入了一个预定义值（由客户指定）。如果 ECU 在客户装配工厂中通讯不成功，则该预定义值仍将存储在 EEPROM 中。这类情况必须被视为抽真空加注或人工排气过程不成功或发生错误。

In case of a successful evacuation and filling or repair bleed process these data are modified by the facility tester. The used data must be agreed and confirmed between the facility supplier and customer.
如果抽真空加注或人工排气过程成功完成，加注或排气设备将对过程字节数据进行修改，这些使用的过程数据必须经过设备供应商和客户同意和确认。

More details to the chapter “Check of the process bytes – optional” can be read in the customer specific documentations.
关于“检查过程字节-可选”更多信息请参考客户规范文件。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

The evaluation of the process byte data has to ensure the following process:
对过程字节数据进行评估以确认下列过程：

1) ABS/ESP® system respectively base brake system was successfully filled by the online evacuation and filling or offline repair bleed process
ABS/ESP®系统和基础制动系统在线上抽真空加注或线下人工排气过程中成功加注。

Written process bytes ok ⇒ **Continue**

Written process bytes nok ⇒ **Abort, print out fault message and send vehicle to repair area**

写入过程字节合格 ⇒ 继续下线检测流程
写入过程字节不合格 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.1.4 Coding (e.g. engine- and gear box variant, vehicle identification number, etc.) - optional
写入变量编码（例如发动机和变速器变量，车架号（VIN）等）– 可选

Coding (e.g. engine- and gear box variant, vehicle identification number, etc.)- optional 写入变量编码（例如发动机和变速器变量，车架号（VIN）等）– 可选	
Tester: 测试设备	ECU:
1. Start of the login procedure for the ABS/ESP® ECU. 开始 ABS/ESP® ECU 登录程序。 2. Start of the coding procedure for the ABS/ESP® ECU. 开始 ABS/ESP® ECU 编码程序。	1. ECU sends the seed, checks the received login key and allows access. ECU 发送种子信息，检查接收到的登录密钥并开放访问权限。 2. ECU stores the coding data in the E^2PROM. ECU 在 E^2PROM 中储存编码数据。

Notes: 注意：

To have writing access to the ABS/ESP® E2PROM, a login procedure has to be executed within the same diagnostic session (otherwise the ECU answers with a negative response „access denied“).
为了取得 ABS/ESP® E2PROM 的写入权限，必须在同一次诊断时间内执行登录程序（否则 ECU 将给予否定回复“访问被拒绝”）。

At first the ECU checks the received key. In case of the key is valid the ECU is unlocked till the customer specific diagnostic session is left (e.g. finished by a received stop communication command, a software reset or a hard key off).
首先，ECU 检查接收到的密钥。如果密钥有效，ECU 将保持解锁直至本次客户指定诊断时间结束（结束原因包括接收到“停止通讯”的指令、软件重启或点火开关关闭）。

After the ECU is unlocked the tester can send the coding data. The ECU validates the data (depending on the data) and stores them in the E2PROM of the ABS/ESP® ECU. After the next ignition cycle this data will be verified (vehicle must be fully assembled) and will become valid.
ECU 解锁后测试设备才能发送编码数据。ECU 对数据进行验证（取决于数据）并保存至 ABS/ESP® ECU 的 E2PROM 中。在下次点火循环中这些数据将在确认（车辆必须装配完整）后开始生效。

The login procedure and used algorithm is customer specific.
登录程序和所用算法为客户指定。

This step is only required, if the coding data are written during the EOL tests.
本步骤仅在下线检测写入变量编码数据时才需要。

More details to the chapter “Coding” can be read in the customer specific documentations.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

关于“写入变量编码”更多信息请参考客户规范文件。

Coding successful ⇒ Continue

Coding not successful ⇒ Abort, print out fault message and send vehicle to repair area

写入变量编码成功 ⇒ 继续下线检测流程
写入变量编码不成功 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.1.5 Lateral (AQS) and longitudinal (AXS) acceleration sensor offset value calibration - optional 侧向(AQS)与纵向(AXS)加速度传感器偏移量校准 - 可选

Lateral (AQS) and longitudinal (AXS) acceleration sensor offset value calibration - optional 侧向(AQS)与纵向(AXS)加速度传感器偏移量校准 - 可选		
Operator: 操作者	Test rig / Tester: 试验台/测试设备	Remarks: 备注
☞ Turn steering wheel to straight ahead position 将方向盘转至正前方位置 Car should be positioned straight ahead on the equipment 车辆在测试设备上应处于直行位置	Calibration of the lateral (AQS) and longitudinal acceleration sensor (AXS) offset values 校准侧向(AQS)与纵向(AXS)加速度传感器偏移量	The requirement of this calibration depends on the mounting tolerance of the ESP system and the used system functions (hill hold control (HHC), etc.). Details are customer specific. 校准要求取决于 ESP 系统的安装公差与系统所具备的功能（坡道驻车控制 HHC 等）。具体请参考客户规范。

<u>Conditions during sensor offset calibration:</u> 传感器偏移校准的前提条件：

- Maximum inclination angle of the test rig within the allowed limits (both alignment directions, for details look in the chart below) 试验台最大倾斜角必须在允许范围内（在两个对正方向上都必须满足，具体请参考下方图表）
- Test rig must be stationary (i.e. not on production line) 试验台必须静止（即不在生产线上）
- Steering wheel in straight ahead position 方向盘转至正前方位置
- Correct tire pressure 胎压正确
- Normal loading condition 正常负载状况
- Low capacity of the vehicle tank allowed (production fuelling)允许车辆油箱低容积（生产中加注燃料）
- Car standing on its wheels 车辆由自身车轮支撑
- Only the driver should be sitting in the car 仅司机一人必须坐于车内
- No additional disturbances such as closing the door or closing the engine hood are allowed 不允许有额外的如关车门或关发动机罩等干扰
- Maximum inclination angle of the ESP® system / bracket within the allowed mounting tolerance ESP®系统与支架的最大倾斜角必须在允许的安装公差之内

The calibration may be done together with the End-of-line test on the rolling road tester, or at a previous assembly station with a suitable flat area (e.g. alignment test rig).
校准可以在转毂测试仪上与下线检测一同进行，或在之前装配台的平坦区域（如对正试验台）上进行。

<u>Tolerance chain of all possible single tolerances:</u> 所有单一公差的公差链

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Tolerance chain of all single tolerances 所有单一公差的公差链			
No.	Kind of the single tolerance 单一公差种类	Max. value (°) 最大值	Description 描述
1	Tolerances of the ESP ECU and the AXS/AQS sensors ESP ECU 和 AXS/AQS 传感器的公差	≤ 2,00	Inaccuracy during the offset calibration in the Bosch assembly plant (ECU level), hydraulic tolerance (tolerance ECU/HU), fixing rubbers (tolerance HU/bracket), signal accuracy of the acceleration sensors (e.g. temperature drift, signal interference) 博世装配工厂的偏移校准（ECU 本身），液压单元公差（ECU 与液压单元间的公差），固定橡胶公差（液压单元与支架间的公差），加速度传感器信号精度（如温度偏移、信号干扰）
2	Mounting tolerance of the ESP unit in the vehicle ESP 单元在车内的安装公差	≤ tbd (customer specific 客户指定)	Design of the bracket, accuracy of the mounting position (e.g. position of the welded mounting bolts) 支架设计，安装位置精度（如焊接固定螺栓的位置）
3	Accuracy of the chassis 底盘精度	≤ tbd (customer specific 客户指定)	Accuracy of the chassis which influence the inclination angle of the vehicle (e.g. different tire pressure, asymmetric suspension FA/RA) 影响车辆倾斜角的底盘精度（如不同的胎压，不对称的悬架前轴/后轴）
4	Maximum inclination angle of the test rig respectively ground during calibration 校准时试验台与地面的最大倾斜角	≤ 0,57	Inclination angle of the vehicle caused by unevenness ground respectively alignment station or test rig during execution of the calibration. 校准时由不平的地面和对正台或试验台引起的车辆倾斜角
---	---	tbd	Sum of all tolerances at a statistic reflection (radical of the square sum); $S_{_statistic_reflection} = \sqrt{s_1^2 + s_2^2 + s_n^2}$ 所有统计公差和（平方和的根）；$S_{_statistic_reflection} = \sqrt{s_1^2 + s_2^2 + s_n^2}$
---	---	tbd	Sum of all single tolerances „Worst Case“; $S_{_all_single_tolerances} = s_1 + s_2 + s_n$) 所有单一公差和“最坏情况”； $S_{_all_single_tolerances} = s_1 + s_2 + s_n$)
---	---	tbd	Average value (= max. value) of the sum of all tolerances at a statistic reflection and the sum of all single tolerances; $s_{_max} = S_{_statistic_reflection} + S_{_all_single_tolerances} / 2$) 所有统计公差和与所有单一公差和的平均值（=最大值）； $s_{_max} = S_{_statistic_reflection} + S_{_all_single_tolerances} / 2$)

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

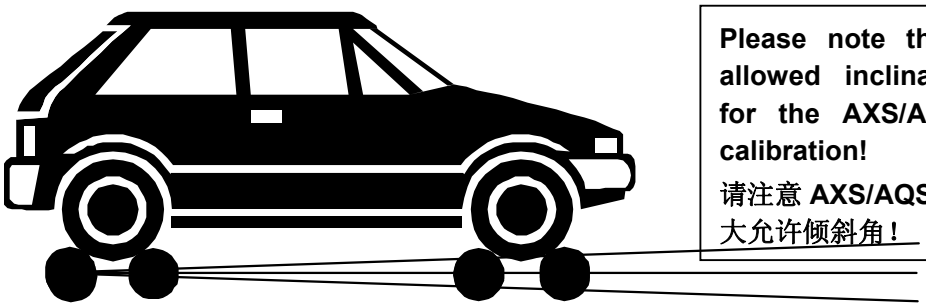
Remarks: 备注：

To guarantee correct ABS / ESP® functions the AXS/AQS sensor inclination angle might not exceed a defined maximal value during execution of the calibration procedure.
为了保证 ABS / ESP®的正常功能，在校准过程中，AXS/AQS 传感器倾斜角不能超过规定的最大值。
This maximal value is the average value of the sum of all tolerances at a statistic reflection and the sum of all single tolerances. In addition the resolution of the customer specific diagnostic function has to be considered.
该最大值为所有统计公差和与所有单一公差和的平均值。此外，客户指定诊断功能的精度也必须考虑。
Especially the tolerance regarding the flatness of the ground where the calibration is executed should be as small as possible to have a bigger range for the assembling, mounting and chassis (tire pressure, suspension, etc.) tolerances.
特别是当公差与执行校准的地面的平坦程度相关时，该公差应当尽可能小，以便为装配公差、安装公差和底盘（胎压、悬架等）公差留出更多的空间。
The ESP® ECU is delivered to the customer with predefined AXS and AQS offset values in the E2PROM. The values show that the sensor is not yet calibrated. As long as these values are stored a fault code is set in the fault code memory. This fault code indicates a missing calibration of the AXS and AQS. During the calibration procedure, the ECU writes the measured values of the AXS and AQS sensor offsets into the E2PROM. From now on, the ECU uses the stored offsets in the E2PROM to correct the measured acceleration values.
ESP® ECU 在发送给客户时 E2PROM 中已经储存了预先定义的纵向/侧向加速度传感器（AXS 与 AQS）偏移值，这些值表明传感器尚未经过校准。只要这些值储存在 E2PROM 中，故障码内存中就会记录有一个相应故障码，该故障码说明 AXS 与 AQS 未经过校准。在校准过程中，ECU 会将测量得到的 AXS 与 AQS 偏移值写入 E2PROM 中。在此之后，ECU 将使用该储存的偏移值来对测得的加速度值进行修正。
It is not possible for the ECU to check whether the calibration conditions described above have been met, so the customer must guarantee the correct conditions during calibration of the sensor in the plant.
由于 ECU 自身无法检查上述的校准条件是否满足，因此必须由客户来保证工厂中校准传感器时能满足这些条件。
If the measured AXS or AQS signal offset value A1 is too large, the ECU will reject the calibration. The ECU will also reject the calibration if during the calibration the spread (difference) of measured values over the calibration time tcal_AXS_AQS is too large. In this case the fault code is still set in the fault code memory.
如果测得的 AXS 或 AQS 信号偏移值 A1 太大，ECU 将拒绝校准。当校准过程中测量值沿校准时间 tcal_AXS_AQS 的散布（差异）过大时，ECU 也会拒绝校准。此时故障码内存中将记录该故障码。
If the ESP® inclusive hydraulic unit is changed in the rework area, a new calibration must be executed. This procedure is also valid for service and workshops at the dealer.
如果在维修区中更换过 ESP®的液压单元，必须重新进行校准。该流程同样适用于服务站和经销商维修车间的情况。
More details to the chapter “Lateral (AQS) and longitudinal (AXS) acceleration sensor offset value calibration - optional” can be read in the customer specific documentations.
关于“ 侧向(AQS)与纵向(AXS)加速度传感器偏移量校准 - 可选”更多信息请参考客户规范文件。
Parameter: 参数

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

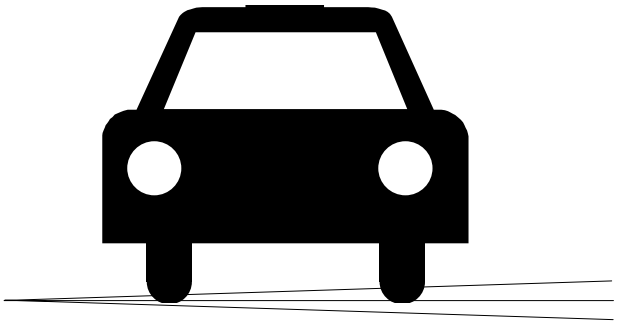
Kind of parameters:	Parameter name:	Unit:	Physical value:
Fault condition 故障条件	AXS or AQS signal offset value A_1 AXS 或 AQS 信号偏移值 A_1	Degrees (°)	> 10
Time	Calibration time $t_{cal_AXS_AQS}$ 校准时间 $t_{cal_AXS_AQS}$	Milliseconds (ms)	≤ 1500

Side view of the vehicle (AXS/AQS sensor element calibration): 汽车侧视图（**AXS/AQS** 校准）



Please note the maximal allowed inclination angle for the AXS/AQS sensor calibration!
请注意 **AXS/AQS** 校准时的最大允许倾斜角！

Front view of the vehicle (AXS/AQS sensor element calibration):汽车前视图（**AXS/AQS** 校准）



Please note the maximal allowed inclination angle for the AXS/AQS sensor calibration!
请注意 **AXS/AQS** 校准时的最大允许倾斜角！

AXS/AQS calibration successful ⇒ **Continue**

AXS/AQS calibration not successful ⇒ **Abort, print out fault message and send vehicle to repair area**

AXS/AQS 校准成功 ⇒ 继续下线检测流程
AXS/AQS 校准不成功 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.1.6 Calibration of the steering angle sensor (SAS) - optional 方向盘转角传感器（SAS）校准 - 可选

Calibration of the steering angle sensor (SAS) – optional **方向盘转角传感器（SAS）校准 - 可选**	
Operator: 操作者	Test rig / Tester: 试验台/测试设备
☞ Turn the steering wheel to straight ahead position 将方向盘转至正前方位置 SAS calibration tolerance measured at the steering wheel: ≤ ± A_2 方向盘处测得的 SAS 校准公差：≤ ± A_2	Calibration (zeroing of the offset) of the steering angle sensor 方向盘转角传感器校准（偏移量调零）

Remarks: 注意：

The calibration may be done together with the End-of-line test on the rolling road tester, or at a previous assembly station (e.g. alignment test rig).
校准可以在转毂测试仪上与下线检测一同进行，或在之前装配台（如对正试验台）上进行。

Following steps have to be done in the described sequence:
下列步骤应当按照规定顺序进行：

1. Initialisation of the SAS by turning of the steering wheel after ignition on (hardware respectively customer specific).
在打开点火开关后，通过转动方向盘来初始化 SAS 传感器（由所用硬件和客户指定）。

2. Turning of the steering wheel to straight ahead position.
The maximal SAS calibration tolerance W2 measured at the steering wheel: ≤ ± A2. These conditions have to be guaranteed for the total calibration time tCal_SAS.
将方向盘转至正前方位置。方向盘处测得的 SAS 校准公差最大值 W2 应当≤ ± A2。这些条件必须在整个校准时间 tCal_SAS 上得到满足。

3. Calibration of the steering wheel sensor with the customer specific diagnostic function.
通过客户指定诊断功能进行方向盘转角传感器校准。

4. Verification of the calibration by evaluation of the fault code memory respectively the customer specific SAS status information. In case of a not successful calibration of the SAS, the calibration procedure has to be executed once again before the process is aborted.
通过评估故障码内存和客户指定 SAS 状态信息来确认校准。如果 SAS 校准失败，在退出之前应尝试再次校准。

In case of a wrong executed or missing calibration an actual customer specific fault code is stored in the fault code memory of the ESP® ECU (ESP® warning lamp on).

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

如果校准错误或未校准，ESP® ECU 的故障码内存中将储存一个真实的客户指定故障码（ESP® 告警灯点亮）。

Parameter: 参数

Kind of parameters:	Parameter name:	Unit:	Physical value:
Condition	SAS calibration tolerance W_2 SAS 校准公差 W_2	Degrees (°)	5
Time	Calibration time t_{Cal_SAS} 校准时间 t_{Cal_SAS}	Milliseconds(ms)	≤ 1500

The requirement to calibrate the steering angle sensor depends on the used hardware (customer specific)!
方向盘转角传感器的校准要求取决于所用硬件（客户指定）。

More details to the chapter “Calibration of the steering angle sensor (SAS) - optional” can be read in the customer specific documentations.
关于“方向盘转角传感器（SAS）校准 - 可选”更多信息请参考客户规范文件。

SAS calibration successful ⇒ Continue

SAS calibration not successful ⇒ Abort, print out fault message and send vehicle to repair area

SAS 校准成功 ⇒ 继续下线检测流程
SAS 校准不成功 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.1.7 Check of the input / output signals 检查输入/输出信号

Check of the input / output signals 检查输入/输出信号		
Tester / ECU 测试设备		Notes: 注意:
Checks respectively activate the input / output signals (channels). 检查并启动输入/输出信号（频道）		The tester checks the input and output signals for plausibility. Some outputs / channels must be activated before they can be tested. 测试设备检查输入输出信号的真实性。有些输出/频道在测试之前必须先启动。
ECU data: ECU 数据	Demanded state of the ECU output: ECU 输出要求值	→ Operator / Tester / Notes: 操作者/ 测试设备 / 注意
Example of I/O check for an ABS system: ABS 系统输入/输出检查实例：		
Hand brake switch: 手刹开关	on	→ Actuate the hand brake (switch)! 拉起手刹
	off	→ Release the hand brake (switch)! 放下手刹
Brake light switch: 制动灯开关	on	→ Apply the brake pedal! 踩下制动踏板
	off	→ Release the brake pedal! 松开制动踏板
ABS lamp: ABS 告警灯	on	Switch on the ABS lamp! 点亮 ABS 告警灯
		→ Check, if the ABS lamp is switched on! 检查 ABS 告警灯是否点亮
	off	Switch off the ABS lamp! 关闭 ABS 告警灯
		→ Check, if the ABS lamp is switched off! 检查 ABS 告警灯是否关闭
Example of the additional part of a I/O check for a ESP® system (ABS I/O check must also be executed): ESP®系统输入/输出检查附加部分实例（ABS 输入/输出检查也必须执行）		
ESP® passive-switch: ESP®功能开关	on	→ Actuate the ESP® passive-switch! 打开开关
	off	→ Release the ESP® passive-switch! 关闭开关
ESP® passive-lamp: ESP®功能开关指示灯	on	Switch on the ESP® passive-lamp! 点亮 ESP®功能开关指示灯
		→ Check, if the ESP® passive- lamp switched on! 检查 ESP®功能开关指示灯是否点亮
	off	Switch off the ESP® passive-lamp! 关闭 ESP®功能开关指示灯
		→ Check, if the ESP® passive- lamp switched off! 检查 ESP®功能开关指示灯是否关闭

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Yaw rate sensor (YRS): 横摆角速度传感器	- 1,2…+ 1,2 °/s	---
Longitudinal acceleration sensor (AXS): 纵向加速度传感器	- 0,384…+ 0,384 m/s^2	Sensors (ALS and AQS) are calibrated; inclination of the ground respectively test rig within the allowed tolerance 传感器已校准；地面与试验台之间倾斜角在允许公差之内。
Lateral acceleration sensor(AQS): 侧向加速度传感器	- 0,384…+ 0,384 m/s^2	
Pressure sensor (PS): 压力传感器	≥ 35 bar	→ Apply the brake pedal! 踩下制动踏板
	- 7,7 ≤ PS ≤ 7,7 bar	→ Release the brake pedal! 松开制动踏板
Steering angle sensor (SAS): 方向盘转角传感器	-5°... + 5°	→ Turn the steering wheel into straight ahead position! 将方向盘转至正前方位置
	SAS ≥ 30°	→ Turn the steering wheel counter clockwise ≥ 30°! 将方向盘逆时针旋转≥ 30°!
	SAS ≤ -30°	→ Turn the steering wheel clockwise ≤ -30°! 将方向盘顺时针旋转≤ -30°!

Notes: 注意：

The check of the components respectively inputs and output signals have to be done step by step depending on the operator (e.g. 1. step: operator applied brake pedal; 2. step: operator released brake pedal). To be checked components respectively inputs at the ABS/ESP® ECU are customer specific (all mentioned tests are only examples).
零部件与输入输出信号的检查必须由操作者按步骤循序渐进来完成（如：第一步，操作者踩下制动踏板；第二步，操作者松开制动踏板）。需要检查的零部件和需要输入 ABS/ESP® ECU 的信号由客户指定（上述测试作为范例仅供参考）。

The actual ECU data must be compared with the demanded data which are stored in tester data base.
ECU 实际数据必须与测试设备数据库中储存的默认数据进行对比。

The operator prompts (”release brake pedal”, ”apply brake pedal”, etc.) must be identified on display.
操作者提示信息（“松开制动踏板”、“ 踩下制动踏板”等）必须在测试设备上显示出来。

Precondition for the described steering angle sensor test is that the SAS calibration was executed at a previous assembly process.
上述方向盘转角传感器测试的前提条件是 SAS 校准已经在之前的装配过程中完成。

More details to the chapter “Check of the inputs and output signals” can be read in the customer specific documentations.
关于“检查输入/输出信号”更多信息请参考客户规范文件。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

The evaluation of the input / output signals has to ensure the following functions and assembly processes:

1) To ensure all electrical connections to the ABS/ESP® ECU
2) To ensure the mechanical connection between the SAS and steering column
3) To ensure the function of the pressure sensors
4) To ensure the function of the brake light switch, etc.

评估输入/输出信号以确保下列功能和装配过程正常：
1）确保所有 ABS/ESP® ECU 的电气连接正常
2）确保 SAS 与转向柱之间的机械连接正常
3）确保压力传感器功能正常
4）确保制动灯开关功能正常

Data correct ⇒ **Continue**

Data not correct ⇒ **Abort, print out fault message and send vehicle to repair area**

数据正确 ⇒ 继续下线检测流程
数据不正确 ⇒ 退出，打印错误信息并将车辆送至维修区

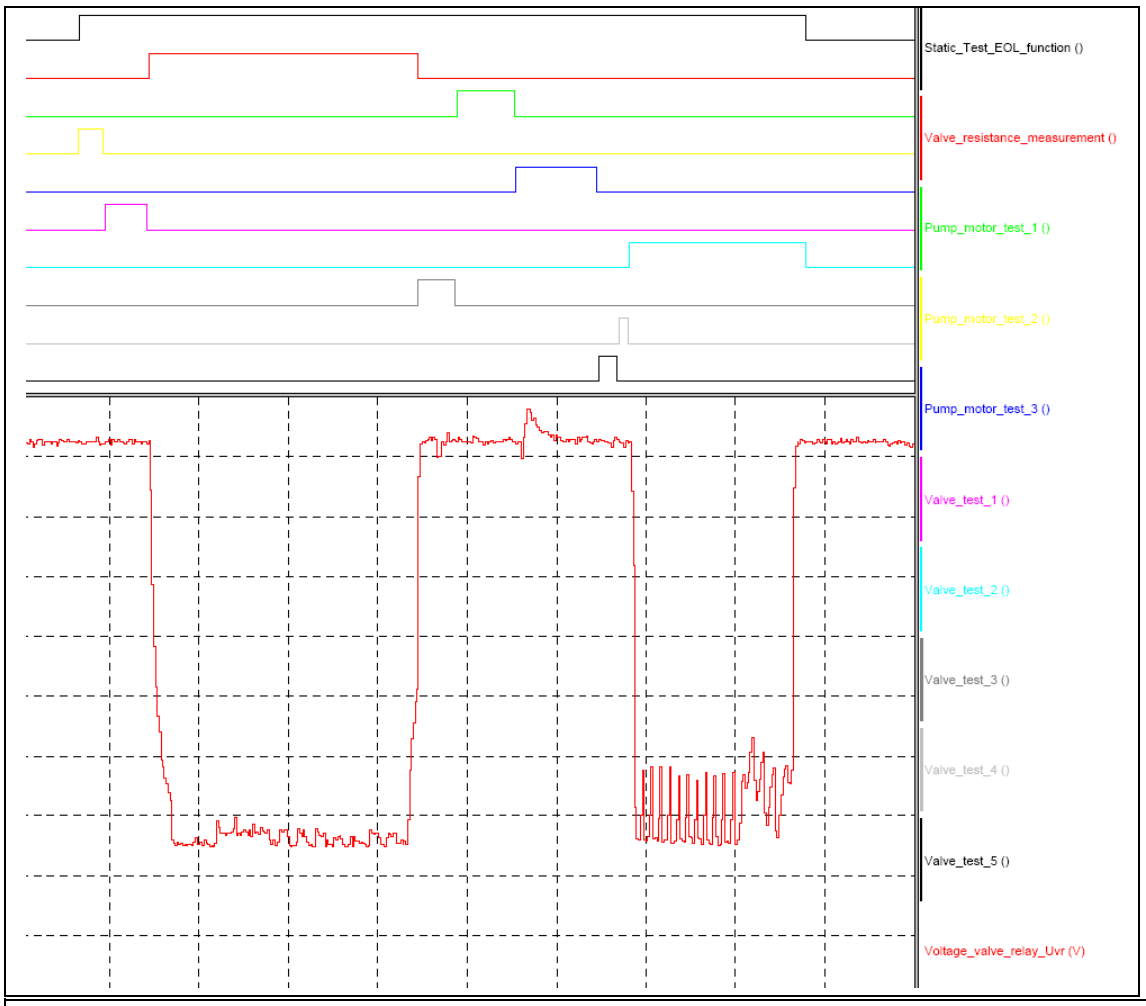
© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.1.8 Static Test - optional 静态测试 – 可选

Static Test		
Operator: 操作者	Tester: 测试设备	Remarks: 备注
→ Don't apply the brake pedal! 禁止踩制动踏板	1. Start of the static test 开始静态测试 2. Evaluation of the fault code memory. 评估故障码内存	The static test doesn't provide any specific results. In case of any detected failure a fault code is set. That's the reason why the fault code memory has to be evaluated after the test is finished. 静态测试并不生成任何具体结果。如果探测到任何故障，将储存故障码。因此，在测试结束后必须评估故障码内存。

Graphical presentation of the static test: 静态测试示意图



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Remarks: 备注：

The static test starts all valve and pump motor tests, which are normally triggered by the ABS/ESP® ECU after ignition on (preconditions: e.g. no special diagnostic mode active, no under voltage, etc.). All these valve and pump motor tests are identical with the ECU self tests in standstill conditions after power on. In addition valve and pump motor tests are executed, which are normally triggered by the ABS / ESP® ECU only at a defined vehicle speed.
静态测试将启动所有阀和泵马达测试，正常情况下 ABS/ESP® ECU 在点火后也将启动这些测试（前提条件：没有启用特殊诊断模式、电压没有过低等等）。所有阀和泵马达测试与 ECU 自检测试基本相同，区别在于前者是在静止条件下进行，而后者是在达到规定车速时进行。

The static test doesn't provide any specific results. After execution of this test the fault code memory has to be evaluated, because in case of a detected fault the corresponding fault code is set in the fault code memory by the static test. For more **details about the used diagnostic function please refer chapter 5.1.**
静态测试并不生成任何具体结果。如果探测到任何故障，故障码内存中将储存相应故障码。因此，在测试结束后必须评估故障码内存。关于这里采用的诊断功能更多信息请参考 5.1 小节。

The **test time t_{ABS} respectively test time t_{ESP}** depends on the used system respectively of the possible test repetitions of each suspicious valve or pump motor test.
测试时间 **t_{ABS}** 和 **t_{ESP}** 取决于所用的系统，以及当某个阀或泵马达测试结果比较可疑时可能进行的重复测试。

In case of an occurred **timeout $t_{_Timeout_St}$** the static test has to be repeated, before the vehicle is indicated as faulty. If this is not possible caused by a limited cycle time within the assembly line, the vehicle has to be marked as faulty car and sent to the rework area. At this location the static test has to be repeated. In case of the timeout occurs again the vehicle has to be investigated for under voltage conditions or wiring harness faults (e.g. loosen ground connection, etc.). This is also valid in case of no fault code is stored.
如果发生超时 **$t_{_Timeout_St}$**，静态测试必须在车辆被报错之前重新进行。如果由于产线上生产循环时间有限而无法重新进行，该车辆必须被标记为故障车辆并送至维修区重新进行静态测试。如果再次发生超时现象，必须检查车辆是否有电压过低或线束故障（如接地线接触不良等）的问题。在没有故障码的情况下也应当进行同样检查。

Test conditions: 测试条件

- Vehicle engine is running, to guarantee a **minimum voltage at the ABS/ESP® ECU ≥ U_1**.
  车辆发动机保持运转以确保 **ABS/ESP® ECU 处最低电压 ≥ U_1**；
- Vehicle stands still.
  车辆静止不动。

Parameter: 参数

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Kind of parameters:	Parameter name:	Unit:	Physical value:
Condition	Minimum voltage U_1 最低电压 U_1	Volt (V)	≥ 12
Time	Timeout $t_{_Timeout_St}$ 超时 $t_{_Timeout_St}$	Milliseconds (ms)	> 10000
	Test time t_{ABS} 测试时间 t_{ABS}		1500 to 4500
	Test time t_{ESP} 测试时间 t_{ESP}		2000 to 6000

More details to the chapter “Static Test - optional” can be read in the customer specific documentations.
关于“静态测试 – 可选”更多信息请参考客户规范文件。

<u>The execution of the static test has to ensure the following functions and assembly processes:</u>
进行静态测试以确保下列功能和装配过程正常：

1) To ensure the electrical function of all valves and the pump motor.
   确保所有阀和泵马达的电气功能正常
2) Start of the valve coil resistance measurement
   开始测量阀线圈阻抗

Static Test ok ⇒ Continue

Static Test nok ⇒ Abort, print out fault message and send vehicle to repair area

静态测试合格 ⇒ 继续下线检测流程
静态测试不合格 ⇒ 退出，打印错误信息并将车辆送至维修区

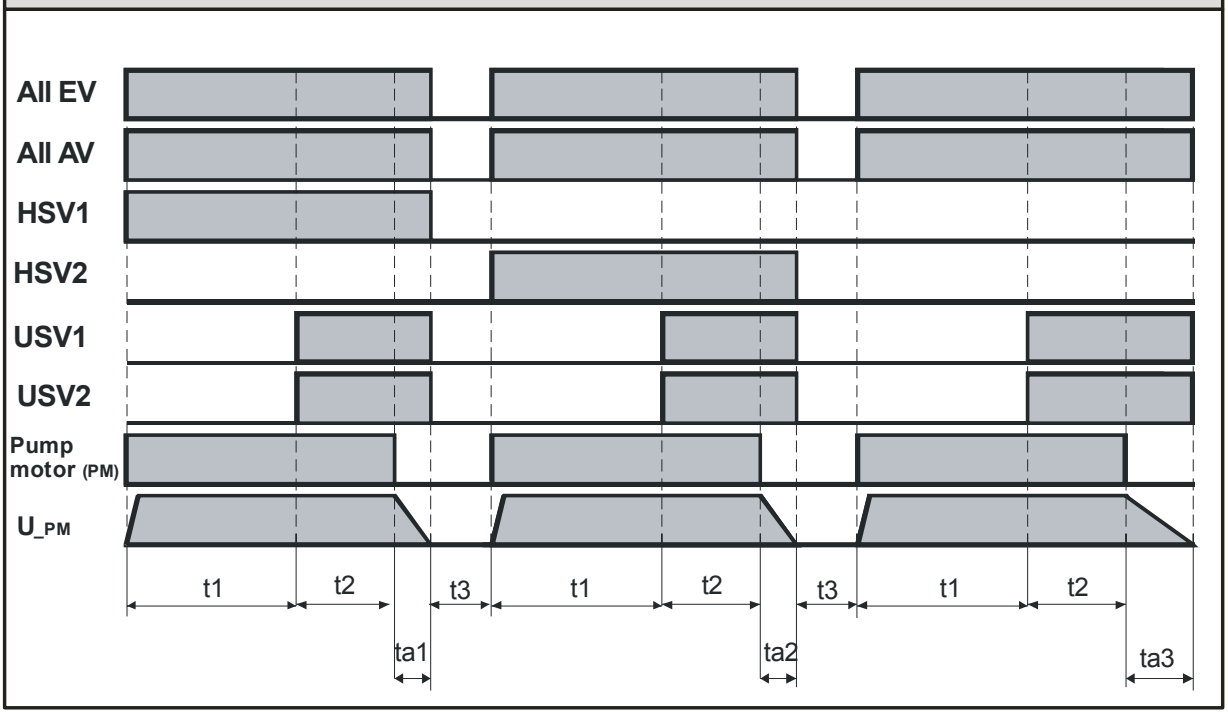
© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.1.9 Pump motor slow-down time test (only ESP® systems) – optional 泵马达减速时间测试（仅限 ESP®系统） – 可选

Pump motor slow-down time test 泵马达减速时间测试		
Operator:操作者	Tester:测试设备	Remarks:备注
→ Don't apply the brake pedal! 禁止踩制动踏板	Start of the pump motor slow-down time test and evaluation of the test results. 开始泵马达减速时间测试并评估测试结果	The diagnostic tester evaluates the ECU test results and indicates if the test is passed successfully. 诊断测试设备评估 ECU 测试结果并显示测试是否顺利通过。

Graphical presentation of the pump motor slow-down time test: 泵马达减速时间测试示意图：

Pump motor slow-down time test



Remarks: 备注：

The test will be used to check the feed rate of the pumps respectively the active pressure increase function as well as the tightness of different valves. This test should be executed before the dynamic ESP® test to detect a not successful filled hydraulic unit. In case of failed test the vehicle has to repair bleed.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

该测试用于检查回流泵的回流速率和主动增压功能以及阀的密封性。该测试应在 ESP® 动态测试之前执行，以检测液压单元加注是否成功。如果测试失败，车辆必须进行人工排气。

After the tester has been started the pump motor slow-down time test , the ESP® hydraulic unit creates in each master cylinder circuit step by step an internal **system pressure of p_1 bar** between the closed inlet and pilot valves as a result of the running pump motor and the opened priming valves. After switching off of the pump motor at the test step 1 (MCC1) and test step 2 (MCC2) the ECU records the **post run times t_{a1} and t_{a2}** of the pump motor. Finally the **post run time t_{a3}** of the pump motor is measured in unpressurized conditions (test step 3). The post run times are measured and calculated from the generated post run voltage after switching off of the pump motor. The post run times t_{a1}, t_{a2} and t_{a3} have to be read by the tester after the pump motor slow-down time test is finished. For more details about the used diagnostic function please refer chapter 5.2.
当测试设备启动泵马达减速时间测试后，ESP® 液压单元将逐步地在每个主缸回路中建立起 **p_1 bar** 的内部系统压力，该压力由运转的泵马达和打开的高压控制阀产生，作用于关闭的进液阀和回路控制阀之间。在测试步骤 1（MCC1）和测试步骤 2（MCC2）中每次断电泵马达后，ECU 都将记录泵马达的滞后运转时间 **t_{a1}** 和 **t_{a2}**。最后 ECU 还将测量泵马达在无压力状态下的滞后运转时间 **t_{a3}**（测试步骤 3）。滞后运转时间是通过测量泵马达断电后的滞后运转电压计算出来的。滞后运转时间 **t_{a1}** 、 **t_{a2}** 和 **t_{a3}** 必须由测试设备在泵马达减速时间测试结束后读取。关于这里采用的诊断功能更多信息请参考 5.2 小节。

<u>Conditions:</u> 条件：

- Vehicle engine is running, to guarantee a **minimum voltage at the ESP® ECU ≥ U_1**.
  车辆发动机保持运转以确保 **ESP® ECU 处最低电压 ≥ U_1**；
- Vehicle stands still.
  车辆静止不动；
- The brake pedal might not applied during the whole running test! The tester has to ensure this by reading and evaluating the BLS status.
  在整个测试运行过程中禁止踩制动踏板！测试设备必须通过读取和评估制动灯开关状态来确保这一点。

<u>Details to the test step 1:</u> 测试步骤 1 具体信息：

After the start of the routine respectively the start of the test step 1 all outlet (AV) and inlet valves (EV) as well as the prime valve 1 (HSV1) and the pump motor are switched on.

After the **start time t_1 ($t_1 \geq t_{1min}$)** is expired both USV valves are switched on to create an internal system pressure in the master cylinder circuit 1 (MCC1). After the **start time t_2 ($t_2 \geq t_{2min}$)** is expired the pump motor is switched off and the post run time t_{a1} is measured.

At the end of the measurement all valves are switched off and after the **waiting time t_3** (time constant in the software) test step 2 is started.
在程序和测试步骤 1 开始之后，所有出液阀（AV）、所有进液阀（EV）以及高压开关阀 1（HSV）和泵马达都将启动。
当启动时间 **t_1 ($t_1 \geq t_{1min}$)** 结束之后，所有回路控制阀都将打开，从而在主缸回路 1（MCC1）中建立一个内部系统压力。当启动时间 **t_2 ($t_2 \geq t_{2min}$)** 结束之后，泵马达将断电，从而测量其滞后运转时间 t_{a1}。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

在测量结束时，所有阀都将断电，测试步骤 2 将在等待时间 **t_3**（软件中的时间常量）后开始。

Details to the test step 2: 测试步骤 2 具体信息：

After the start of the routine respectively the start of the test step 2 all outlet (AV) and inlet valves (EV) as well as the prime valve 2 (HSV2) and the pump motor are switched on.

After the **start time t_1 ($t_1 \geq t_{1min}$)** is expired both USV valves are switched on to create an internal system pressure in the master cylinder circuit 2 (MCC2). After the **start time t_2 ($t_2 \geq t_{2min}$)** is expired the pump motor is switched off and the post run time t_{a2} is measured.

At the end of the measurement all valves are switched off and after the **waiting time t_3** (time constant in the software) test step 3 is started.

在程序和测试步骤 2 开始之后，所有出液阀（AV）、所有进液阀（EV）以及高压开关阀 2（HSV）和泵马达都将启动。
当启动时间 **t_1 ($t_1 \geq t_{1min}$)** 结束之后，所有回路控制阀都将打开，从而在主缸回路 2（MCC2）中建立一个内部系统压力。当启动时间 **t_2 ($t_2 \geq t_{2min}$)** 结束之后，泵马达将断电，从而测量其滞后运转时间 t_{a2}。
在测量结束时，所有阀都将断电，测试步骤 3 将在等待时间 **t_3**（软件中的时间常量）后开始。

Details to the test step 3: 测试步骤 3 具体信息：

After the start of the routine respectively the start of the test step 3 all outlet (AV) and inlet valves (EV) and the pump motor are switched on. At this test step no prime valve is switched on.

After the **start time t_1 ($t_1 \geq t_{1min}$)** is expired both USV valves are switched on (no system pressure is created). After the **start time t_2 ($t_2 \geq t_{2min}$)** is expired the pump motor is switched off and the post run time t_{a3} (unpressured conditions) is measured.

At the end of the measurement all valves are switched off.

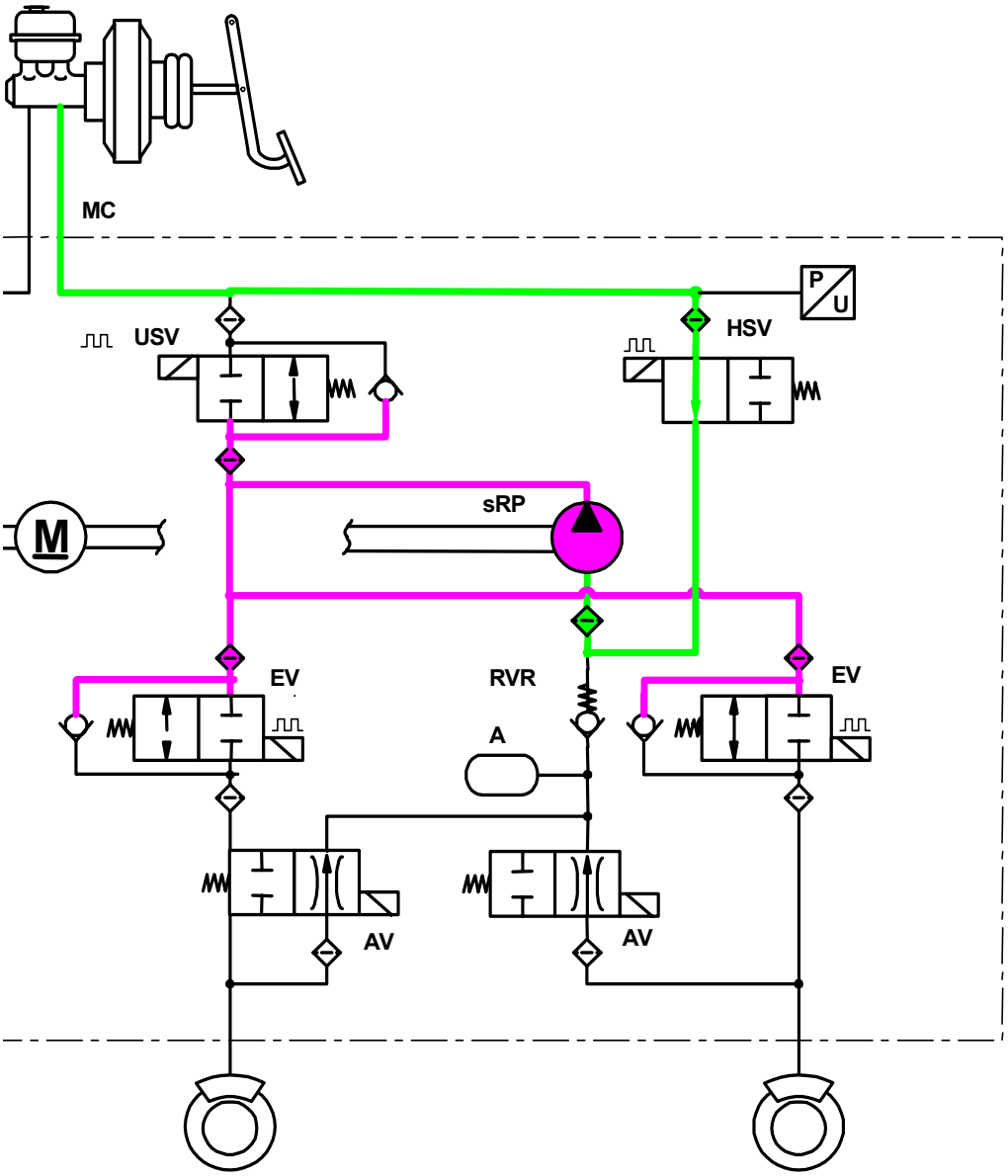
After test step 3 is finished the tester has to request the routine results of all 3 test steps with a separate service.

在程序和测试步骤 3 开始之后，所有出液阀（AV）、所有进液阀（EV）和泵马达都将启动。在步骤 3 中没有高压开关阀启动。
当启动时间 **t_1 ($t_1 \geq t_{1min}$)** 结束之后，所有回路控制阀都将打开（不建立系统压力）。当启动时间 **t_2 ($t_2 \geq t_{2min}$)** 结束之后，泵马达将断电，从而测量其滞后运转时间 t_{a3}。
在测试步骤 3 结束后，测试设备应通过单独服务来获取全部 3 个测试步骤的程序运行结果。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Details to the internal system pressure of the test step 1 & 2: 测试步骤 1&2 的内部系统压力具体信息



By closing of the EV and USV valve, opening of the HSV valve and switching on of the pump motor a system internal pressure is created. This system pressure (p1 bar) is limited by the maximal allowed activation current for the USV valve (Details to the valve protection measurements can be read in the chapter 6). To avoid an influence of a possible EV leakage on the wheel circuits, all AV valves are switched on.

通过关闭 EV 和 USV、打开 HSV 和启动泵马达，建立一个系统内部压力，该系统压力（p1 bar）是由 USV 最大允许启动电流决定的（关于阀保护措施具体请参考第 6 章）。为了避免潜在的 EV 泄漏对轮缸回路造成影响，所有 AV 都必须打开。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Parameter: 参数：

Kind of parameters:	Parameter name:	Unit:	Physical value:
Condition 状态	Minimum voltage U_1 最低电压 U_1	Volt (V)	12
	System pressure p_1 系统压力 p_1	Bar	110 ± 20
Time constant in the software 软件中的时间常量	Waiting time t_3 等待时间 t_3	Milliseconds (ms)	1500
Start of routine 启动程序	Start time t_1 启动时间 t_1		$\geq t_{1min}$
	Start time t_2 启动时间 t_2		$\geq t_{2min}$
Minimum requirement for the start of routine 启动程序最低要求	Start time t_{1min} 启动时间 t_{1min}		400
	Start time t_{2min} 启动时间 t_{2min}		200
Routine results 程序运行结果	Post run time t_{a1} 滞后运行时间 t_{a1}		xx
	Post run time t_{a2} 滞后运行时间 t_{a2}		xx
	Post run time t_{a3} 滞后运行时间 t_{a3}		xx
Evaluation parameter 评估参数	Difference time $t_{a_delta_1}$ 差异时间 $t_{a_delta_1}$		tbd
	Difference time $t_{a_delta_2}$ 差异时间 $t_{a_delta_2}$		tbd
	Post run time $t_{a1/a2_min}$ 滞后运行时间 $t_{a1/a2_min}$		tbd
	Post run time $t_{a1/a2_max}$ 滞后运行时间 $t_{a1/a2_max}$		tbd
	Post run time t_{a3_min} 滞后运行时间 t_{a3_min}		tbd
	Post run time t_{a3_max} 滞后运行时间 t_{a3_max}		tbd
Time 时间	Total test time $t_{total_test_time}$ 总测试时间 $t_{total_test_time}$		6500

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Remarks: 备注：

The given total test time $t_{total_test_time}$ of the pump motor slow-down time test depends on the selected start times (t_1 & t_2) and the test conditions. For the given time following was assumed:
$t_{total_test_time} = 3^* t_{1min} + 3^* t_{2min} + 2 * t_3 + 3 * t_{a3_max}$;
for t_{a3_max} = 3 * 600 ms e.g. if the ESP® hydraulic unit is completely empty (not filled).
泵马达减速时间测试给定的总测试时间 $t_{total_test_time}$ 取决于所选择的启动时间（t_1 & t_2）和测试条件。在给定时间内假定下式成立：
$t_{total_test_time} = 3^* t_{1min} + 3^* t_{2min} + 2 * t_3 + 3 * t_{a3_max}$;
例如，当 ESP®液压单元完全抽空（未加注）时，t_{a3_max} = 3 * 600 ms。

Different causes e.g. the ESP® ECU supply voltage, the state of filling (filled / not filled) of the ESP® hydraulic unit or the allowed valve leakages influence the post rum times (t_{a1} & t_{a2} & t_{a3}), therefore the customer has to define the evaluation parameter (**difference time $t_{a_delta_1}$, difference time $t_{a_delta_2}$, post run time $t_{a1/a2_min}$, post run time $t_{a1/a2_max}$, post run time t_{a3_min} and post run time t_{a3_max}**) by doing a statistic judgement.
影响泵马达滞后运行时间（t_{a1} & t_{a2} & t_{a3}）的因素很多，包括 ESP® ECU 供电电压、ESP®液压单元的加注状态（加注/未加注）或阀泄漏允许量，因此客户应当通过统计学判断来定义评估参数（**差异时间 $t_{a_delta_1}$、差异时间 $t_{a_delta_2}$、滞后运行时间 $t_{a1/a2_min}$、滞后运行时间 $t_{a1/a2_max}$、滞后运行时间 t_{a3_min} 和滞后运行时间 t_{a3_max}**）。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Recommended evaluation logic: 推荐的评估逻辑
$t_{a1/a2_min} \leq t_{a1} \leq t_{a1/a2_max}$
$t_{a1/a2_min} \leq t_{a2} \leq t_{a1/a2_max}$
$t_{a3_min} \leq t_{a3} \leq t_{a3_max}$
$t_{a3} - t_{a1} \geq t_{delta_1}$
$t_{a3} - t_{a2} \geq t_{delta_1}$
$t_{a1} - t_{a2} \leq \pm t_{delta_2}$

More details to the chapter “pump motor slow-down time test (only ESP® systems)” can be read in the customer specific documentations.
关于“泵马达减速时间测试（仅限 ESP®系统）”更多信息请参考客户规范文件。

<u>The execution of the pump motor slow-down time test has to ensure the following functions and assembly processes:</u>
执行泵马达减速时间测试以确保下列功能和装配过程正常：

1) To ensure the pump flow rate respectively a active ESP® pressure increase function (system internal pressure increase) at vehicle standstill conditions
   确保车辆静止状况下泵的流速和 ESP®主动增压功能（系统内部增压）正常
2) To ensure a sufficient filled secondary and priming circuit of the ESP® system
   确保 ESP®系统第二回路和高压开关回路已充分加注

Pump motor slow-down time test ok ⇒ Continue

Pump motor slow-down time test nok ⇒ Abort, print out fault message and send vehicle to repair area

泵马达减速时间测试合格 ⇒ 继续下线检测流程
泵马达减速时间测试不合格 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.1.10 Evaluation and erasing of the fault code memory 评估及清除故障码内存

Evaluation and erasing of the fault code memory 评估及清除故障码内存

Tester / ECU 测试设备/ECU

Erase the fault code memory. Immediately after execution of the erasing command the ECU executes the so-called „good check“. In case of actual faults are existing they will be stored again.
清除故障码内存。在清除指令执行完毕后，ECU 立即执行“合格检查”。如果存在真实故障，将再次设置故障码。

Recommend **waiting time t_{good_check}** till the fault code memory can be evaluated again. Alternate an ECU reset function can be executed. 在再次评估故障码内存之前，推荐的等待时间为 **t_{good_check}**。作为替代方案，可以重启 ECU。

Evaluation of the fault code memory. Following strategy is recommended:

- If system free of faults (see 1. example),
  continue with the EOL tests.
- If system not free of faults (see 2. example),
  stop the EOL tests, print out fault message and send vehicle to repair area.

评估故障码内存。推荐采用下列措施：

- 如果系统无故障（见例 1），继续下线检测；
- 如果系统有故障（见例 2），停止下线检测，打印错误信息并将车辆送至维修区。

Example: 范例	1. diagnostic trouble code (DTC): 第 1 个故障码	Status of 1. DTC: 第 1 个 DTC 状态	1.+n. DTC: 第 1+n 个故障码	Status of 1+n. DTC: 第 1+n 个 DTC 状态	Evaluation: 评估
1)	---	---	---	---	Ok 合格
2)	WSS front left, line fault 前左 WSS 线路故障	Actual 真实	---	---	Nok 不合格

Remarks: 备注：

The ECU outputs the entire contents of the fault code memory. These data include different information. The number of the maximal storable fault codes is customer specific.
Following data of the received fault information have to be evaluated:
ECU 输出故障码内存的全部内容，这些数据中包含了各种不同的信息。最多可储存的故障码数量由客户指定。
采集的故障信息中应当评估的数据如下：

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

- Number of DTC 故障码数量
- Status of DTC (actual/historic) 故障码状态（真实/历史）

The tester has to evaluate the fault data and to transfer the number of DTC into a failure text and to print out a failure protocol. The status information of a DTC shows if a fault is actual or historic.
测试设备需要评估故障数据，将故障码数量转换为故障文本，并打印出故障协议。故障码的状态信息表明该故障是真实或是历史故障。

The waiting time t_{good_check} after execution of the good check is required to ensure, that all fault paths are tested by the ECU and in case of they are still active (actual), they will be stored again. The time which is required for the renewed fault storing depends on the monitoring (filter) time of each individual fault.
合格性检查结束后的等待时间 t_{good_check} 是用来确保 ECU 测试过所有的故障路径，如果有故障仍然存在（真实），该故障码将被再度储存。这个用于更新存储故障的时间长短是由每个单独故障所需的监测（筛选）时间决定的。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Parameter: 参数

Kind of parameters:	Parameter name:	Unit:	Physical value:
Time	Waiting time t_{good_check}	Milliseconds (ms)	5000

More details to the chapter "Evaluation and erasing of the fault code memory" can be read in the customer specific documentations.
关于"评估及清除故障码内存" 更多信息请参考客户规范文件。

The evaluation of the fault code memory has to ensure the following functions and assembly processes:
评估故障码内存以确保下列功能和装配过程正常：

1) To ensure that the ABS/ESP® system is free of faults before the dynamic Tests are executed on a 2-axle brake dynamometer (low speed) respectively on a roller test bench with 4 independent and detachable rolls (high speed)
必须确保在双轴制动测力计（低速）或 4 滚筒测试台（高速）上执行动态测试之前，ABS/ESP®系统无故障。

ABS/ESP® system free of faults ⇒ Continue

ABS/ESP® system not free of faults ⇒ Abort, print out fault message and send vehicle to repair area

ABS/ESP®系统无故障 ⇒ 继续下线检测流程
ABS/ESP®系统有故障 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2 Dynamic tests on a 2-axle brake dynamometer (low speed) 双轴制动测力计动态测试（低速）

4.2.1 Wheel speed sensor correlation, signal quality and WSS11i tests (optional) 轮速传感器关联性、信号质量和 WSS 11i 测试（可选）

Wheel speed sensor tests (correlation and signal quality) 轮速传感器测试（关联性与信号质量）									
Test rig speed $V_{_Test}$: 试验台速度			ECU data to be evaluated: 待评估的 ECU 数据				Proposal for evaluation: 评估建议值	Test time: 测试时间
1. test step: wheel speed sensor correlation test 测试步骤：轮速传感器关联性测试								
Roll FL	On	x km/h	wheel FL		xx	km/h	$V_{FL} \geq V_{roll_on}$	$t_{_WSS_correlation}$
Roll FR	Off	y km/h	wheel FR		xx	km/h	$V_{FR} \leq V_{roll_off}$	
Roll RL	Off	y km/h	wheel RL		xx	km/h	$V_{RL} \leq V_{roll_off}$	
Roll RR	Off	y km/h	wheel RR		xx	km/h	$V_{RR} \leq V_{roll_off}$	
Roll FL	On	x km/h	wheel FL		xx	km/h	$V_{FL} \geq V_{roll_on}$	
Roll FR	On	x km/h	wheel FR		xx	km/h	$V_{FR} \geq V_{roll_on}$	
Roll RL	Off	y km/h	wheel RL		xx	km/h	$V_{RL} \leq V_{roll_off}$	
Roll RR	Off	y km/h	wheel RR		xx	km/h	$V_{RR} \leq V_{roll_off}$	
Roll FL	On	x km/h	wheel FL		xx	km/h	$V_{FL} \geq V_{roll_on}$	
Roll FR	On	x km/h	wheel FR		xx	km/h	$V_{FR} \geq V_{roll_on}$	
Roll RL	On	x km/h	wheel RL		xx	km/h	$V_{RL} \geq V_{roll_on}$	
Roll RR	Off	y km/h	wheel RR		xx	km/h	$V_{RR} \leq V_{roll_off}$	
2. test step: wheel speed sensor signal quality test 测试步骤：轮速传感器信号质量测试								
Roll FL	On	x km/h	wheel FL	max:	xx	km/h	$V_{FLmax} \leq V_{FLØ} + V_{WSS_tolerance}$	$t_{_WSS_signal_quality}$
				min:	xx	km/h	$V_{FLmin} \geq V_{FLØ} - V_{WSS_tolerance}$	
				Ø	xx	km/h	$V_{_Test} - V_{tolerance} \leq V_{FLØ} \leq V_{_Test} + V_{tolerance}$	
Roll FR	On	x km/h	wheel FR	max:	xx	km/h	$V_{FRmax} \leq V_{FRØ} + V_{WSS_tolerance}$	
				min:	xx	km/h	$V_{FRmin} \geq V_{FRØ} - V_{WSS_tolerance}$	
				Ø	xx	km/h	$V_{_Test} - V_{tolerance} \leq V_{FRØ} \leq V_{_Test} + V_{tolerance}$	
Roll RL	On	x km/h	wheel RL	max:	xx	km/h	$V_{RLmax} \leq V_{RLØ} + V_{WSS_tolerance}$	
				min:	xx	km/h	$V_{RLmin} \geq V_{RLØ} - V_{WSS_tolerance}$	
				Ø	xx	km/h	$V_{_Test} - V_{tolerance} \leq FR_{LØ} \leq V_{_Test} + V_{tolerance}$	
Roll RR	On	x km/h	wheel RR	max:	xx	km/h	$V_{RRmax} \leq V_{RRØ} + T_{WSS_tolerance}$	
				min:	xx	km/h	$V_{RRmin} \geq V_{RRØ} - V_{tolerance}$ WSS_tolerance	
				Ø	xx	km/h	$V_{_Test} - V_{tolerance} \leq FR_{RØ} \leq V_{_Test} + V_{tolerance}$	

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

3. test step (optional): WSSi additional information „rotation direction” 测试步骤（可选）：WSSi 附加信息“旋转方向”					
Roll FL	On	x km/h	rotation direction FL	$RD_{FL} = RD_{_demand}$	$t_{_WSS_rotation_direction}$
Roll FR	On	x km/h	rotation direction FR	$RD_{FR} = RD_{_demand}$	
Roll RL	On	x km/h	rotation direction RL	$RD_{RL} = RD_{_demand}$	
Roll RR	On	x km/h	rotation direction RR	$RD_{RR} = RD_{_demand}$	
4. test step (optional): WSSi additional information „mounting position” 测试步骤（可选）：WSSi 附加信息“安装位置”					
Roll FL	On	x km/h	mounting position FL	$MP_{FL} = MP_{_demand}$	$t_{_WSS_mounting_position}$
Roll FR	On	x km/h	mounting position FR	$MP_{FR} = MP_{_demand}$	
Roll RL	On	x km/h	mounting position RL	$MP_{RL} = MP_{_demand}$	
Roll RR	On	X km/h	mounting position RR	$MP_{RR} = MP_{_demand}$	

Remarks: 备注：

1. test step: wheel speed sensor correlation test: 测试步骤：轮速传感器关联性测试

To check the correlation of the wheel speed sensors (connection of the WSS with the related wheel) the test rig switches on the rolls step by step with the **test speed $V_{_Test}$**. In parallel the tester takes the wheel speed data of the ECU and compares them with the **speed $V_{_roll_on}$** und **$V_{_roll_off}$** during the **test time $t_{_WSS_correlation}$**.
试验台以 **$V_{_Test}$** 的测试速度依次启动转毂，以检查轮速传感器关联性（各 WSS 与相应车轮的联系）。同时，测试设备采集 ECU 中的轮速数据并与测试时间 **$t_{_WSS_correlation}$** 内的速度 **$V_{_roll_on}$** 和 **$V_{_roll_off}$** 进行对比。

2. test step: wheel speed sensor signal quality test: 测试步骤：轮速传感器信号质量测试

During the signal quality test all rolls are switched on (this test step will be executed immediately after the correlation test). The tester starts the in the ECU software included WSS test routine for the **test time $t_{_WSS_signal_quality}$** (time depends on the test speed and wheel circumference), which picks up with the **sampling time $t_{_sampling}$** all sensor speed data. The selected test time has to cover a complete revolution of a wheel.
在信号质量测试中，所有转毂都将启动（该测试步骤在关联性测试结束后立即开始）。测试设备启动 ECU 软件内置的 WSS 测试程序，测试时间为 **$t_{_WSS_signal_quality}$**（时长取决于测试速度与车轮周长），在采样时间 **$t_{_sampling}$** 时采集所有传感器的速度数据。所选择的测试时间长度必须能够保证车轮旋转过完整一周。

The WSS test routine stores the **minimum, maximum and average speed data (V_{FLmin}, V_{FLmax}, $V_{FLØ}$, etc.)** of each wheel, which were measured during the selected test time. After the end of the test is reached, the routine outputs these data on request of the tester.
The test equipment evaluates these data and compares them with the calculated limit values from the average speed data and the **tolerance $V_{WSS_tolerance}$**. Additional the tester has to check if the measured average speed of each wheel is within a defined speed range. This is done by comparing of the individual average wheel speed and the test rig test speed ± **tolerance $V_{tolerance}$**. **For more details about the evaluation please refer the chapter 4.2.1.4. Details about the used diagnostic function can be read in the chapter 5.3.**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

WSS 测试程序储存每个车轮的**最小、最大和平均速度数据**（**V_{FLmin}, V_{FLmax}, $V_{FLØ}$** 等等），这些数据是在测试时间内测得的。在测试结束时，程序将应测试设备要求输出这些数据。
测试设备评估这些数据，并将其与由平均速度数据和**公差 $V_{WSS_tolerance}$** 计算得出的极限值进行对比。此外，测试设备应当检查所测得的各车轮平均速度是否在规定的速度范围之内，这是通过对比单个车轮平均速度与试验台测试速度**±公差 $V_{tolerance}$** 来实现的。**关于评估更多信息请参考 4.2.1.4 小节，关于所用诊断功能的信息请参考 5.3 小节。**

<u>3. & 4. test step (optional): WSSi rotation direction and mounting position test:</u> 测试步骤（可选）：WSSi 旋转方向和安装位置测试

These additional and optional test steps are only required for vehicles with intelligent wheel speed sensors (e.g. Bosch WSS11i). These tests are directly executed after the standard correlation and signal quality tests. To ensure that the WSS weren't wrongly mounted the tester reads and checks the information "rotation direction" and "mounting position". These data have to be read with the corresponding diagnostic function during one revolution of the wheel. At first the information "rotation direction of all wheels" within the **test time $t_{_WSS_rotation_direction}$** is read and compared with the expected value **rotation direction demand RD_{demand}**. During the following **test time $t_{_WSS_mounintg_position}$** (time depends on the test speed and wheel circumference) the information "mounting position of all wheels" is read 3 times, at the beginning, in the middle and at the end of the test time. The read ECU data are compared with the expected **mounting position demand MP_{demand}**. The requirement to read this data only 3 times is sufficient, because sporadic occurred fault information is stored in the ECU for the **storing time t_{St}**. During this time the stored data can be read via the corresponding diagnostic service. **For more details about the evaluation please refer the chapter 4.2.1.5 and 4.2.1.6.**
这些附加和可选的测试步骤仅适用于装有智能轮速传感器（如博世 WSS11i）的车辆。这些测试将在标准的关联性测试和信号质量测试结束后立刻进行。为了保证 WSS 安装正确，测试设备将读取并检查"旋转方向"和"安装位置"信息。这些数据应当在车轮单次旋转中由相应的诊断功能读取。首先，读取**测试时间 $t_{_WSS_rotation_direction}$** 内的"所有车轮旋转方向"信息，并与期望值**旋转方向要求 RD_{demand}** 进行对比。在紧接着的测试时间 **$t_{_WSS_mounintg_position}$**（时长取决于测试速度与车轮周长）内，读取"所有车轮安装位置"信息 3 次，分别是在测试时间的最初、中间和最后阶段。读取的 ECU 数据将与期望值**安装位置要求 MP_{demand}** 进行对比。考虑到偶发故障信息将在**存储时间 t_{St}** 内保存在 ECU 中，读取数据 3 次已经足够。在这段时间内，储存的数据可以通过相应的诊断服务来读取。**关于评估更多信息请参考 4.2.1.5 和 4.2.1.6 小节**

<u>Parameter:</u> 参数：

Kind of parameters:	Parameter name:	Unit:	Physical value:
Time constant in the software 软件中的时间常量	Sampling time $t_{_sampling}$ 采样时间	Milliseconds (ms)	5
	Storing time t_{St} 存储时间		3000
Time 时间	Test time $t_{_WSS_correlation}$ 测试时间		tbd
	Test time $t_{_WSS_signal\ quality}$ 测试时间		tbd
	Test time $t_{_WSS_rotation_direction}$ 测试时间		tbd
	Test time $t_{_WSS_mounting_position}$ 测试时间		tbd

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Condition 条件	Lower test speed $V_{_min}$ 最低测试速度	Kilometre per hour Km/h)	≥ 3
	Upper test speed $V_{_max}$ 最高测试速度		≤ 15
Evaluation parame-ter 评估参数	Test speed $V_{_roll_on}$ 测试速度		$= V_{Test} - T_{Vvalid}$
	Test speed $V_{_roll_off}$ 测试速度		1
	tolerance $V_{WSS_tolerance}$ 公差	Percent (%)	tbd
	tolerance $V_{tolerance}$ 公差		10
	Rotation direction RD_{demand} 旋转方向要求	---	forward
	Mounting position MP_{demand} 安装位置要求		tbd

Notes to the given parameters: 给定参数注意事项：

The test times depends on the test rig speed, the times to accelerate the rolls and the used tire sizes. These times are to be defined and specified by the customer.
For the definition of the test times $t_{_WSS_signal_quality}$ and $t_{_WSS_mounitng_position}$ following formula is valid:
测试时间取决于试验台速度、转毂加速时间和使用的轮胎尺寸。这些时间必须由客户指定。
在定义测试时间 $t_{_WSS_signal_quality}$ 和 $t_{_WSS_mounitng_position}$ 时，可以参考下列公式：

Test time= wheel circumference / test rig speed 测试时间=车轮周长/试验台速度

Example: 范例

Test rig speed 试验台速度: 4.5 km/h (1,25 m/s)

Wheel circumference 车轮周长: 2 Meter

Test time 测试时间 = 1,6 seconds (2m / 1,25 m/s)

→ recommended test time with some safety: 2 seconds
推荐安全测试时间：2 秒

To ensure that all important wheel speed information (wheel speed signals respectively mounting position (air gap) information – only vehicles with WSS11i) are available, the **test speed $V_{_Test}$** might not below the **lower test speed $V_{_min}$**.
为了确保采集到所有重要的车轮速度信息（车轮速度信号与安装位置（气隙）信息 – 仅限装有 WSS11i 的车辆），测试速度 **$V_{_Test}$** 不应低于最低测试速度 **$V_{_min}$**。

To ensure the highest sampling rate per tooth respectively speed resolution, the **test speed $V_{_Test}$** might not exceed the **upper test speed $V_{_max}$**.
为了确保每个齿的采样率和速度的精度达到最高，测试速度 **$V_{_Test}$** 不应超过最高测试速度 **$V_{_max}$**。

As low as possible the test speed is selected, even more is the sampling rate per tooth. Generally following reference values are valid:

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

测试车速选择的越低，每个齿的采样率就越高。通常有下列对应关系：

Speed (km/h) 车速	Samplings per tooth 每齿采样数
2,5	12
5	6
10	3
15	**2**

Details to all 4 test steps and evaluations can be seen in the graphics on the following pages.
关于所有 4 个测试步骤及其评估的细节请参考下页示意图。

<u>The execution of the wheel speed sensor tests have to ensure the following functions and assembly processes:</u>
进行轮速传感器测试以确保下列功能和装配过程正常：

1) To ensure the correct correlation between the wheels speed sensors and the wheels
   确保轮速传感器和车轮之间关联性正确
2) To ensure the signal quality regarding missing tooth respectively single division error
   确保信号质量，没有缺齿和孤峰错误。
3) Correct installation (evaluation of the additional WSSi information “rotation direction”)
   正确安装（附加的 WSSi “旋转方向” 信息评估）
4) To ensure the correct air gap respectively magnetic field intensity (evaluation of the additional WSSi information “mounting position”)
   确保气隙和磁场强度正确（附加的 WSSi “安装位置” 信息评估）

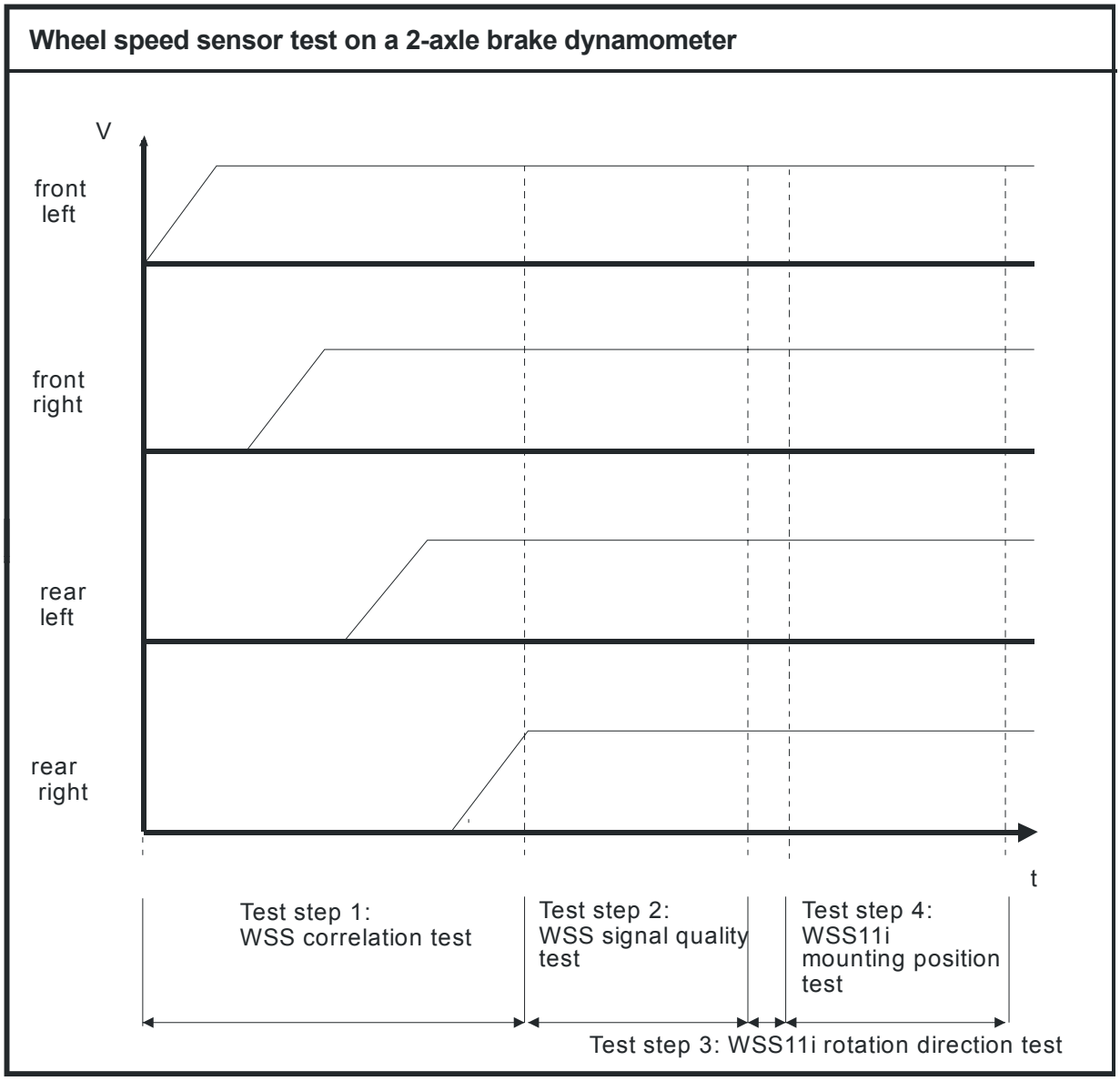
No WSS failure detected ⇒ Continue

WSS failure detected ⇒ Failure message, abort, prints out fault message and send vehicle to repair area

未检测到轮速传感器故障 ⇒ 继续下线检测流程
检测到轮速传感器故障 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.1.1 Graphical presentation of the entire wheel speed sensor tests (test step 1-4) 轮速传感器整个测试示意图（测试步骤 1-4）

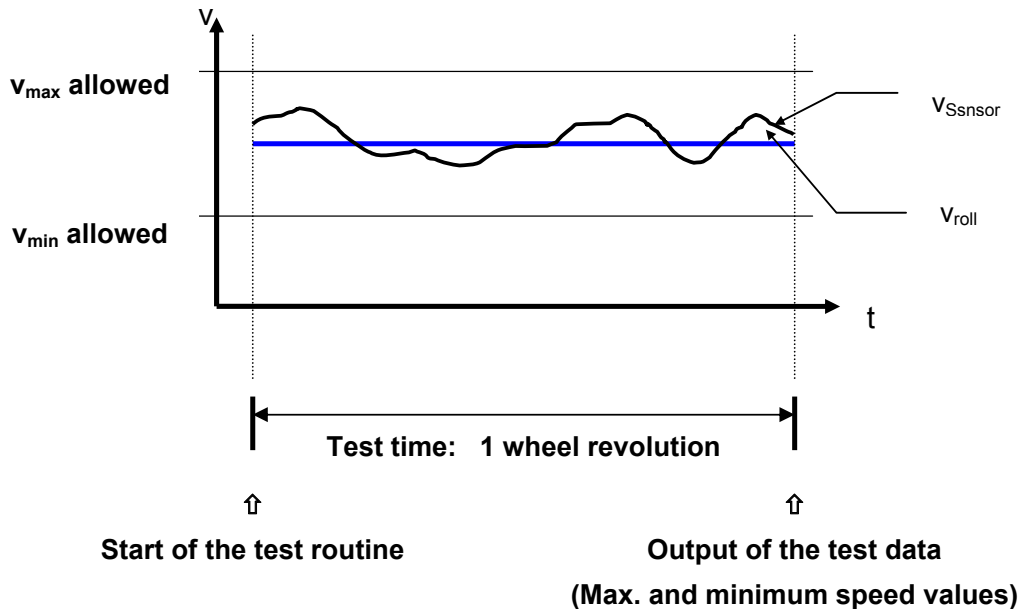


© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.1.2 Evaluation diagrams of the WSS signal quality test WSS 信号质量测试评估图

Wheel speed sensor test (signal quality test) 轮速传感器测试（信号质量测试）

Test diagram (Example at one wheel): 测试图（单个车轮范例）



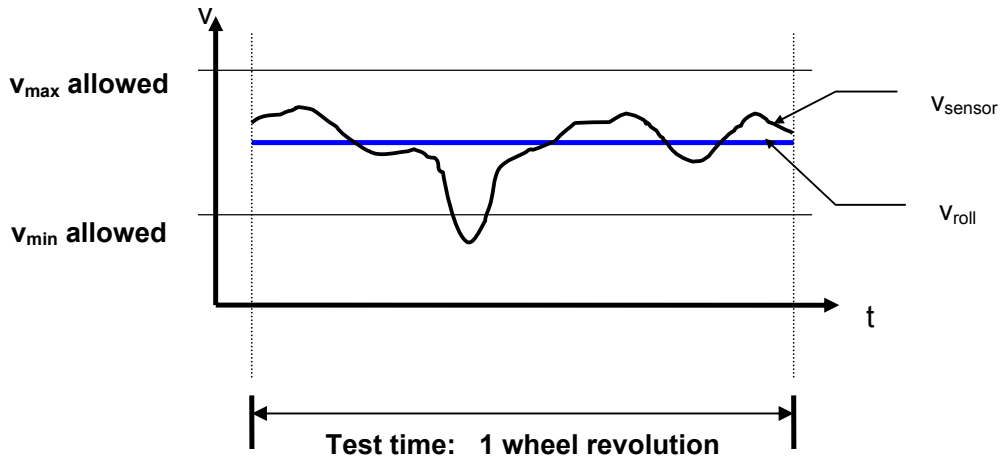
Remarks: 备注

The test rig turns on the individual rolls by external drives step by step while the tester requests the speed data of the sensors and checks the correct correlation.
The tester then runs a function (routine) that checks the sensor quality over a minimum of one revolution for all wheels. The ECU internally picks up each sensor speed and stores the minimum, maximum and average (optional) speed data within the selected test time. In order to identify faults of the sensor, toothed wheels, air gaps or eccentricities the tester has to evaluate whether the speeds are within a certain limit. These limits have to be defined to cover all WSS signal tolerances. The tolerance of test bench must be hold as small as possible. The selected test rig speed has to be within the function speed of the system.
试验台在外部驱动作用下依次启动各转毂，同一时间内测试设备采集传感器的速度数据并检查关联性是否正确。
测试设备随后运行程序，检查在所有车轮旋转至少一周情况下的传感器信号质量。ECU 将采集各传感器速度并储存所选测试时间内的最小、最大和平均（可选）速度数据。为了识别传感器、齿圈、气隙或偏心等故障，测试设备应当评估速度是否处于极限之内。这些极限值在定义之初就应考虑到将所有 WSS 信号公差包括在内。测试台的公差必须控制得尽可能小。所选择的试验台速度必须处于系统的运行速度范围之内。

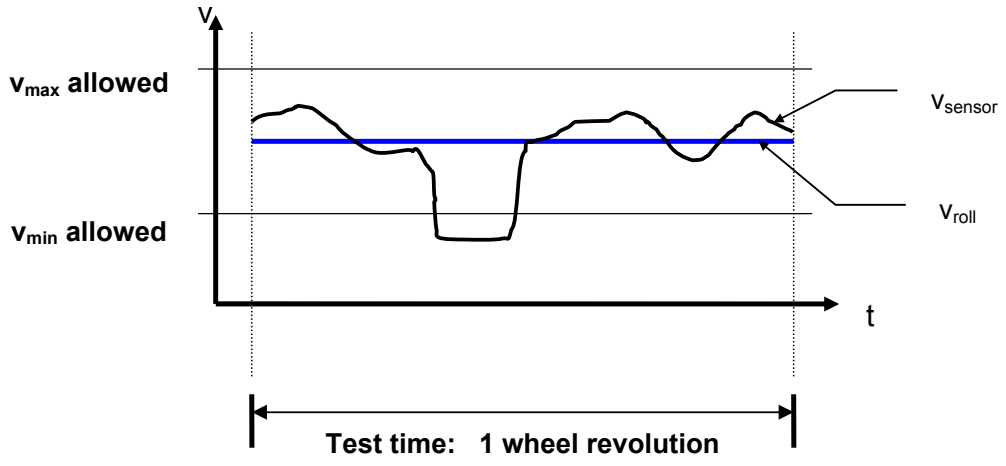
© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Fault simulation: 故障模拟

Fault simulation: faulty tooth wheel 故障模拟：齿圈故障



Fault simulation: Air gap to big or eccentricity 故障模拟：气隙过大或偏心



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.1.3 Exemplary determination of the allowed WSS signal tolerance WSS 信号公差允许值定义示范

Influences on the WSS signal quality test: 对 WSS 信号质量测试的影响因素	Nominal tolerance in %: 名义公差	Real tolerance in %: 实际公差	Remarks: 备注
Constant velocity respectively repetition accuracy of the test rig speed 试验台速度的恒定速度和循环精度	3,00	2,00	The nominal test rig tolerance has to be taken out of the test rig documentation. The real test rig tolerance should be measured and then used as test tolerance. The definition of this tolerance is up the customer responsibility. 试验台名义公差由试验台相关文件提供。必须测量试验台实际公差并将其作为测试公差。该公差定义应由客户指定。
Signal tolerance (WSS and tooth wheel) 信号公差（WSS 和齿圈）	5,00	2,50	The real signal tolerance is normally better than the specified nominal value by Bosch. Consequently this real signal tolerance has to be defined by a statistic evaluation and then be used as test tolerance. The definition of this tolerance is up the customer responsibility. 信号公差实际值在正常情况下要优于博世指定的名义值。因此，应当用统计学评估方法来定义该信号公差实际值，并将其作为测试公差。该公差定义应由客户指定。
Total tolerance of the sum of all single tolerances (SST) 所有单一公差和的总公差（SST）	---	4,50	Maximal tolerance of the sum of all single tolerances. 所有单一公差和的最大公差
Total tolerance of the radical of the square sum of all single tolerances (RSS) 所有单一公差平方和的平方根的总公差（RSS）	---	3,20	Maximal tolerance of the statistic evaluation. 统计学评估的最大公差
Tolerance $V_{WSS_tolerance}$: 公差 **$V_{WSS_tolerance}$**	---	**3,85**	To guarantee the most exact test of the WSS signals, not the nominal tolerance should be used, but the average value of total tolerance of the sum of all single tolerances (SST) and total tolerance of the radical of the square sum of all single tolerances (RSS). 为了确保 WSS 信号测试结果精确，应使用 SST 和 RSS 两者平均值，而不应使用名义公差。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.1.4 Evaluation proposal for the WSS test (signal quality regarding single pitch deviation, eccentricity, etc.): WSS 测试（测试信号质量包括孤峰偏离、偏心等）的评估建议

1. step: Comparison of the maximal measured wheel speeds (V_{FLmax}, etc.) with the average wheel speeds ($V_{ØFL,}$ etc.) plus the **tolerance $V_{WSS_tolerance}$**:
步骤：将最大轮速测量值（V_{FLmax} 等）与平均轮速（$V_{ØFL}$ 等）加上公差 **$V_{WSS_tolerance}$** 的和进行比较：

step 1.1: $V_{FLmax} \leq V_{ØFL} + V_{WSS_tolerance}$
step 1.2: $V_{FRmax} \leq V_{ØFR} + V_{WSS_tolerance}$
step 1.3: $V_{RLmax} \leq V_{ØRL} + V_{WSS_tolerance}$
step 1.4: $V_{RRmax} \leq V_{ØRR} + V_{WSS_tolerance}$

2. step: Comparison of the minimal measured wheel speeds (V_{FLmin}, etc.) with the average wheel speeds ($V_{ØFL,}$ etc.) minus the **tolerance $V_{WSS_tolerance}$**:
步骤：将最小轮速测量值（V_{FLmin} 等）与平均轮速（$V_{ØFL}$ 等）减去公差 **$V_{WSS_tolerance}$** 的差进行比较：

step 2.1: $V_{FLmin} \geq V_{ØFL} - V_{WSS_tolerance}$
step 2.2: $V_{FRmin} \geq V_{ØFR} - V_{WSS_tolerance}$
step 2.3: $V_{RLmin} \geq V_{ØRL} - V_{WSS_tolerance}$
step 2.4: $V_{RRmin} \geq V_{ØRR} - V_{WSS_tolerance}$

3. step: To ensure, that all wheels are within a plausible speed during this test, the measured average speed of each wheel has additional to be compared with the test rig speed V_{test} plus respectively minus the **tolerance $V_{tolerance}$**:
步骤：为了确保测试中所有车轮速度真实，测量的各车轮平均速度必须与试验台速度 V_{test} 加上/减去公差 **$V_{tolerance}$** 的和/差进行一次额外对比：

step 3.1 $V_{test} – V_{tolerance} \leq V_{ØFL} \leq V_{test} + V_{tolerance}$
step 3.2: $V_{test} - V_{tolerance} \leq V_{ØFR} \leq V_{test} + V_{tolerance}$
step 3.3: $V_{test} - V_{tolerance} \leq V_{ØRL} \leq V_{test} + V_{tolerance}$
step 3.4: $V_{test} - V_{tolerance} \leq V_{ØRR} \leq V_{test} + V_{tolerance}$

<u>Example:</u> 范例：

Measured minimal, maximal and average speeds (km/h) of each wheel:
各车轮的最小、最大和平均车速测量值(km/h)：

V_{FLmin}	V_{FLmax}	V_{FRmin}	V_{FRmax}	V_{RLmin}	V_{RLmax}	V_{RRmin}	V_{RRmax}
4,4	4,6	4,5	4,6	4,2	4,5	4,4	4,65
$V_{ØFL}$	$V_{ØFR}$	$V_{ØRL}$	$V_{ØRR}$				
4,48	4,48	4,48	4,48				

Test rig speed (V_{test}): 4,5 km/h
试验台速度(V_{test}): 4,5 km/h

<u>Judgement:</u> 判断：

1.1 step: V_{FLmax} (4,6 ≤ 4,652) = ok
1.2 step: V_{FRmax} (4,6 ≤ 4,652) = ok
1.3 step: V_{RLmax} (4,3 ≤ 4,652) = ok
1.4 step: V_{RRmax} (4,65 ≤ 4,652) = ok
2.1 step: V_{FLmin} (4,4 ≥ 4,308) = ok

3.1 step: $V_{ØFL}$ (4,032 ≤ 4,48 ≤ 4,928) = ok
3.2 step: $V_{ØFR}$ (4,032 ≤ 4,48 ≤ 4,928) = ok
3.3 step: $V_{ØRL}$ (4,032 ≤ 4,48 ≤ 4,928) = ok
3.4 step: $V_{ØRR}$ (4,032 ≤ 4,48 ≤ 4,928) = ok

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

2.2 step: V_{FRmin} (4,5 ≥ 4,308) = ok
2.3 step: V_{RLmin} (4,2 ≥ 4,308) = nok
2.4 step: V_{RRmin} (4,4 ≥ 4,308) = ok

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.1.5 Evaluation of the additional information "rotation direction" of a intelligent WSS (e.g. Bosch WSS11i) – optional 智能 WSS（如博世 WSS11i）附加信息"旋转方向"的评估–可选

Description „RD_{Demand}": 描述	Judgement: 判断
Condition not correct (measured value not valid) 条件不正确（测量值无效）	nok
Rotation direction forward 旋转方向向前	**ok**
Rotation direction backward 旋转方向向后	nok

Remarks: 备注：

This diagnostic service is sent by the tester during the wheel is turned by the test rig. The ECU outputs the actual identified rotation direction. An additional rearward drive test is not required, but it can be executed instead of the forward drive test.
该诊断服务是由测试设备在试验台转动车轮过程中发送的。ECU 输出识别的实际旋转方向。附加的后向驱动测试是可选测试，当后向驱动测试执行后，前向驱动测试就不必再执行。

4.2.1.6 Evaluation of the additional information "mounting position" of a intelligent WSS (e.g. Bosch WSS11i) – optional 智能 WSS（如博世 WSS11i）附加信息"安装位置"的评估–可选

Description „MP_{demand}": 描述	Judgement: 判断	Possible fault causes: 可能故障原因
Mounting position of the WSS ok WSS 安装位置合格	**Ok 合格**	**---**
Mounting position of the WSS near of the limits WSS 安装位置接近极限	Conditional failure* 故障取决于具体情况	Magnetic field intensity (mT) ≤ 5,2 till 9,6 (typical 7,2) caused by: • Sensor neck or bracket damaged or burred • Dirt between sensor neck and the bracket • Mounting screw loosen • Axle or WSS bracket have not the right dimension stability • Weak magnetic field intensity of the tooth wheel 磁场强度（mT）介于 5,2 至 9,6 之间（典型值 7,2），原因如下： • 传感器颈部或支架损坏 • 传感器颈部和支架间有脏物 • 安装螺栓松动 • 轴或 WSS 支架尺寸稳定性不正确 • 齿圈磁场强度不足
Faulty mounting position of the WSS WSS 安装位置错误	Failure 故障	Magnetic field intensity (mT) ≤ 0,9 till 2,6 (typical 1,4) caused by: • Sensor neck or bracket damaged or burred • Dirt between sensor neck and the bracket • Mounting screw loosen • Axle or WSS bracket have not the right dimension stability • Weak magnetic field intensity of the tooth wheel 磁场强度（mT）介于 0,9 至 2,6 之间（典型值 1,4），原因如下： • 传感器颈部或支架损坏

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

		• 传感器颈部和支架间有脏物 • 安装螺栓松动 • 轴或 WSS 支架尺寸稳定性不正确 • 齿圈磁场强度不足
Conditions not fulfilled (measured value not valid) 条件不满足（测量值无效）	Failure 故障	• Electrical defect (e.g. WSS line interrupted, etc.) • Info not available, because $\mathbf{V}_{_min} > V_{_Test} > \mathbf{V}_{_max}$ • 电气故障（如 WSS 断线等） • 由于 $\mathbf{V}_{_min} > V_{_Test} > \mathbf{V}_{_max}$ 导致信息无法取得

Remarks: 备注：

During the **test time** $\mathbf{t}_{_WSS_mounting_position}$ the tester has to evaluate the mounting position information 3 times. The first time directly after the start of the test time, the second at the middle and the third time at the end of the test. The evaluation of 3 times is enough because a sporadic occurring mounting position fault information is stored for the **storing time** $\mathbf{t_{St}}$, so that this information can be dependably read via the diagnostics. Please not that the test time must be defined to cover one wheel revolution.
在测试时间 $\mathbf{t}_{_WSS_mounting_position}$ 内，测试设备应当评估安装位置信息 3 次。在测试时间一开始立即进行第 1 次评估，在中间阶段和测试最后分别进行第 2 次和第 3 次评估。考虑到偶发故障信息将在**存储时间** $\mathbf{t_{St}}$ 内保存在 ECU 中，读取数据 3 次已经足够，因此该信息可以放心地通过诊断功能来读取。请注意测试时间长度应保证能够使车轮旋转过完整一周。

* Depending on the design of the air gap (between WSS and tooth wheel) the information “mounting position of the WSS near of the limits“ have to be judged as good or ok case. In case of the range of all possible air gap tolerances allows judging the information “mounting position of the WSS near of the limits as not good or nok case; Bosch recommends strongly doing that, because of more safeness.
* 根据气隙（介于 WSS 与齿圈之间）的设计，应判断“WSS 安装位置接近极限”信息为良好或合格。但当考虑到所有可能的气隙公差范围时，也允许判断“WSS 安装位置接近极限”信息为非良好或不合格。为了安全起见，博世强烈推荐采取后者。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.2 Base brake- and ABS test 基础制动测试和 ABS 测试

Base brake test (inclusive the optional leakage test) and ABS test (function "pressure decrease and increase" and brake pipe correlation test) 基础制动测试（包括可选泄漏测试）和 ABS 测试（"增减压"功能和制动管路关联性测试）				
Operator: 操作者	Test rig: 试验台	To be evaluated brake forces: 待评估制动力	Demanded brake forces: 要求的制动力	Measure sample: 测量样本
→ Don't apply Brake pedal! 禁止踩制动踏板	Measuring of the freewheeling forces 测量自由转动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \le F_1$ $F_{FR} \le F_1$ $F_{RL} \le F_2$ $F_{RR} \le F_2$	M1
→ Apply the brake pedal till the expected brake forces are reached! 踩下制动踏板直至达到期望的制动力	Pressure increase at all wheels 所有车轮增压 Measuring of the brake forces 测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_3 \le F_{FL} \le F_4$ $F_{FL} - F_{brake\ tolerance} \le F_{FR} \le F_{FL} + F_{Brake\ tolerance}$ $F_5 \le F_{RL} \le F_6$ $F_{RL} - F_{Brake\ tolerance} \le F_{RR} \le F_{RL} + F_{_Brake_tolerance}$	M2
→ Keep the brake pedal constant! 保持制动踏板不动	Measuring of the brake forces before and after the **measure time t_9** → leakage test (optional) 在**测量时间 t_9** 前后测量制动力→泄漏测试（可选）	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \le F_9$ $F_{FR} \le F_9$ $F_{RL} \le F_9$ $F_{RR} \le F_9$	M3.1 M3.2
→ Keep the brake pedal constant! 保持制动踏板不动	Pressure decrease FL Measure brake forces 前左轮减压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \le F_7$ $F_3 \le F_{FR} \le F_4$ $F_5 \le F_{RL} \le F_6$ $F_5 \le F_{RR} \le F_6$	M4
→ Keep the brake pedal constant! 保持制动踏板不动	Pressure increase FL Pressure decrease FR Measure brake forces 前左轮增压，前右轮减压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \ge F_3$ $F_{FR} \le F_7$ $F_5 \le F_{RL} \le F_6$ $F_5 \le F_{RR} \le F_6$	M5
→ Keep the brake pedal constant! 保持制动踏板不动	Pressure increase FR Measure brake forces 前右轮增压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \ge F_3$ $F_{FR} \ge F_3$ $F_5 \le F_{RL} \le F_6$ $F_5 \le F_{RR} \le F_6$	M6

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

→ Keep the brake pedal constant! 保持制动踏板不动	Pressure decrease RL Measure brake forces 后左轮减压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \geq F_3$ $F_{FR} \geq F_3$ $F_{RL} \leq F_8$ $F_5 \leq F_{RR} \leq F_6$	M7
→ Keep the brake pedal constant! 保持制动踏板不动	Pressure increase RL Pressure decrease RR Measure brake forces 后左轮增压，后右轮减压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \geq F_3$ $F_{FR} \geq F_3$ $F_{RL} \geq F_5$ $F_{RR} \leq F_8$	M8
→ Keep the brake pedal constant! 保持制动踏板不动	Pressure increase RR Measure brake forces 后右轮增压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel FL: xx N Wheel FR: xx N	$F_{FL} \geq F_3$ $F_{FR} \geq F_3$ $F_{RL} \geq F_5$ $F_{RR} \geq F_5$	M9
→ Release the brake pedal! 松开制动踏板	Measuring of the freewheeling forces 测量自由转动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_1$ $F_{FR} \leq F_1$ $F_{RL} \leq F_2$ $F_{RR} \leq F_2$	M10

Remarks: 备注

The base brake and ABS tests are executed directly after the WSS tests. As first step during the base brake test (inclusive the optional leakage test) the freewheeling and base brake forces are measured. In addition the optional leakage of the entire brake system inclusive the tightness of the AV valves of the ABS / ESP® system are tested.
在 WSS 测试后立即进行基础制动测试和 ABS 测试。基础制动测试（包括可选泄漏测试）的第一步是测量自由转动力和基础制动力。附加的泄漏测试（可选）将检查整个制动系统包括 ABS / ESP®的 AV 阀在内的密封性。

After the test rig rolls are switched on the freewheeling forces are measured, afterwards the operator is asked to apply the brake pedal immediately. As soon as the expected brake forces are reached, the operator has to keep the brake pedal constant and shouldn't release it till all brake forces of the base brake and the optional leakage test are measured. At the optional leakage test all EV and USV valves are switched on for a short time, to reduce the influence of the operator on this test.
当试验台转毂启动后，首先测量自由转动力，测量完后要求操作者立即踩下制动踏板。当制动力达到期望值后，操作者必须保持制动踏板不动，直至测量完基础制动测试和可选泄漏测试要求的所有制动力为止。在可选泄漏测试中所有 EV 和 USV 阀都将短暂打开，以降低操作者个人因素对试验的影响。

The ABS test is executed immediately after the base brake and optional leakage test. The operator has to keep the brake pedal still constant. The tester activates the corresponding valves and the pump motor step by step for the ABS test. In addition the tester measures the corresponding brake forces at each step and wheel.
ABS 测试将在基础制动测试和可选泄漏测试后立即进行。操作者仍然必须保持制动踏板不动。测试设备将依次启动相应的电磁阀和泵马达。另外，测试设备将在每一步骤中测量各车轮相应的制动力。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

All mentioned force values and tolerances can be read in table in the chapter 4.2.3.4.
上述所有力与公差值请参考 **4.2.3.4** 小节中列表。

The operator has to apply, keep constant and release the brake pedal after he was requested by the tester!
在测试设备发出请求后，操作者必须相应地踩下、保持不动、松开制动踏板。

The respective brake forces differences are calculated by the tester and compared with the fixed limits.
测试设备计算出制动力差异并将其与固定极限值进行对比。

To reduce the entire cycle time of the ABS test, the test cycle respectively corresponding valve activation for each wheel can overlap.
为了降低 ABS 测试整个循环时间，测试循环和各车轮相应的阀动作可以相互重叠。

The base brake test has to be executed at all vehicles independent of the installed control system. The test is required to ensure the function and tightness of base brake.
在所有车辆上都必须进行基础制动测试，这与车辆安装的控制系统无关。该测试用于确保基础制动系统的功能和密封性正常。

The ABS test can be used as well as for vehicles with an ABS or ESP® system. In case of a 3 channel ABS system is used, the described test procedure for the wheel RL is valid for the total rear axle. The described test step for the wheel RR can be removed, but only in case of a 3 channel system.
ABS 测试亦适用于装有 ABS 或 ESP®系统的车辆。如果 ABS 系统为 3 通道系统，上述后左轮测试步骤可以用于整个后轴，而后右轮测试步骤可以去掉，但这仅限于 3 通道系统。

The base brake test, the optional leakage and ABS test have to ensure the following functions and assembly processes:
执行基础制动测试、可选泄漏测试和 ABS 测试以确保下列功能和装配过程正常：

1) To ensure the function respectively tightness of the base brake system inclusive ABS / ESP® tightness (to exclude AV leakage)
   确保基础制动系统功能和密封性正常，且 ABS / ESP®（排除 AV 阀泄漏）密封性正常。
2) To ensure the correlation of the brake pipes between the ABS / ESP® hydraulic ports and each wheel
   确保 ABS / ESP®液压单元的各个轮缸回路螺纹孔通过制动管路关联到正确的相应车轮
3) To ensure the function of the EV and AV valves after brake filling process (e.g. to exclude an impairment of function caused by contamination, etc.)
   在加注制动系统（例如：为了消除污染物对系统功能造成的损害）后，确保 EV 和 AV 阀功能正常
4) To ensure delivery rate of both pumps
   确保两个泵单元的泵液速度正常

ABS test ok ⇒ **Continue**

ABS Test nok ⇒ **Failure message, abort, print out fault message and send vehicle to repair area**

ABS 测试合格 ⇒ 继续下线检测流程

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

ABS 测试不合格 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.2.2.1 Test sequence of the ABS test ABS 测试序列

Start mes-sage:	Operator prompt: 操作者提示信息	Activation before the time: 时间前动作	Time:	Activation after the time: 时间后动作	Measure sample: 测量样本
---	Don't apply Brake pedal! 禁止踩下制动踏板	---	---	---	M1
---	Apply the brake pedal till the ex-pected brake forces are reached! 踩下制动踏板直至达到制动力期望值	---	---	---	M2
1	Keep the brake pedal constant! 保持制动踏板不动	Pump motor on 启动泵马达	---	---	---
2		Inlet valves of the front axle on 启动前轴进液阀	---	Inlet valves of the rear axle on 启动后轴进液阀	M3.1
3		Both USV (only ESP® systems) on 启动所有 USV(仅限 ESP®系统)	t9	---	M3.2
4		Both USV (only ESP® systems) off 断电所有 USV(仅限 ESP®系统)	---	---	---
5		AVFL on 启动前左 AV	t5	AVFL off 断电前左 AV	M4
6		EVFL off 断电前左 EV	t1.3	EVFL on 启动前左 EV	---
7		AVFR on 启动前右 AV	t5	AVFR off 断电前右 AV	M5
8		EVFR off 断电前右 EV	t1.3	EVFR on 启动前右 EV	M6
9		AVRL on 启动后左 AV	t5	AVRL off 断电后左 AV	M7
10		EVRL off 断电后左 EV	t1.4	EVRL on 启动后左 EV	---
11		AVRR on 启动后右 AV	t5	AVRR off 断电后右 AV	M8
12		EVRR off 断电后右 EV	t1.4	EVRR on 启动后右 EV	M9
13	Release the brake pedal! 松开制动踏板	Inlet valves of the front axle off 断电前轴进液阀	---	Inlet valves of the rear axle off 断电后轴进液阀	---
14		---	t8	Pump motor off 断电泵马达	M10

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Remarks: 备注：

As first step all test rig rolls are switched on and then the freewheeling forces are measured (measure sample M0) at all wheels. These values are subtracted from the afterwards measured brake forces. As next step the operator is prompt to apply the brake pedal till the expected brake force range is reached (M2). Afterwards the operator has to keep the brake pedal constant.
首先试验台所有转毂都将启动，并测量所有车轮的自由转动力（测量样本 M0）。这些值将从后面测得的制动力中减去。下一步操作者将被提示踩下制动踏板直至达到制动力期望范围（M2），接下来操作者必须保持踩住制动踏板不动。

Then the tester switches on the pump motor (start message 1) and all inlet valves (start message 2). At ESP® systems both USV valves are additionally switched on (start message 3), to reduce the influence of the operator on this test. After this step the brake forces are measured before and after the time t9 at all wheels. With this step the tightness of the base brake system is tested respectively ensured (M3.1 & M3.2). At the end of this step both USV valves are switched off (start message 4). Afterwards the pressure in the wheel FL is released by switching on of the outlet valve for the time t5 (start message 5). After this test step the brake forces are measured at all wheels to ensure the pressure decrease at the wheel FL (M4).
随后测试设备启动泵马达（启动信息 1）和所有进液阀（启动信息 2），如果是 ESP®系统还需要启动所有 USV 阀（启动信息 3）以降低操作者个人因素对测试的影响。该步骤完成后，测量时间 t9 前后所有车轮的制动力，从而检查并确认基础制动系统的密封性（M3.1 & M3.2）。在这一步的最后，断电所有 USV 阀（启动信息 4）。接下来通过在 t5 的时间内启动出液阀来释放前左轮的压力（启动信息 5）。之后，测量所有车轮的制动力以确保前左轮减压（M4）。

As next step the inlet valve is immediately opened for the time t1.3 to increase the pressure in the wheel FL. In parallel the pressure in the wheel FR is released by switching on of the outlet valve for the time t5 (start messages 6 & 7). After this test step the brake forces are measured at all wheels again to ensure the pressure increase at the wheel FL and the pressure decrease at the wheel FR (M5).
下一步，进液阀将立即打开 t1.3 的时间以给前左轮增压。同时，通过在 t5 的时间内启动出液阀来释放前右轮的压力（启动信息 6&7）。在这一测试步骤之后，再度测量所有车轮的制动力以确保前左轮增压和前右轮减压（M5）。

Thereupon the pressure of the wheel FR is increased again by opening of the inlet valve for the time t1.3 (start message 8). In addition the brake forces are measured (M6).
因此，通过在 t1.3 的时间内打开进液阀，前右轮的压力将再度增大（启动信息 8）。另外还要测量制动压力（M6）。

The test at the rear axle is done in the same way (Start messages 9 till 12; measure samples M7 till M9).
后轴测试方法与前轴相同（启动信息 9 至 12；测量样本 M7 至 M9）。

At the end of the ABS test the operator is prompt to release the brake pedal. Afterwards all inlet valves are immediately switched off (start message 13). The pump motor is switched off after the time t8 (start message 14). This is required to discharge the accumulator chambers. In parallel the brake forces of all wheels are measured again and then they are compared with the freewheeling forces which were measured at the beginning of the base brake test (M10).
在 ABS 测试最后操作者将被提示松开制动踏板，随后所有进液阀将被立即断电（启动信息 13）。泵马达将在时间 t8 后断电（启动信息 14），这是为了排空蓄液仓。同时，再度测量所有车轮的制动力，并与基础制动测试初期测得的自由转动力进行对比（M10）。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Details about the used diagnostic function can be read in the chapter 5.5.
关于采用的诊断功能具体信息请参考 **5.5** 小节。

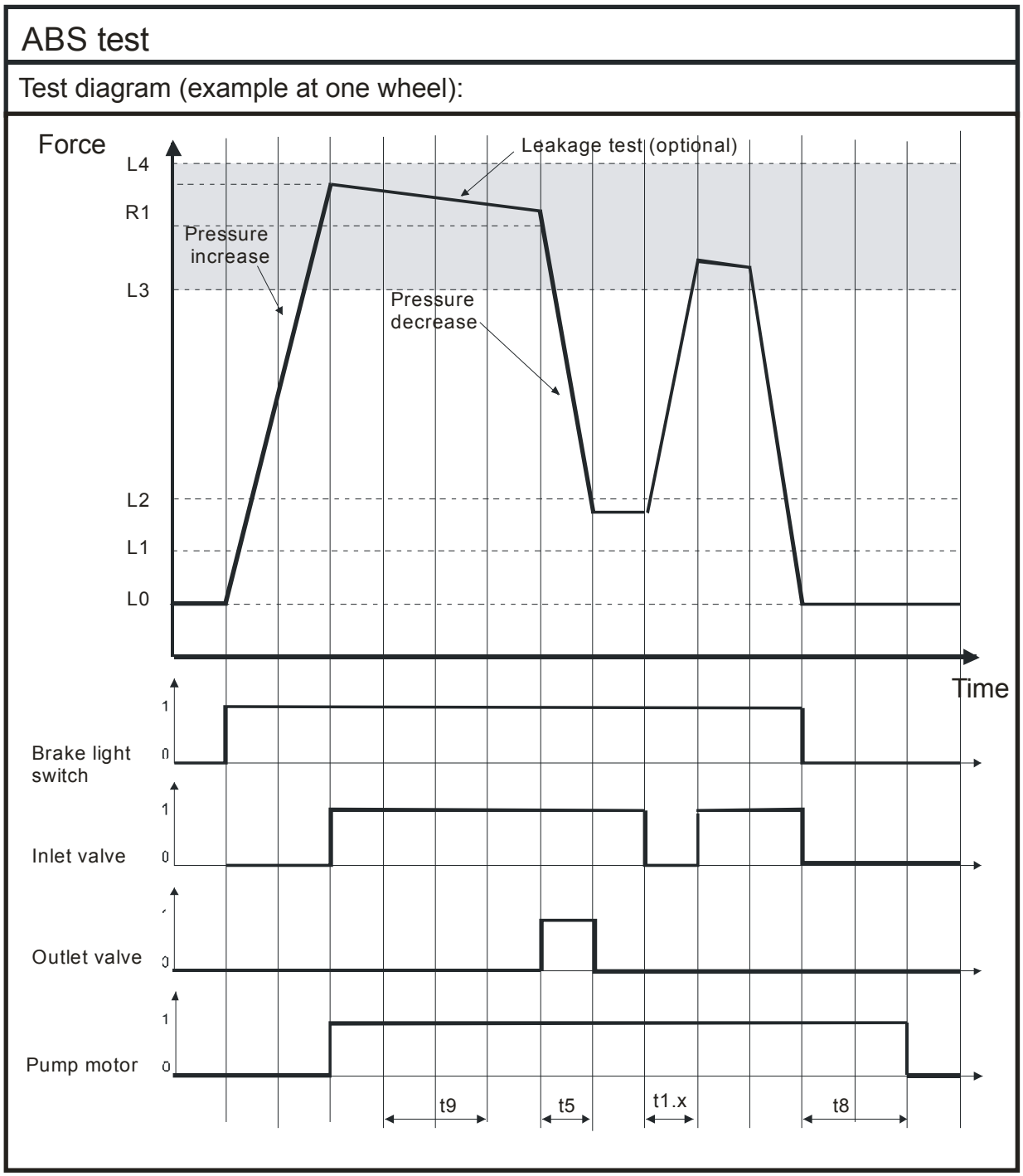
Parameter: 参数：

See chapter 4.2.3.4.
参考 4.2.3.4 小节

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.2.2.2 Diagram of the ABS test (example at one wheel) ABS 测试示意图（以单个车轮为例）



Remarks: 备注

- At first step the test rig (tester) measures the freewheeling forces. These values are compared with the maximal allowed limit (L1).
  试验台（测试设备）首先测量自由转动力，这些值将与最大允许极限（L1）进行对比。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

- Afterwards the operator is requested to apply the brake pedal and keep constant the entire test cycle.
  随后，操作者将被要求踩下制动踏板，并在整个测试循环中保持踩住不动。

- With applying of the brake pedal the brake forces are increased by the operator. The brake forces which have to be reached have to be within the defined limits (L3/L4). Afterwards the pump motor and all inlet valves are switched on.
  当踩下制动踏板时，操作者增大制动力，最终达到的制动力必须在规定的极限值之内（L3/L4）。之后，启动泵马达和所有进液阀。

- Now the optional leakage test is executed for the time t9. The measured brake forces before and after this time might not exceed the maximal allowed brake force difference (R1).
  现在在 t9 的时间内执行可选的泄漏测试。在该时间前后分别测得的制动力相比之下不应超过制动力最大允许差异值（R1）。

- As next step the tester switches on the corresponding outlet valve for the time (t2) to create a pressure decrease at the corresponding wheel. In addition the brake forces are measured and compared with the defined maximal allowed limit (L2).
  下一步，测试设备在 t2 的时间内启动出液阀，从而在相对应的车轮内实现减压。另外还将测量制动力并将其与规定的最大允许极限（L2）进行对比。

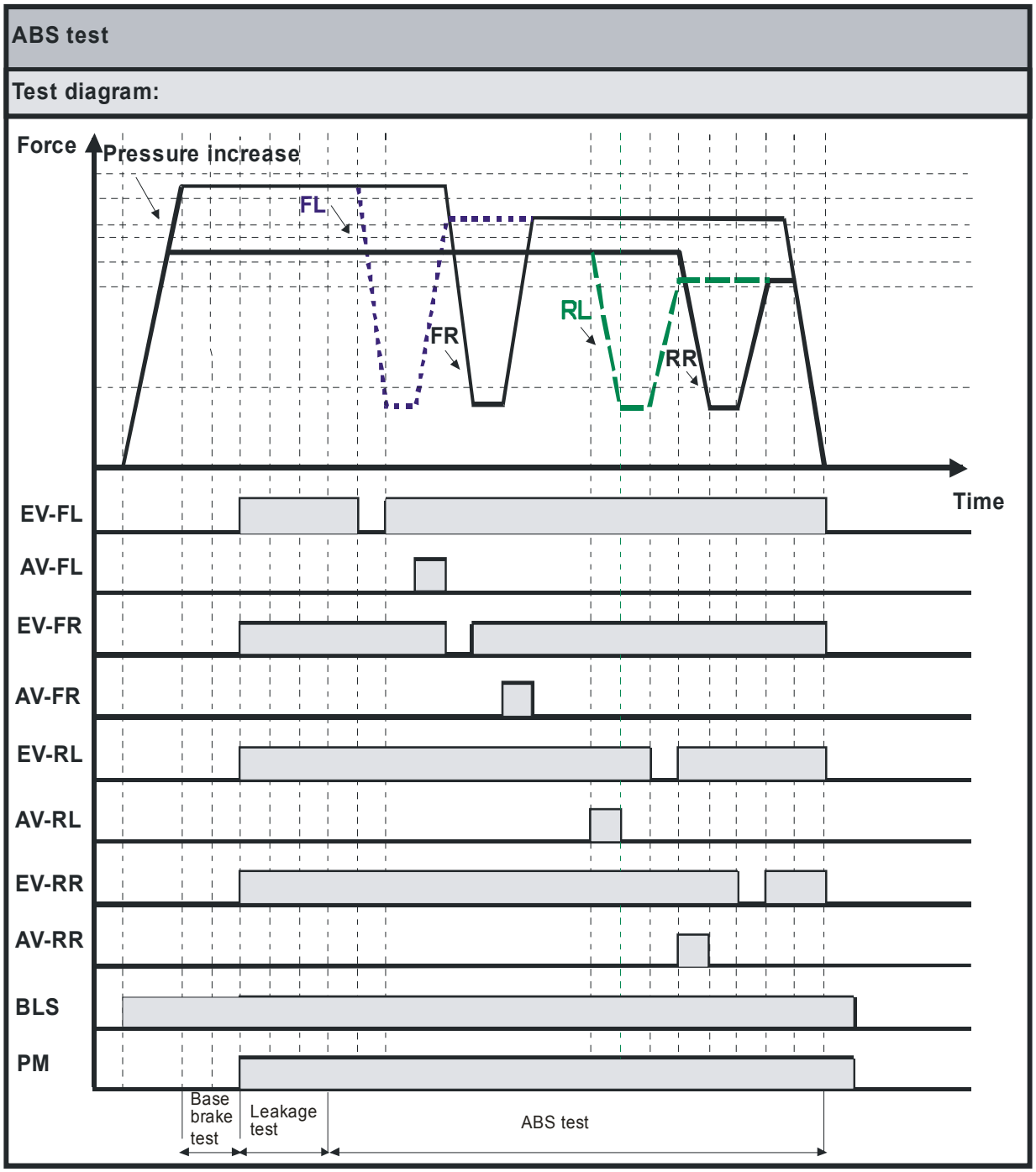
- Immediately after the previous test step is finished, the tester switches off the corresponding inlet valve for the time (t1.x) to create a pressure increase at the corresponding wheel. In addition the brake forces are measured and compared with the defined maximal allowed limit (L3).
  在上一个测试步骤结束后，测试设备将立即断电进液阀达 t1.x 的时间以在相对应的车轮内实现增压。另外还将测量制动力并将其与规定的最大允许极限（L3）进行对比。

- As soon as the complete test is finished the operator is requested to release the brake pedal. Afterwards all inlet valves are opened respectively switched off. To discharge the accumulator chamber, the pump motor is switched off first after the time t8.
  当整个测试一结束，操作者将被要求松开制动踏板。随后所有进液阀将断电打开。为了排空蓄液仓，泵马达将在时间 t8 后首先断电。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.2.2.3 Diagram of the ABS test at all wheels 所有车轮的 ABS 测试示意图



Remarks: 备注：

Please note the maximal allowed valve activation times. More information about these times can be read in the chapter 6.
请注意阀启动最大允许时间。关于这些时间更多信息请参考第 6 章。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.2.4 Proposal of a evaluation logic for the optional leakage and ABS test 可选泄漏测试和 ABS 测试的评估逻辑建议

a) Evaluation of the freewheeling force before the wheels are braked (exemplary for the wheel FL):
在对车轮制动前评估自由转动力（以前左轮为例）：

If
F_FL ≤ F1

then F_FL = F_FL_freewheel and execute test step b)

else Stop process and output “Freewheel force at the wheel FL nok”

如果 F_FL ≤ F1，则 F_FL = F_FL_freewheel 并执行测试步骤 b)；
否则停止流程并输出“前左轮自由转动力不合格”。

b) Evaluation base brake test (exemplary for the front axle):
评估基础制动测试（以前轴为例）：

F_FL_corr = F_FL – F_FL_freewheel
F_FR_corr = F_FR – F_FR_freewheel

If
F3 ≤ F_FL_corr ≤ F4
and
F_FL_corr – F_brake_tolerance ≤ F_FR_corr ≤ F_FL_corr + F_brake_tolerance

then execute test step c)

else Stop process and output “Base brake test at the front axle nok”

如果 F_FL = F_FL_freewheel 且 F_FL_corr – F_brake_tolerance ≤ F_FR_corr ≤ F_FL_corr + F_brake_tolerance，则执行测试步骤 c)；
否则停止流程并输出“前轴基础制动测试不合格”。

c) Evaluation of the optional leakage test (exemplary for the wheel FL):
评估可选泄漏测试（以前左轮为例）：

If
F_FL_1 (before the measure time) – F_FL_2 (after the measure time) ≤ F9

then execute test step d)

else Stop process and output “Leakage test at the wheel FL nok”

如果 F_FL_1 (before the measure time) – F_FL_2 (after the measure time) ≤ F9，
则执行测试步骤 d）；
否则停止流程并输出“前左轮泄漏测试不合格”。

d) Evaluation ABS test „pressure decrease“ (exemplary for the wheel FL):
评估 ABS 测试“减压”（ 以前左轮为例）：

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

If
F_FL_corr ≤ F7

then execute test step e)

else Stop process and output “ABS pressure decrease at the wheel FL nok”

如果 F_FL_corr ≤ F7，
则执行测试步骤 e）；
否则停止流程并输出“前左轮 ABS 减压不合格”。

e) Evaluation ABS test „pressure increase“ (exemplary for the wheel FL):
评估 ABS 测试“增压”（ 以前左轮为例）：

If
F_FL_corr ≥ F3

then execute test step f)

else Stop process and output “ABS pressure increase at the wheel FL nok”

如果 F_FL_corr ≥ F3，
则执行测试步骤 f）；
否则停止流程并输出“前左轮 ABS 增压不合格”。

f)) Evaluation of the freewheeling force after the wheels are released (exemplary for the wheel FL):
在放松车轮后评估自由转动力（以前左轮为例）：

If
F_FL ≤ F1

then output “Base brake test FA and ABS test at the wheel FL ok”

else output “Freewheel force at the wheel FL after the ABS test nok”

如果 F_FL ≤ F1，
则输出“前轴基础制动测试和前左轮 ABS 测试合格”；
否则输出“ABS 测试之后前左轮自由转动力不合格”。

Example: 范例：

Variables for evaluation*: 评估变量*：
F1 = 250N
F3 = 800N
F4 = 2000N
F7 = 350N
F9 = 200N
F_brake_tolerance = 25%
Measure time t9: 1000ms

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

*All described variables for evaluation are examples and must be finally defined by the customer!
*所有上述评估变量只是范例，最终还需由客户定义。

a) Measured brake forces (N) of each wheel at the test step „freewheeling force before the wheels are braked“:
测试步骤“车轮制动前的自由转动力”中测得的各车轮制动力（N）：

F_FL	F_FR	F_RL	F_RR
150	155	105	100

b) Measured brake forces (N) of each wheel at the test step „base brake test“:
测试步骤“基础制动测试”中测得的各车轮制动力（N）：

F_FL	F_FR	F_RL	F_RR
1550	1575	650	655

c) Measured brake forces (N) of each wheel at the test step „optional leakage test“:
测试步骤“可选泄漏测试”中测得的各车轮制动力（N）：

F1_FL	F1_FR	F1_RL	F1_RR
1500	1570	645	645
	Measure time t9		
F2_FL	F2_FR	F2_RL	F2_RR
1450	1560	640	645

d) Measured brake forces (N) of each wheel at the test step „pressure decrease“:
测试步骤“减压”中测得的各车轮制动力（N）：

F_FL	F_FR	F_RL	F_RR
350	1560	640	645

e) Measured brake forces (N) of each wheel at the test step „pressure increase“:
测试步骤“增压”中测得的各车轮制动力（N）：

F_FL	F_FR	F_RL	F_RR
1350	1560	640	645

f) Measured brake forces (N) of each wheel at the test step „freewheeling force after the wheels are released“:
测试步骤“放松车轮后的自由转动力”中测得的各车轮制动力（N）：

F_FL	F_FR	F_RL	F_RR
150	155	105	100

<u>Judgement of each test step for the wheel FL respectively FA:</u>
前左轮和前轴的每个测试步骤判断值

a) Freewheeling at the wheel FL before the base brake test: 150 ≤ 250 **= ok**
基础制动测试前的前左轮自由转动力：150 ≤ 250 **= ok**
b) Base brake test front axle: 800 ≤ 1400 ≤ 2000 && 1050 ≤ 1425 ≤ 1750 **= ok**
前轴基础制动测试：800 ≤ 1400 ≤ 2000 && 1050 ≤ 1425 ≤ 1750 **= ok**
c) Optional leakage test at the wheel FL: 50 ≤ 200 **= ok**
前左轮可选泄漏测试：50 ≤ 200 **= ok**
d) ABS test „pressure decrease FL“: 200 ≤ 350 **= ok**
ABS 测试“前左轮减压”：200 ≤ 350 **= ok**
e) ABS test „pressure increase FL“: 1200 ≥ 800 **= ok**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

ABS 测试“前左轮增压”：1200 ≥ 800 **= ok**

f) Freewheeling at the wheel FL after the ABS test: 150 ≤ 250 **= ok**
ABS 测试后的前左轮自由转动力：150 ≤ 250 **= ok**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.3 ESP® test ESP®测试

ESP® test (function test „active pressure increase“) ESP®测试 (功能测试“主动增压”)				
Operator: 操作者	Test rig: 试验台	To be evaluated brake forces: 待评估制动力	Demanded brake forces: 要求的制动力	Measure sample: 测量样本
→ Don't apply Brake pedal! 禁止踩制动踏板	Measurement of the free-wheeling forces 测量自由转动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_1$ $F_{FR} \leq F_1$ $F_{RL} \leq F_2$ $F_{RR} \leq F_2$	M0
→ Don't apply Brake pedal! 禁止踩制动踏板	Pressure increase FL Measure brake forces 前左轮增压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \geq F_3$ $F_{FR} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M1+n
→ Don't apply Brake pedal! 禁止踩制动踏板	Pressure decrease FL Measure brake forces 前左轮减压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FR} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M2
→ Don't apply Brake pedal! 禁止踩制动踏板	Pressure increase FR Measure brake forces 前右轮增压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FR} \geq F_3$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M3+n
→ Don't apply Brake pedal! 禁止踩制动踏板	Pressure decrease FR Measure brake forces 前右轮减压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M4
→ Don't apply Brake pedal! 禁止踩制动踏板	Pressure increase RL Measure brake forces 后左轮增压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \geq F_5$ $F_{RR} \leq F_8$	M5+n
→ Don't apply Brake pedal! 禁止踩制动踏板	Pressure decrease RL Measure brake forces 后左轮减压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M6
→ Don't apply Brake pedal! 禁止踩制动踏板	Pressure increase RR Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \geq F_5$	M7+n

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

	后右轮增压，测量制动力			
→ Don't apply Brake pedal! 禁止踩制动踏板	Pressure decrease RR Measure brake forces 后右轮减压，测量制动力	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M8

Remarks: 备注：

The ESP® test is executed directly after the WSS tests. Due to the fact that all rolls are running, the freewheeling forces can be measured immediately (measure sample M0). Afterwards the tester executes step by step the required valve and pump motor activations and measures the brake forces at all wheels (M1 till M8).
During the entire ESP® test the brake pedal might not applied by the operator. To ensure that the tester should cyclic evaluate the brake light switch status between the individual pressure decrease cycles.
ESP®测试在 WSS 测试后立即进行。考虑到所有转毂还在旋转，可以立即测量自由转动力（测量样本 M0）。随后测试设备将依次启动必要的阀和泵马达，并测量所有车轮的制动力（M1 至 M8）。
在整个 ESP®测试中，操作者不得踩下制动踏板。为了确保这一点，测试设备应当在各减压循环之间周期性地评估制动灯开关状态。

All mentioned force values and tolerances can be read in the table in the chapter 4.2.3.4.
上述力和公差值请参考 4.2.3.4 小节中列表。

Details to ensure the tightness and the function of base brake system can be read in the chapter 4.2.2.
关于"确保基础制动系统密封性和功能正常"具体信息请参考 4.2.2 小节。

The ESP® Test has to ensure the following functions and assembly processes:
执行 ESP®测试以确保下列功能和装配过程正常：

1) To ensure the correlation of the brake pipes between the ESP® hydraulic ports and each wheel
   确保 ESP®液压单元的各个轮缸回路螺纹孔通过制动管路关联到正确的相应车轮
2) To ensure the tightness respectively function of the EV and AV valves after brake filling process (e.g. to exclude an impairment of function caused by contamination, etc.)
   在加注制动系统（例如：为了消除污染物对系统功能造成的损害）后，确保 EV 和 AV 阀密封性和功能正常
3) To ensure delivery rate of both pumps respectively the active ESP® pressure increase function at all wheels
   确保两个泵单元的泵液速度以及 ESP®在所有车轮上的主动增压功能正常

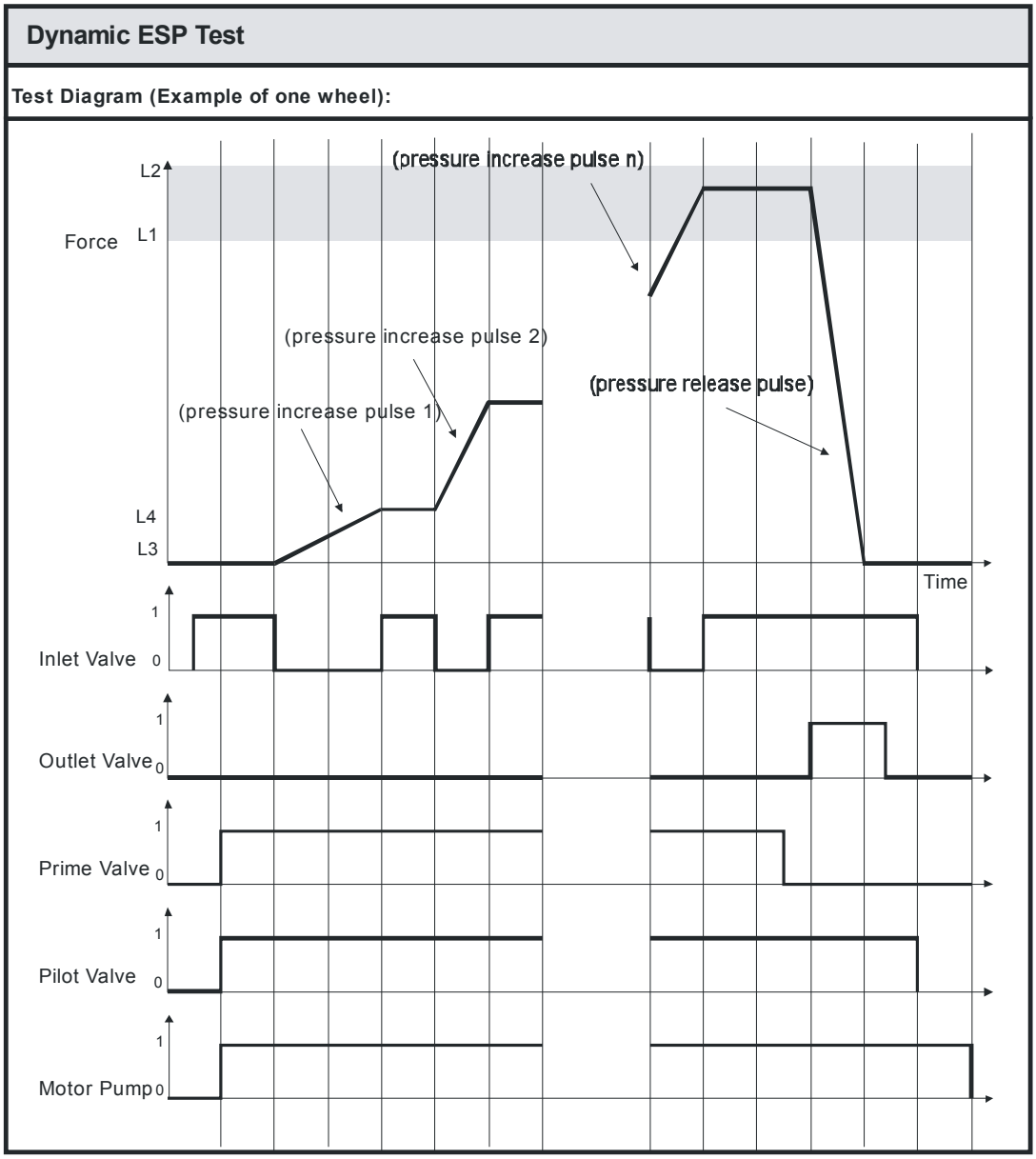
ESP® test ok ⇒ Continue

ESP® test nok ⇒ Failure message, abort, print out fault message and send vehicle to repair area

ESP®测试合格 ⇒ 继续下线检测流程
ESP®测试不合格 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.3.1 Schematically sequence exemplary at one wheel 单一车轮的测试序列示意图



Remarks: 备注：

The hydraulic unit creates a pressure by running of a loop of pressure increase pulses. This loop is executed as long as the defined limit (L1) is reached. The limit (L2) should not be exceeded.The first pressure increase pulse is longer than the following. This is required to reduce the air gap between the brake disk and pads.

液压单元通过运行单个循环的增压脉冲来产生制动压力。该循环执行时间应当达到规定极限 L1，但不超过极限 L2。第一个增压脉冲的时间将比之后的增压脉冲长，这是为了降低制动盘和摩擦垫片之间的气隙。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

After the tester executed a pressure release function, the tester measures the brake forces and compares them with maximal allowed limits (L3/L4).
在减压功能执行完后，测试设备将测量制动力并将其与最大允许极限（L3/L4）进行对比。

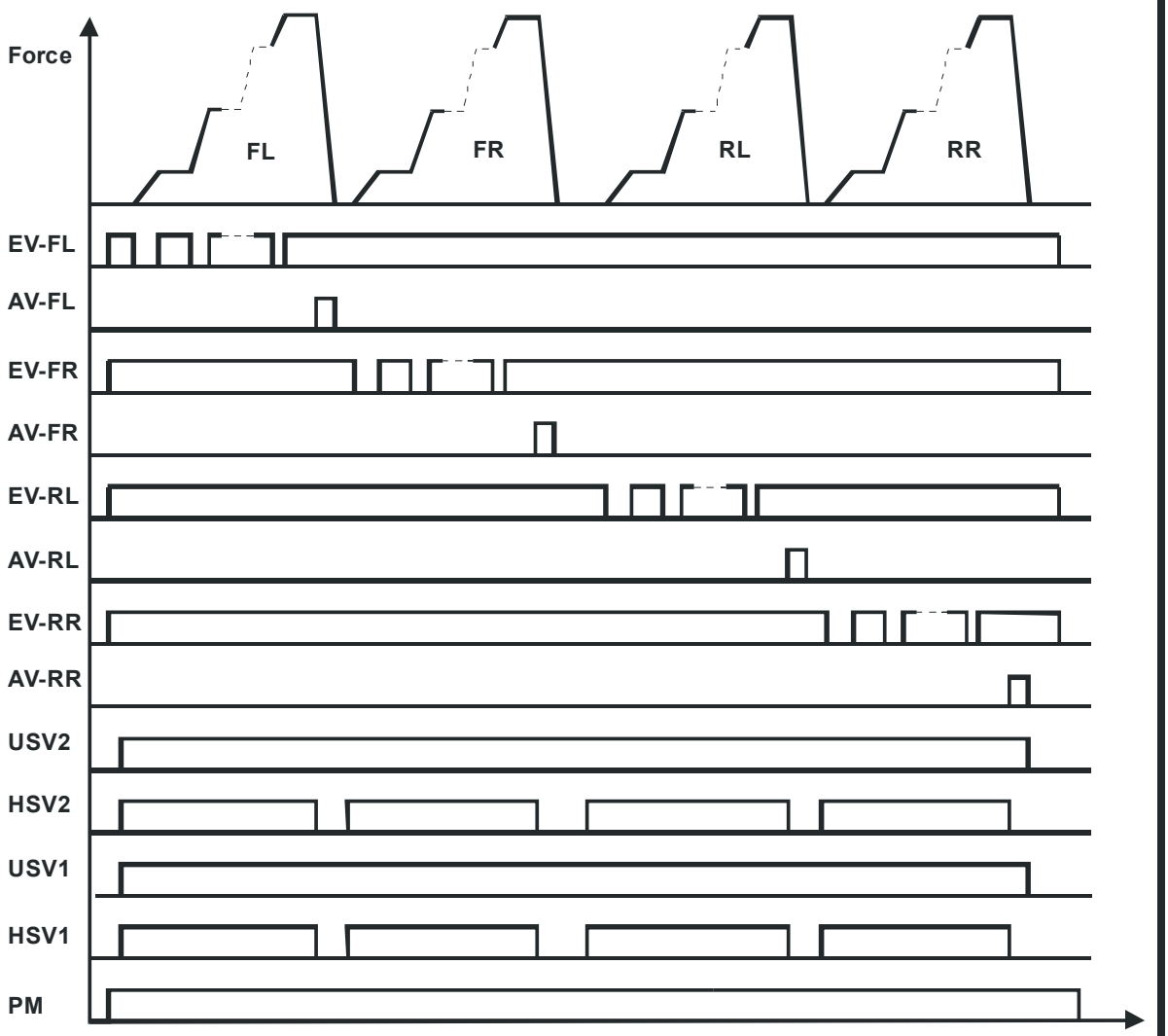
The diagram shows the test cycle at one wheel. The total test cycle includes this test at each wheel.
该示意图显示了单个车轮的测试循环。整个测试循环包括每个车轮的测试。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.3.2 Graphical presentation of the entire ESP® tests on a 2 axle brake dynamometer 基于双轴制动测力计的 ESP®完整测试示意图

ESP test

Test diagram:



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.3.3 Test sequence for the ESP® test ESP®测试序列

Start mes-sage: 启动信息	Operator prompt: 操作者提示信息	Activation before the time 时间前动作	Time:	Activation after the time: 时间后动作	Measure sample: 测量样本
--	Don't apply Brake pedal! 禁止踩制动踏板！	---	---	---	M0
1		Pump motor on 启动泵马达	---	---	---
2		Inlet valves front axle on 启动前轴进液阀	---	Inlet valves rear axle on 启动后轴进液阀	---
3		USV1 and USV2 on	---	HSV1 and HSV2 on	---
4		EVFL off	t1.1	EVFL off	M1
5.1 – 5.n		EVFL off	t2	EVFL off	M1+n
6		HSV1 & HSV2 off	---	---	---
7		AVFL on	t5	AVFL off	M2
8		HSV1 & HSV2 on	---	---	---
9		EVFR off	t1.1	EVFR on	M3
10.1 – 10.n		EVFR off	t2	EVFR on	M3+n
11		HSV1 & HSV2 off	---	---	---
12		AVFL on	t5	AVFL off	M4
13		HSV1 & HSV2 on	---	---	---
14		EVRL off	t1.2	EVRL on	M5
15.1 – 15.n		EVRL off	t2	EVRL on	M5+n
16		HSV1 & HSV2 off	---	---	---
17		AVRL on	t5	AVRL off	M6
18		HSV1 & HSV2 on	---	---	---
19		EVRR off	t1.2	EVRR on	M7
20.1 – 20.3		EVRR off	t2	EVRR on	M7+n
21		HSV1 & HSV2 off	---	---	---
22		AVRR on	t5	AVRR off	M8
23		USV1 & USV2 off	---	---	---

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

24		Inlet valves front axle off 断电前轴进液阀	---	Inlet valves rear axle off 断电后轴进液阀	---
25		---	t8	Pump motor off 断电泵马达	---

Remarks: 备注：

The operator might not apply the brake pedal during the entire ESP® Test, because the brake (system) pressure is active created by the ESP® hydraulic unit.
由于系统制动压力是由 ESP®液压单元主动产生的，操作者在整个 ESP®测试过程中不得踩下制动踏板。

After the freewheeling forces are measured (measure sample M0) at all wheels, the tester switches on the pump motor (start message 1) at first, then all inlet valves (start message 2) and all USV respectively HSV valves (start message 3). To get a sufficient brake force increase at the wheel FL, the tester runs a number of pressure increase pulses. The first pulse should be longer than the following. This is required to reduce the air gap between the brake pads and disk (start message 4, time t1.1). Afterwards the tester runs a number of loops with a smaller pressure increase time (start message 5.1 till 5.n, time t2). This is required to control the brake force in very small steps. After the expected brake force is reached, the tester switches off both HSV valves (start message 6) at first, then the pressure of the wheel FL is released by switching on of the outlet valve (start message 7, time t5).
在测完所有车轮的自由转动力（测量样本 M0）后，测试设备首先启动泵马达（启动信息 1），然后再启动所有进液阀（启动信息 2）、USV 和 HSV 阀（启动信息 3）。为了保证前左轮制动力充分增大，测试设备将运行若干个增压脉冲。第一个增压脉冲的时间应当长于之后的脉冲，这是为了降低摩擦垫片和制动盘之间的气隙（启动信息 4，时间 t1.1）。随后，测试设备运行若干个增压时间相对较短的循环（启动信息 5.1 至 5.n，时间 t2），从而以极小的幅度来控制制动压力。在达到预期的制动力后，测试设备将首先断电所有 HSV 阀（启动信息 6），然后再通过启动出液阀（启动信息 7，时间 t5）来释放前左轮的压力。

Between and after each individual test step all brake forces are measured to ensure the active pressure increase respectively pressure release function at the wheel FL (M1 - M1+n, M2).
测量每个单独测试步骤之中和之后的所有制动力，以确保前左轮的主动增压和减压功能正常（M1 - M1+n, M）。

To be able to execute the same procedure at the wheel FR both HSV valves have to be switched on again (start message 8). Afterwards the pressure is increased at first, then released at the wheel FR (start messages 9-12).
当在前右轮执行相同步骤时，所有 HSV 阀必须再次启动（启动信息 8）。之后，对前右轮先增压再减压（启动信息 9-12）。

Between and after each individual test step all brake forces are measured to ensure the active pressure increase respectively pressure release function at the wheel FR (M3 – M3+n, M4).
测量每个单独测试步骤之中和之后的所有制动力，以确保前右轮的主动增压和减压功能正常（M3 – M3+n, M4）。

The test at the rear axle is done in the same way (start messages 13 till 22). To ensure the active pressure increase respectively pressure release function at the rear axle all brake forces are measured between and after each individual test step (M5 – M5+n, M6 respectively. M7 – M7+n, M8).

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

后轴测试方法与前轴相同（启动信息 13 至 22）。为了确保后轴的主动增压和减压功能正常，应测量每个单独测试步骤之中和之后的所有制动力（M5 – M5+n, M6 和 M7 – M7+n, M8）。

At the end of the ESP® tests all activated valves are switched off (start messages 23-24). Then the pump motor is switched off after the time t8 (start message 25).
在 ESP®测试结束时，断电所有已启动的阀（启动信息 23-24）。然后在 t8 的时间后断电泵马达（启动信息 25）。

Details about the used diagnostic function can be read in the chapter 5.5.
关于采用的诊断功能具体信息请参考 5.5 小节。

Parameter: 参数

See chapter 4.2.3.4. 参考 4.2.3.4 小节

Important remark: 重要备注：

At ESP® systems pilot valves without pressure limitation are used. To protect these systems against over pressure, the pressure is limited by controlling of the pilot valve current. This current corresponds to the maximal system pressure (fluctuation by the allowed tolerances possible). More details can be read in the chapter 6.
ESP®系统中使用的回路控制阀没有限压。为了避免系统压力过大，将通过调节回路控制阀的电流来限制压力。该电流 与最大系统压力相对应（可能在允许公差范围内波动）。更多信息请参考第 6 章。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.3.4 Parameter (Base brake, ABS and ESP® test) 参数（基础制动测试、ABS 测试和 ESP®测试）

Kind of parameter:	Parameter name:		Unit:	Physical value:
Time 时间	1. Pressure increase pulse FA (ESP® test) 前轴增压脉冲（ESP®测试）	t1.1	Milliseconds (ms)	120
	1. Pressure increase pulse RA (ESP® Test) 后轴增压脉冲（ESP®测试）	t1.2		80
	General pressure increase pulse FA (ABS test) 常规前轴增压脉冲（ABS 测试）	t1.3		60
	General pressure increase pulse RA (ABS test) 常规后轴增压脉冲（ABS 测试）	t1.4		40
	Pressure increase pulse for the loop (general) 循环内的增压脉冲（常规）	t2		20
	Pressure decrease pulse (general) 减压脉冲（常规）	t5		300
	Post run time of the pump motor 泵马达滞后运转时间	t8		2000
	Measure time EV leakage test EV 泄漏测试测量时间	t9		1000
Number 数量	Number of loops for the pressure increase 增压循环数量	x1	---	5
Evaluation Parameter 评估参数	Freewheeling force FA 前轴自由转动力	F1	Newton (N)	250
	Freewheeling force RA 后轴自由转动力	F2		150
	Force after pressure increase min. value FA 前轴增压后的制动力最小值	F3		800
	Force after pressure increase max. value FA 前轴增压后的制动力最大值	F4		2000
	Force after pressure increase min. value RA 后轴增压后的制动力最小值	F5		400
	Force after pressure increase max. value RA 后轴增压后的制动力最大值	F6		1200
	Force after pressure decrease value FA 前轴减压后的制动力值	F7		350
	Force after pressure decrease value RA 后轴减压后的制动力值	F8		250
	Force after EV leakage test max. value EV 泄漏测试后的制动力最大值	F9 (R1)		200
	Tolerance $F_{_brake_tolerance}$ 公差	---	Percent (%)	25

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Remarks: 备注：

The mentioned times are proposed times such as the activation times for the pressure increase and release cycles, the stabilisation and measure times. All these times depend on the vehicle (e.g. size of brake, orifices in the hydraulic unit, etc.) respectively the test rig, therefore these times must be confirmed respectively modified during trials at the customer plant.
上述时间值为建议值，例如增压减压循环的启动时间、稳定时间和测量时间等。所有这些时间值都取决于车辆（如制动系统尺寸、液压单元螺纹孔径等）和试验台，因此必须在客户工厂中通过试验确认或调整。

Also the mentioned evaluation parameter have to be defined by a statistic evaluation with the first tested vehicles .The definition of all these data is up the customer responsibility.
同时，上述的评估参数必须在最初测试车辆上通过统计学评估来定义。所有数据的定义取决于客户。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.2.3.5 Proposal of a evaluation logic for the ESP® test ESP®测试的评估逻辑建议

a) Evaluation of the freewheeling force before the wheels are braked (exemplary for the wheel FL):
在对车轮制动前评估自由转动力（以前左轮为例）：

If
F_FL ≤ F1

then F_FL = F_FL_freewheel and execute test step b)

else Stop process and output “Freewheel force at the wheel FL nok”

如果 F_FL ≤ F1，则 F_FL = F_FL_freewheel 并执行测试步骤 b)；
否则停止流程并输出“前左轮自由转动力不合格”。

b) Evaluation ESP® test „pressure increase“ (exemplary for the wheel FL):
评估 ESP®测试“增压”（以前左轮为例）：

F_FL_corr = F_FL – F_FL_freewheel

If
F_FL_corr ≥ F3

then execute test step c)

else Stop process and output “ESP® pressure increase at the wheel FL nok”

如果 F_FL_corr ≥ F3，则执行测试步骤 c)；
否则停止流程并输出“前左轮 ESP®增压不合格”。

c) Evaluation ESP® test „pressure decrease“ (exemplary for the wheel FL):
评估 ESP®测试“减压”（以前左轮为例）：

If
F_FL_corr ≤ F7

then output “ESP® test at the wheel FL ok”

else output “ESP® pressure decrease at the wheel FL nok”

如果 F_FL_corr ≤ F7，则输出“前左轮 ESP®测试合格”；
否则输出“前左轮 ESP®减压不合格”。

Example: 范例：

Variables for evaluation*: 评估变量*：
F1 = 250N
F3 = 800N
F7 = 350N

*All described variables for evaluation are examples and must be finally defined by the customer!

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

*所有上述评估变量只是范例，最终还需由客户定义。

a) Measured brake forces (N) of each wheel at the test step „freewheeling force before the wheels are braked“:
测试步骤“车轮制动前的自由转动力”中测得的各车轮制动力（N）：

F_FL	F_FR	F_RL	F_RR
150	155	105	100

b) Measured brake forces (N) of each wheel at the test step „ESP® test – pressure increase FL“:
测试步骤“ESP®测试-前左轮增压”中测得的各车轮制动力（N）：

F_FL	F_FR	F_RL	F_RR
1350	170	115	105

c) Measured brake forces (N) of each wheel at the test step „ESP® test – pressure decrease FL“:
测试步骤“ESP®测试-前左轮减压”中测得的各车轮制动力（N）：

F_FL	F_FR	F_RL	F_RR
350	175	120	115

<u>Judgement of each test step for the wheel FL:</u>
前左轮每个测试步骤判断值

a) Freewheeling at the wheel FL before the ESP® test: 150 ≤ 250 **= ok**
ESP®测试前的前左轮自由转动力：150 ≤ 250 **= ok**
b) ESP® test „pressure increase FL“: 1200 ≥ 800 **= ok**
ESP®测试“前左轮增压”：1200 ≥ 800 **= ok**
c) ESP® test „pressure decrease FL“: 200 ≤ 350 **= ok**
ESP®测试“前左轮减压”：200 ≤ 350 **= ok**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.3 Alternative: Dynamic test on a 4 roller test bench with 4 independent rolls (high speed) 替代方案： 在 4 滚筒测试台上进行 4 轮独立旋转动态测试（高速）

4.3.1 Wheel speed sensor quality tests (during acceleration of the vehicle – rotation direction: forward) 轮速传感器质量测试（在车辆加速过程中 – 旋转方向：向前）

Wheel speed sensor test (signal quality test) 轮速传感器测试（信号质量测试）			
Test speed $V_{_Test}$: 测试速度 $V_{_Test}$	ECU data to be evaluated: 待评估 ECU 数据	Proposal for evaluation: 评估建议	Test time: 测试时间
1. test step: Signal quality test 测试步骤：信号质量测试			
$V_{_min} \le V_{_Test} \le V_{_max}$	Wheel FL ΔV_max xx km/h Wheel FR ΔV_max xx km/h Wheel RL ΔV_max xx km/h Wheel RR ΔV_max xx km/h	$\Delta V_{FL_max} \le \Delta V_{WSS_test_accelerated_vehicle}$ $\Delta V_{FR_max} \le \Delta V_{WSS_test_accelerated_vehicle}$ $\Delta V_{RL_max} \le \Delta V_{WSS_test_accelerated_vehicle}$ $\Delta V_{RR_max} \le \Delta V_{WSS_test_accelerated_vehicle}$	$t_{_WSS_signal_quality_accelerated_vehicle}$
2. test step (optional): WSSi additional information „rotation direction“ 测试步骤（可选）：WSSi 附加信息“旋转方向”			
$V_{_min} \le V_{_Test} \le V_{_max}$	Rotation direction FL Rotation direction FR Rotation direction RL Rotation direction RR	$RD_{FL} = RD_{_demand}$ $RD_{FR} = RD_{_demand}$ $RD_{RL} = RD_{_demand}$ $RD_{RR} = RD_{_demand}$	$t_{_WSS_rotation\ direction}$
3. test step (optional): WSSi additional information „mounting position” 测试步骤（可选）：WSSi 附加信息“安装位置”			
$V_{_min} \le V_{_Test} \le V_{_max}$	Mounting position FL Mounting position FR Mounting position RL Mounting position RR	$MP_{FL} = MP_{_demand}$ $MP_{FR} = MP_{_demand}$ $MP_{RL} = MP_{_demand}$ $MP_{RR} = MP_{_demand}$	$t_{_WSS_mounting_position_accelerated_vehicle}$

Remarks: 备注：

1. test step: signal quality test: 测试步骤：信号质量测试

The signal quality test is executed during the acceleration phase before the ESP® or ABS test. As soon as the **lower speed limit $V_{_min}$** is reached, the tester starts a routine within the ECU software for the **test time$_{_WSS_signal_quality_accelerated_vehicle}$** (test time depends on the test speed and wheel circumference). The routine measures and compares with the **sampling time $t_{_sampling}$** the wheel speeds of all wheels within the selected test time or till the **stop speed V_{stop}** is reached. The routine stores the maximal calculated **difference speed per wheel (ΔV_{XX_max}/wheel$_{xx}$)** and outputs (have to

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

be requested by the tester) these data after the selected test time or the stop speed is reached. The tester evaluates these data and compares them with the defined **maximal difference speed $\Delta V_{WSS_test_accelerated_vehicle}$**. The maximal allowed value "maximal difference speed" depends on the deceleration respectively acceleration, WSS, tooth wheel and ECU tolerances. This value must be defined by the customer. **For more details see chapter 4.3.1.3**.
信号质量测试在 ESP® 或 ABS 测试之前的加速阶段中进行。当加速到**最低测试速度 $V_{_min}$** 时，测试设备启动 ECU 软件内置程序，测试时间为 **$t_{_WSS_signal_quality_accelerated_vehicle}$**（时长取决于测试速度与车轮周长）。在所选测试时间内或在达到**停止车速 V_{stop}** 前，该程序将测量所有车轮速度，并与**采样时间 $t_{_sampling}$** 时进行对比。测试程序还将储存**每个车轮的最大速度差（ΔV_{XX_max}/wheel$_{xx}$）**，并在所选测试时间结束或达到停止车速后输出这些数据（应测试设备要求）。测试设备评估这些数据，并将其与定义的**最大速度差异 $\Delta V_{WSS_test_accelerated_vehicle}$** 进行对比。"最大速度差异"允许的上限值取决于加减速度、WSS、齿圈和 ECU 公差，且必须由客户定义。**更多信息请参考 4.3.1.3 小节。**

Details about the used diagnostic function can be read in the chapter 5.4.
关于采用的诊断功能具体信息请参考 5.4 小节。

<u>2. & 3. test step (optional): WSSi rotation direction and mounting position test:</u>
测试步骤（可选）：WSSi 旋转方向和安装位置测试

These additional and optional test steps are only required for vehicles with intelligent wheel speed sensors (e.g. Bosch WSS11i). These tests are directly executed after the signal quality tests. To ensure that the WSS weren't wrongly mounted the tester reads and checks the information "rotation direction" and "mounting position". These data have to be read with the corresponding diagnostic function during one revolution of the wheel. At first the information "rotation direction of all wheels" within the **test time $t_{_WSS_rotation_direction}$** is read and compared with the expected value **rotation direction demand RD_{demand}**. During the following **test time $t_{_WSS_mounintg_position}$** (time depends on the test speed and wheel circumference) the information "mounting position of all wheels" are read 3 times, at the beginning, in the middle and at the end of the test time. The read ECU data are compared with the expected **mounting position demand MP_{demand}**. The requirement to read this data only 3 times is sufficient, because sporadic occurred fault information is stored in the ECU for the **storing time t_{St}**. During this time the stored data can be read via the corresponding diagnostic service. **For more details about the evolution please refer the chapter 4.2.1.5 and 4.2.1.6.**
这些附加和可选的测试步骤仅适用于装有智能轮速传感器（如博世 WSS11i）的车辆。这些测试将在信号质量测试结束后立刻进行。为了保证 WSS 安装正确，测试设备将读取并检查"旋转方向"和"安装位置"信息。这些数据应当在车轮单次旋转中由相应的诊断功能读取。首先，读取**测试时间 $t_{_WSS_rotation_direction}$** 内的"所有车轮旋转方向"信息，并与期望值**旋转方向要求 RD_{demand}** 进行对比。在紧接着的测试时间 **$t_{_WSS_mounintg_position}$**（时长取决于测试速度与车轮周长）内，读取"所有车轮安装位置"信息 3 次，分别是在测试时间的最初、中间和最后阶段。读取的 ECU 数据将与期望值**安装位置要求 MP_{demand}** 进行对比。考虑到偶发故障信息将在**存储时间 t_{St}** 内保存在 ECU 中，读取数据 3 次已经足够。在这段时间内，储存的数据可以通过相应的诊断服务来读取。**关于评估更多信息请参考 4.2.1.5 和 4.2.1.6 小节**

<u>Parameter:</u> 参数：

Kind of parameter:	Parameter name:	Unit:	Physical value:
Time constant in	Sampling time $t_{_sampling}$ 采样时间 $t_{_sampling}$	Milliseconds	5

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

the Software 软件中时间常量	Storing time t_{St} 存储时间 t_{St}	(ms)	3000
Time 时间	Test time $t_{_WSS_signal_quality_accelerated_vehicle}$ 测试时间 $t_{_WSS_signal_quality_accelerated_vehicle}$		tbd
	Test time $t_{_WSS_rotation\ direction}$ 测试时间 $t_{_WSS_rotation\ direction}$		≤ 100
	Test time $t_{_WSS_mounting_position_accelerated_vehicle}$ 测试时间 $t_{_WSS_mounting_position_accelerated_vehicle}$		tbd
Condition 条件	Lower test speed $V_{_min}$ 最低测试速度 $V_{_min}$	Kilometre per hour (Km/h)	≥ 3
	Upper test speed $V_{_max}$ 最高测试速度 $V_{_max}$		≤ 15
Evaluation parameter 评估参数	Maximal difference speed $\Delta V_{WSS_test_accelerated_vehicle}$ 最大速度差异 $\Delta V_{WSS_test_accelerated_vehicle}$		tbd
	Rotation direction RD_{demand} 旋转方向 RD_{demand}	---	forward
	Mounting position MP_{demand} 安装位置 MP_{demand}		tbd

Notes to the given parameters: 给定参数注意事项：

The test times depends on the start speed and the used tire sizes. These times are to be defined and specified by the customer.
For the definition of the test times $t_{_WSS_signal_quality_accelerated_vehicle}$ and $t_{_WSS_mounitng_position_accelerated_vehicle}$ following formula is valid:
测试时间取决于启动速度和使用的轮胎尺寸。这些时间必须由客户指定。
在定义测试时间 $t_{_WSS_signal_quality_accelerated_vehicle}$ 和 $t_{_WSS_mounitng_position_accelerated_vehicle}$ 时，可以参考下列公式：

test time = wheel circumference / start speed 测试时间=车轮周长/启动速度

Example: 范例：

Wheel circumference 车轮周长: 2 Meter

Start speed of the WSS-Signal quality test WSS 信号质量测试启动速度: 5 km/h (1,39 m/s)

Test time 测试时间= 1,44 seconds (2 m / 1,39 m/s)

→ recommended test time with some safety 推荐安全测试时间: 1,8 seconds

Start speed mounting position 安装位置测试启动速度: 12 km/h (3,33 m/s)

Test time 测试时间 = 0,60 seconds (2 m / 3,33 m/s)

→ recommended test time with some safety 安装位置测试启动速度: 0,75 seconds

To ensure that all important wheel speed information (wheel speed signals respectively mounting position (air gap) information – only vehicles with WSS11i) are available, the **test speed $V_{_Test}$** might not below the **lower test speed $V_{_min}$** respectively not exceed the **upper test speed $V_{_max}$**.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

为了确保采集到所有重要的车轮速度信息（车轮速度信号与安装位置（气隙）信息 – 仅限装有 WSS11i 的车辆），**测试速度 $V_{_Test}$** 应高于**最低测试速度 $V_{_min}$**，并不超过**最高测试速度 $V_{_max}$**。

Details to all 3 test steps and evaluations can be seen in the graphics on the following pages.
关于所有 3 个测试步骤及其评估的细节请参考下页示意图。

<u>The execution of the wheel speed sensor tests have to ensure the following functions and assembly processes:</u>
进行轮速传感器测试以确保下列功能和装配过程正常：

1) To ensure the signal quality regarding missing tooth respectively single division error
   确保信号质量，没有缺齿和孤峰错误。
2) Correct installation (evaluation of the additional WSSi information “rotation direction”)
   正确安装（附加的 WSSi “旋转方向” 信息评估）
3) To ensure the correct air gap respectively magnetic field intensity (evaluation of the additional WSSi information “mounting position”)
   确保气隙和磁场强度正确（附加的 WSSi “安装位置” 信息评估）

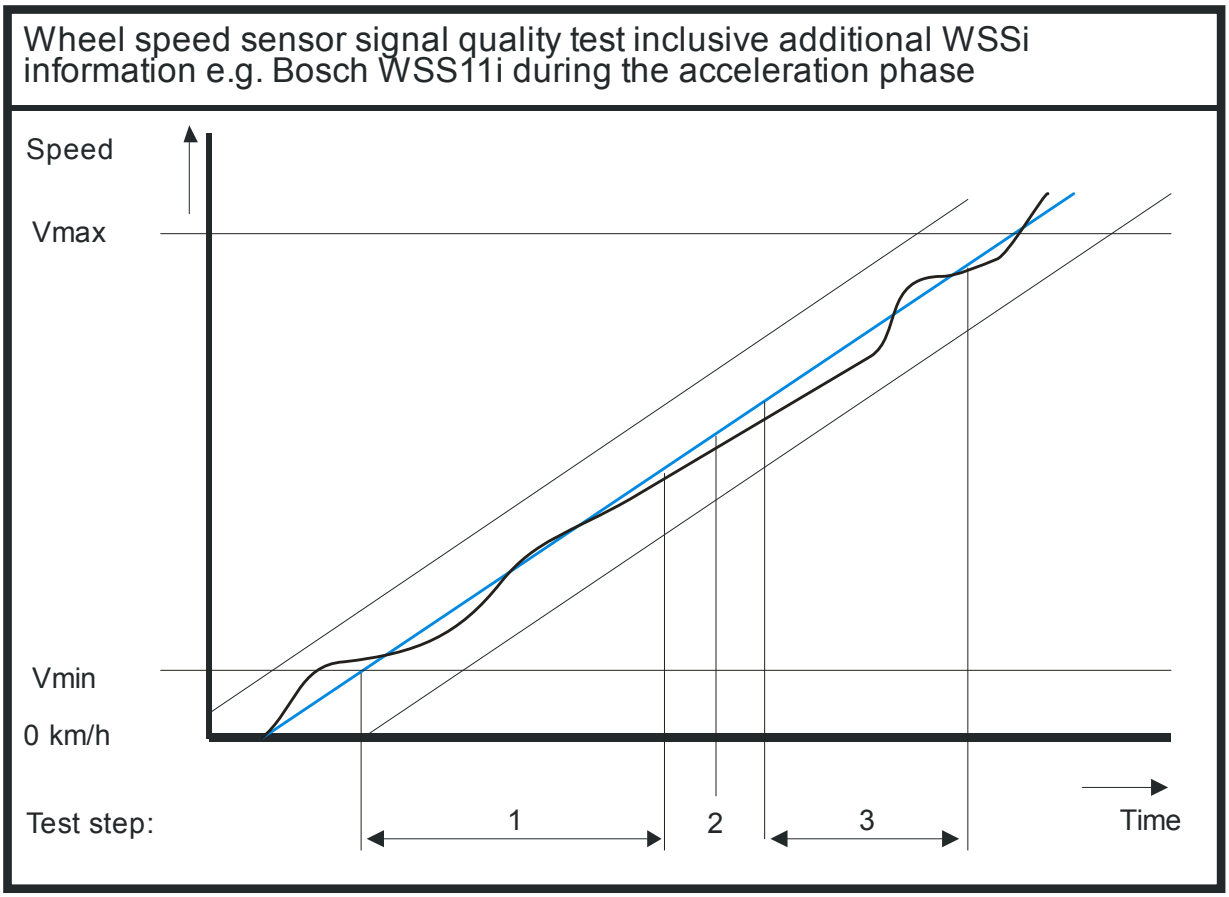
No WSS failure detected ⇒ Continue

WSS failure detected ⇒ Failure message, abort, print out fault message and send vehicle to repair area

未检测到轮速传感器故障 ⇒ 继续下线检测流程
检测到轮速传感器故障 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.3.1.1 Graphical presentation of the entire wheel speed senor tests inclusive the additional WSSi information e.g. Bosch WSS11i during the acceleration phase 加速阶段进行的轮速传感器完整测试（包括 WSSi 附加信息，如博世 WSS11i）示意图

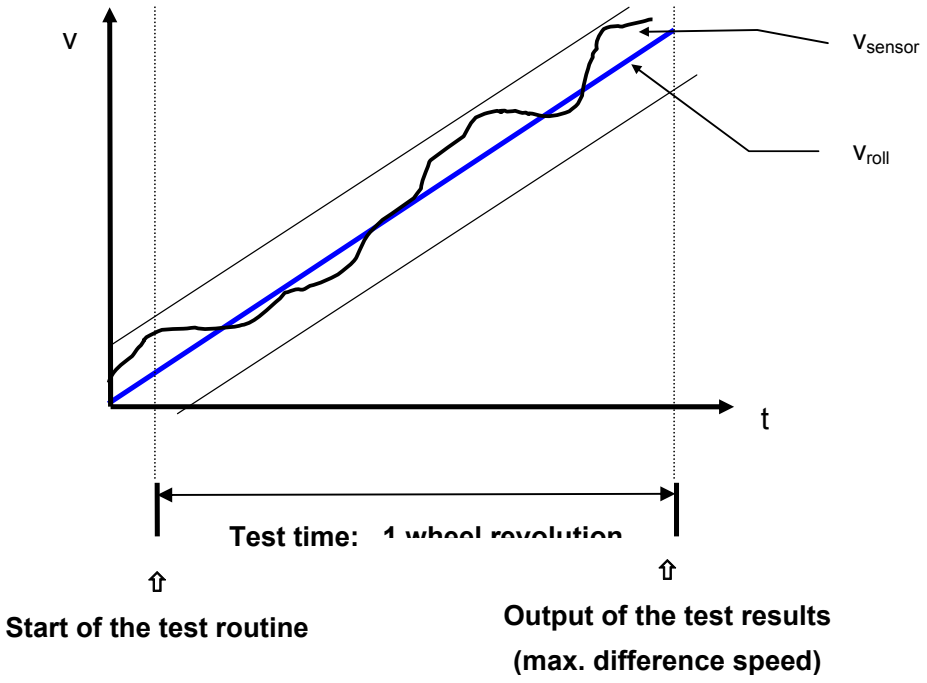


© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.3.1.2 Evaluation diagrams to the WSS signal quality test (test step 1) during the acceleration phase 加速阶段进行的 WSS 信号质量测试（测试步骤 1）评估图

Wheel speed sensor test (signal quality test) during acceleration
加速时的轮速传感器测试（信号质量测试）

Test diagram (exemplary at one wheel): 测试图（以单个车轮为例）



Remarks: 备注：

After the routine is started by tester, the ECU starts to sample cyclic all wheel speeds and calculates the difference speed values as long as one of the stop criteria is reached.
在测试设备启动测试程序后，ECU 开始循环采样所有车轮速度，并计算速度差异值直至满足任一停止标准时才停止。

As long as the calculated difference speed is negative or zero the routine will output “zero”. This indicates a good WSS signal (normal or good case).
如果计算出的速度差异值为负值或零值，程序将输出“0”，这表明 WSS 信号良好（正常或良好情况）。

In case of the calculated difference speed is positive, the routine starts to summarize these positive difference speed values as long as there is again a negative or zero value. The result of the summarised value will be compared with the initial (zero) or stored maximal difference speed value. If the actual calculated difference speed value is larger than the stored or initial value, this value will be stored as new maximal calculated difference speed. At the end of the routine the stored values of all wheels are outputted as routine results. A stored value > zero + ECU and WSS tolerances indicates a abnormally deceleration (e.g. caused by a bad WSS signal --> bad case).

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

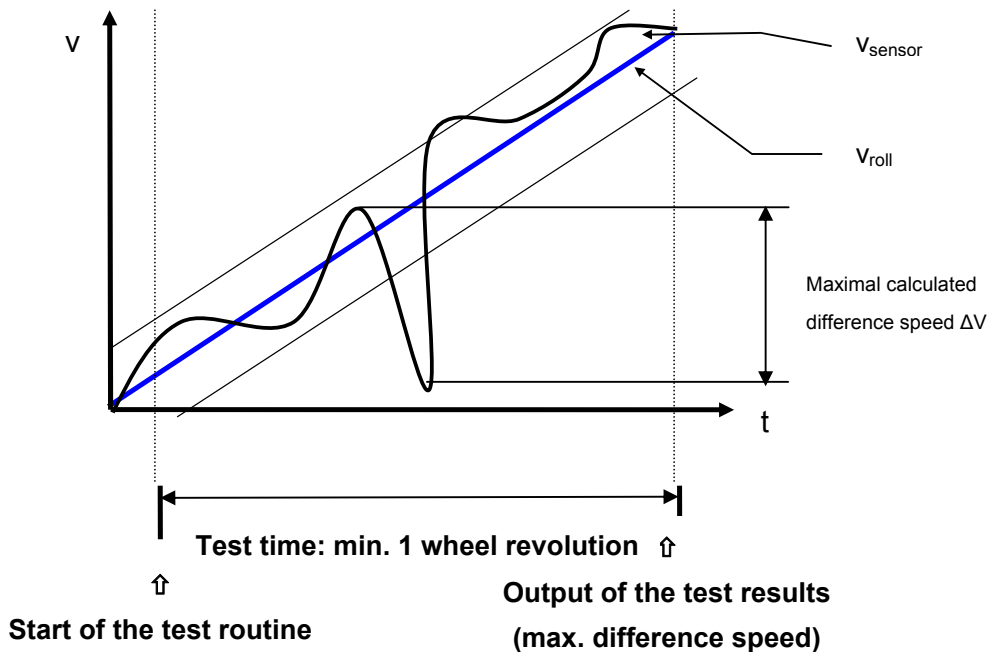
如果计算出的速度差异值为正值，程序将开始统计这些正速度差异值直至负值或零值再次出现。统计结果将与初始值（0）或预先储存的最大速度差异值进行对比，如果计算出的实际速度差异值大于储存值或初始值，则该实际值将作为新的最大速度差异值被储存。在程序结束时，所有车轮的储存值将作为程序结果输出。当储存值> 0 + ECU & WSS 公差时，表明减速过程异常（例如由 WSS 不良信号引起-->不良情况）。

The tester has to compare these data (maximal calculated difference speed per wheel) with the defined limits. These limits have to include all ECU, WSS and tooth wheel tolerances. In addition all fluctuations during a normal acceleration of the vehicle have to be considered.
测试设备应当将这些数据（计算出的各车轮最大速度差异值）与规定的极限相比较。这些极限必须包括所有的 ECU、WSS 和齿圈公差。此外，车辆正常加速过程中的波动也应当予以考虑。

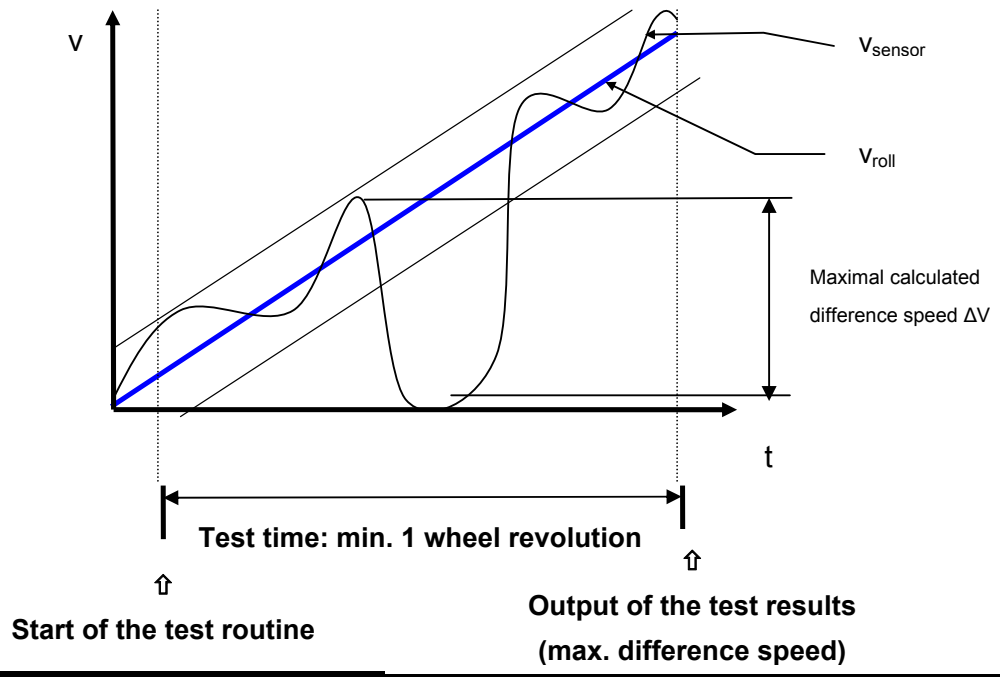
© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Failure simulation: 故障模拟

Failure simulation: faulty tooth wheel e.g. missing tooth 故障模拟：齿圈故障 (如缺齿)



Failure simulation: Air gap to big or eccentricity 故障模拟：气隙过大或偏心



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.3.1.3 Test step 1: Exemplary definition of the maximal allowed difference speed 测试步骤1：最大允许速度差异定义示范

Measurement results of vehicle: 车辆测量结果	ΔV_{FL_max}	ΔV_{FR_max}	ΔV_{RL_max}	ΔV_{RR_max}
1	0,05625	0	0,05625	0,05625
2	0,11250	0,05625	0,11250	0
3	0,05625	0,05625	0,05625	0,05625
4	0	0	0,11250	0,11250
5	0	0,11250	0	0,05625
Average (AV) 平均值（AV）	**0,05344**			
Standard deviation (S) 标准差（S）	**0,04270**			
Maximal difference speed 最大速度差 $\Delta V_{WSS_test_accelerated_vehicle}$ (AV + 5*S)	**0,26695**			

Note: 注意：

At a „good“ vehicles (no tooth wheel faults, etc.) and a equable acceleration (best case) the ECU will output 0 km/h (initial value) as measured maximal difference speed for each wheel. Caused by the fact that in reality the speed fluctuates on the roller test bench and different normal tolerances (e.g. ECU ± 2 bit, etc.) influences the outputted maximal difference speed per wheel, the maximal allowed tolerance “maximal speed difference $\Delta V_{WSS_test_accelerated_vehicle}$” is > 0 km/h. The right value (tolerance) has to be defined by a statistic evaluation. The definition of this value is up the customer responsibility.
当“良好”车辆（如无齿圈错误等）以平稳的加速度（最佳状况）进行测试时，ECU 将输出 0km/h（初始值）作为测得的各车轮最大速度差异值。考虑到实际中轮毂试验台上会有速度波动，并且存在各种常规公差（如 ECU ± 2 bit 等）对输出的各车轮最大速度差异值造成干扰，因此实际上最大允许公差“最大速度差异值 $\Delta V_{WSS_test_accelerated_vehicle}$” > 0 km/h。正确的公差值应当通过统计学评估来定义，且定义应当由客户完成。

4.3.1.4 Evaluation proposal for the WSS test (signal quality regarding single pitch deviation, eccentricity, etc.): WSS 测试（测试信号质量包括孤峰偏离、偏心等）评估建议：

Comparison of the maximal measured difference speed (ΔVxx_{max}, etc.) with the maximal allowed difference speed $\Delta V_{WSS_test_accelerated_vehicle}$ (calculated from the measured average value (AV) plus 5* standard deviation (S), values are defined by a statistic evaluation):
对比测得的最大速度差异值（ΔVxx_{max} 等）和最大允许速度差异值 $\Delta V_{WSS_test_accelerated_vehicle}$（由平均值 AV 加上 5*标准差 S 计算得出，通过统计学评估来定义）：

$\Delta V_{FLmax} \leq \Delta V_{WSS_test_accelerated_vehicle}$
$\Delta V_{FRmax} \leq \Delta V_{WSS_test_accelerated_vehicle}$
$\Delta V_{RLmax} \leq \Delta V_{WSS_test_accelerated_vehicle}$
$\Delta V_{RRmax} \leq \Delta V_{WSS_test_accelerated_vehicle}$

Example: 范例：

Measured maximal difference speeds (km/h) of each wheel: 测得的各车轮最大速度差异值（km/h）

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

ΔVFLmax	ΔVFR_{max}	ΔVRL_{max}	ΔVRR_{max}
0,05625	0	0,05625	3,60

Judgement: 判断值：

ΔV_{FLmax} (0,05625 ≤ 0,26695) = ok
ΔV_{FRmax} (0 ≤ 0,26695) = ok
ΔV_{RLmax} (0,05625 ≤ 0,26695) = ok
ΔV_{RRmax} (3,60 ≤ 0,26695) = nok

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.3.2 Base brake test and dynamic ABS test (function test „brake pipe correlation, pressure increase and release“) 基础制动测试和动态 ABS 测试（“制动管路关联性，增压和减压”功能测试）

Base brake and ABS test (function test „brake pipe correlation, pressure increase and release“) 基础制动测试和 ABS 测试（“制动管路关联性，增压和减压”功能测试）					
Operator: 操作者	Test rig: 试验台	To be evaluated brake forces: 待评估制动力		Demanded brake forces: 要求的制动力	Measure sample: 测量样本
→ Don't apply Brake pedal! 禁止踩制动踏板	Measuring of the free-wheeling decelerations 测量自由转动的减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{FR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RL} \leq \Delta V_{freewheeling_max}$	ΔV1 (V1.1-V1.2)
→ Apply the brake pedal till the ex-pected decelera-tion is reached! 踩下制动踏板直至达到期望减速度	Pressure increase at all wheels 所有车轮增压 Measuring of all decel-erations 测量所有减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_brakes}$ $\Delta V_{FL} - V_{brakes_tolerance} \leq \Delta V_{FR} \leq \Delta_{FL} + V_{brakes_tolerance}$ $\Delta V_{RL} \geq \Delta V_{RA_brakes}$ $\Delta_{RL} - V_{brakes_tolerance} \leq \Delta V_{RR} \leq \Delta_{RL} + V_{brakes_tolerance}$	ΔV2 (V2.1-V2.2)
→ Keep the brake pedal constant! 保持制动踏板不动	Pressure increase FL 前左轮增压 Measuring of all decel-erations 测量所有减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq (\Delta V_{unpressurised} + \Delta V_{freewheeling_FL})$ $\Delta V_{FR} \geq \Delta V_{FA_brakes}$ $\Delta V_{RL} \geq \Delta V_{RA_brakes}$ $\Delta V_{RR} \geq \Delta V_{RA_brakes}$	ΔV3 (V3.1-V3.2)
→ Keep the brake pedal constant! 保持制动踏板不动	Pressure decrease FL 前左轮减压 Pressure increase FR 前右轮增压 Measuring of all decel-erations 测量所有减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{FR} \leq (\Delta V_{unpressurised} + \Delta V_{freewheeling_FR})$ $\Delta V_{RL} \geq \Delta V_{RA_brakes}$ $\Delta V_{RR} \geq \Delta V_{RA_brakes}$	ΔV4 (V4.1-V4.2)
→ Keep the brake pedal constant! 保持制动	Pressure decrease FR 前右轮减压 Pressure increase RL	Wheel FL: Wheel FR:	Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{FR} \geq \Delta V_{FA_pressure_increase_brakes}$	ΔV5 (V5.1-V5.2)

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

踏板不动	后左轮增压 Measuring of all decel-erations 测量所有减速度	Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h	$\Delta V_{RL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RL})$ $\Delta V_{RR} \geq \Delta V_{RA_brakes}$	
→ Keep the brake pedal constant! 保持制动踏板不动	Pressure decrease RL 后左轮减压 Pressure increase RR 后右轮增压 Measuring of all decel-erations 测量所有减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{FR} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{RL} \geq \Delta V_{RA_pressure_increase_brakes}$ $\Delta V_{RR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RR})$	ΔV6 (V6.1-V6.2)
→ Keep the brake pedal constant! 保持制动踏板不动	Pressure decrease RR 后右轮减压 Measuring of all decel-erations 测量所有减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{FR} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{RL} \geq \Delta V_{RA_pressure_increase_brakes}$ $\Delta V_{RR} \geq \Delta V_{RA_pressure_increase_brakes}$	ΔV7 (V7.1-V7.2)
→ Re-lease the brake pedal! 松开制动踏板	Measuring of the free-wheeling decelerations 测量自由转动的减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{FR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RL} \leq \Delta V_{freewheeling_max}$	ΔV8 (V8.1-V8.2)

Remarks: 备注：

The base brake and ABS tests are executed directly after the WSS tests. As first step during the base brake test the freewheeling decelerations are measured. After the final test speed is reached the rolls are detached at first then the freewheeling decelerations are measured. Afterwards the operator is asked to apply the brake pedal immediately. As soon as the expected decelerations are reached, the operator has to keep the brake pedal constant and shouldn't release it till the deceleration values of the base brake test are measured.
在 WSS 测试后立即进行基础制动测试和 ABS 测试。基础制动测试的第一步是测量自由转动的减速度。在达到最终测试速度后，首先断开转毂以测量自由转动减速度，而后要求操作者立即踩下制动踏板。当减速度达到期望值后，操作者必须保持制动踏板不动，直至基础制动测试的减速度值测量完为止。

The ABS test is executed immediately after the base brake test. The operator has to keep the brake pedal still constant. The tester activates the corresponding valves and the pump motor step by step for the ABS test. In addition the tester measures the corresponding decelerations at each step and wheel.
ABS 测试将在基础制动测试后立即进行。操作者仍然必须保持制动踏板不动。测试设备将在 ABS 测试中依次启动相应的电磁阀和泵马达。另外，测试设备将在每一步骤中测量各车轮相应的减速度。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

The operator has to apply, keep constant and release the brake pedal after he was requested by the tester!
在测试设备发出请求后，操作者必须相应地踩下、保持不动、松开制动踏板。

The respective wheel decelerations are calculated by the tester and compared with the fixed limits.
测试设备计算出各车轮减速度并将其与固定极限值进行对比。

To reduce the entire cycle time of the ABS test, the test cycle respectively corresponding vale activation for each wheel can overlap.
为了降低 ABS 测试整个循环时间，测试循环和各车轮相应的阀动作可以相互重叠。

The base brake test has to be executed at all vehicles independent of the installed control system. The test is required to ensure the function and tightness of base brake.
The ABS test can be used as well as for vehicles with an ABS or ESP® system. In case of a 3 channel ABS system is used, the described test procedure for the wheel RL is valid for the total rear axle. The described test step for the wheel RR can be removed, but only in case of a 3 channel system.
在所有车辆上都必须进行基础制动测试，这与车辆安装的控制系统无关。该测试用于确保基础制动系统的功能和密封性正常。
ABS 测试亦适用于装有 ABS 或 ESP®系统的车辆。如果 ABS 系统为 3 通道系统，上述后左轮测试步骤可以用于整个后轴，而后右轮测试步骤可以去掉，但这仅限于 3 通道系统。

The base brake and ABS test have to ensure the following functions and assembly processes:
执行基础制动测试和 ABS 测试以确保下列功能和装配过程正常：

1) To ensure the function of the base brake system
   确保基础制动系统功能正常
2) To ensure the correlation of the brake pipes between the ABS / ESP® hydraulic ports and each wheel
   确保 ABS/ESP®液压单元的各个轮缸回路螺纹孔通过制动管路关联到正确的相应车轮
3) To ensure the function of the EV and AV valves after brake filling process (e.g. to exclude a impairment of function caused by contamination, etc.)
   在加注制动系统（例如：为了消除污染物对系统功能造成的损害）后，确保 EV 和 AV 阀密封性和功能正常
4) To ensure delivery rate of both pumps 确保两个泵单元的泵液速度
5) To ensure the correct correlation between the wheels speed sensors and the wheels
   确保轮速传感器与车轮关联性正确

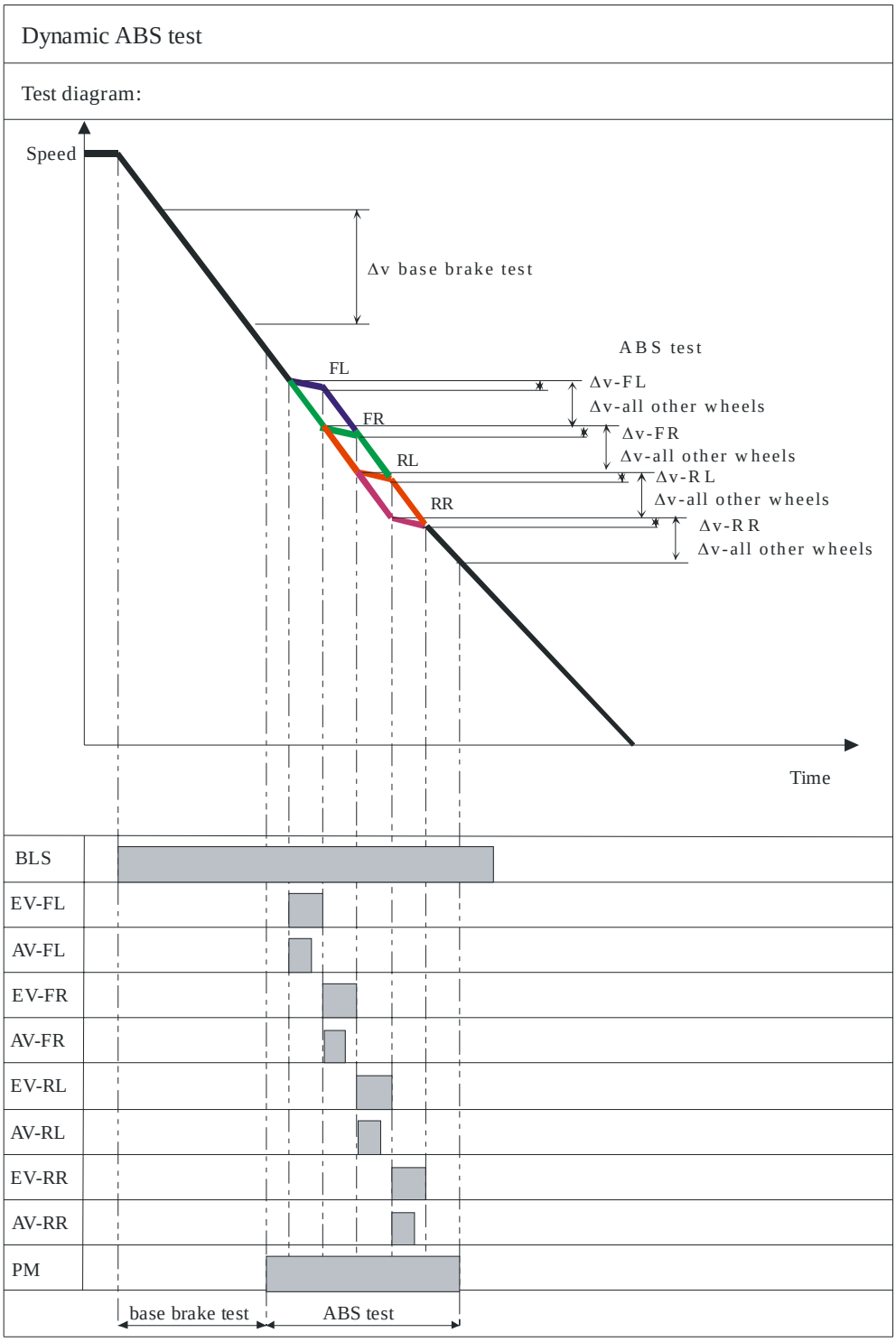
ABS test ok ⇒ **Continue**

ABS Test nok ⇒ **Failure message, abort, print out fault message and send vehicle to repair area**

ABS 测试合格 ⇒ **继续下线检测流程**
ABS 测试不合格 ⇒ **退出，打印错误信息并将车辆送至维修区**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.3.2.1 Diagram of the dynamic ABS test at all wheels 所有车轮的动态 ABS 测试示意图



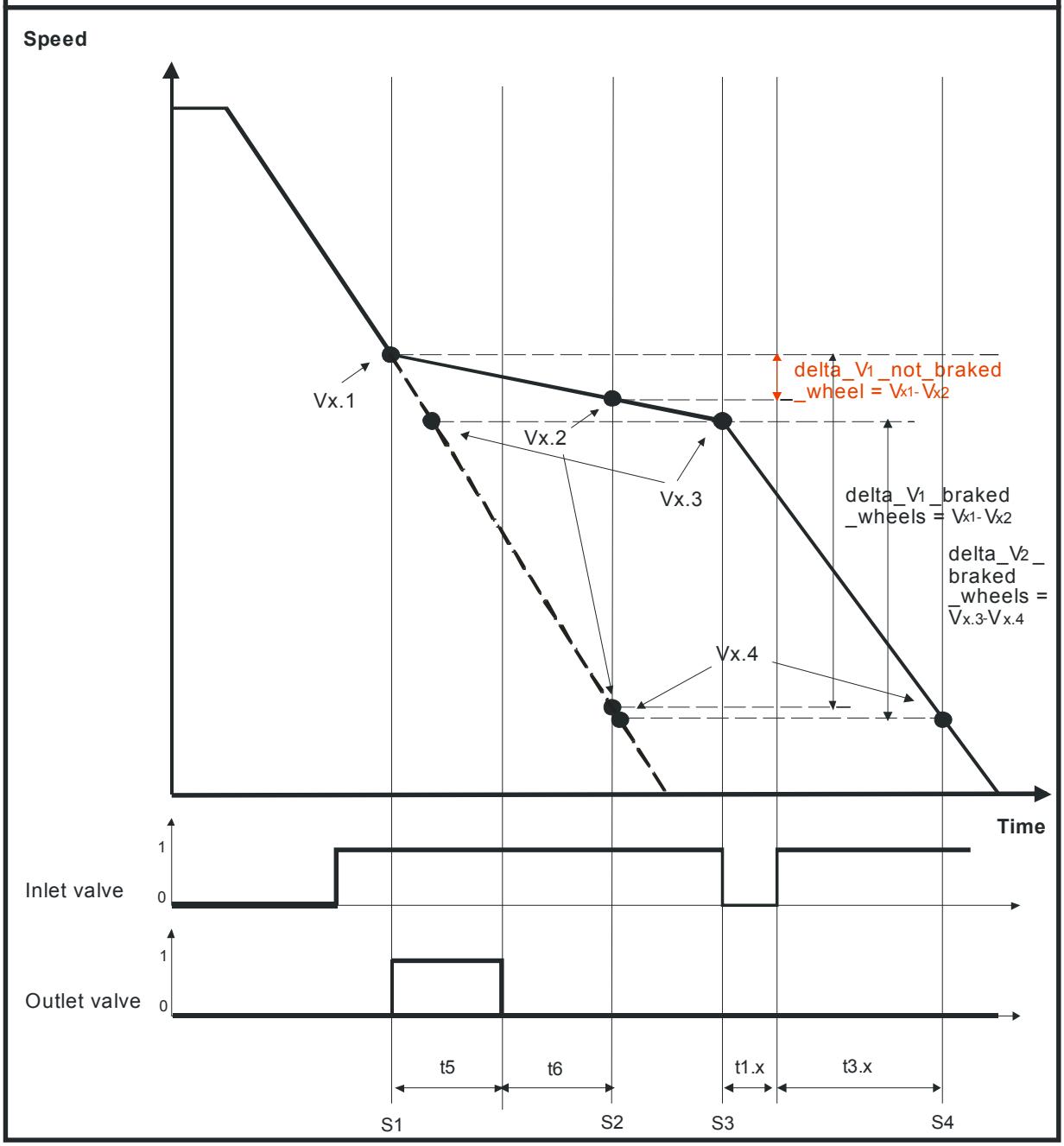
Remarks: 备注

Please note the maximal allowed valve activation times. More information about these times can be read in the chapter 6. 请注意阀启动最大允许时间。关于这些时间更多信息请参考第 6 章。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.3.2.2 Diagram of the dynamic ABS test (exemplary at one wheel) 动态 ABS 测试示意图（以单个车轮为例）

Dynamic ABS test
Test diagram (exemplary at one whee):



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Remarks: 备注

- The diagnostic tester sends the request (S1) to reduce the wheel pressures.
  诊断设备发送请求（S1）降低车轮压力。

- The ECU takes the first speed sample (Vx.1) and executes then the requested pressure decrease for the defined time (t5).
  ECU 采集第一个速度样本（Vx.1）并在随后规定时间（t5）内执行减压请求。

- The ECU takes the second speed sample (Vx.2) after the defined stabilisation time (t6) is expired.
  ECU 在规定的稳定时间（t6）结束后采集第二个速度样本（Vx.2）。

- The ECU takes these two speed samples, calculates the differences speed for the pressure decrease cycle (Vx.1-Vx.2 = ΔV1/wheelxx) and outputs the result if the tester request it (S2).
  ECU 采集这两个速度样本，计算减压循环的速度差异值（Vx.1-Vx.2 = ΔV1/wheelxx），并在诊断设备请求（S2）时输出该结果。

- The diagnostic tester sends the request (S3) to increase the wheel pressure again.
  诊断设备发送请求（S3）再次增加车轮压力。

- The ECU takes the third speed sample (Vx.3) and executes then the requested pressure increase for the defined time (t1.x).
  ECU 采集第三个速度样本（Vx.3）并在随后规定时间（t1.x）内执行增压请求。

- The ECU takes the fourth speed sample (Vx.4) after the defined stabilisation time (t3.x) is expired.
  ECU 在规定的稳定时间（t3.x）结束后采集第四个速度样本（Vx.4）。

- The ECU takes these two speed samples, calculates the differences speed for the pressure increase cycle (Vx.3-Vx.4 = ΔV2/wheelxx) and outputs the result if the tester request it (S4).
  ECU 采集这两个速度样本，计算增压循环的速度差异值（Vx.3-Vx.4 = ΔV2/wheelxx），并在诊断设备请求（S4）时输出该结果。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.3.2.3 Test sequence for the dynamic ABS test 动态 ABS 测试序列

Start message: 启动信息	Operator prompt: 操作者提示信息	Measure sample: 测量样本	Activation before the time 1: 时间 1 前动作	Time 1: 时间 1	Activation before the time 2: 时间 2 前动作	Time 2: 时间 2	Measure sample: 测量样本
1	Don't apply the brake pedal! 禁止踩下制动踏板	V1.1	---	---	---	---	V1.2
2	Apply the brake pedal till the expected decelerations are reached! 踩下制动踏板直至达到减速度期望值	V2.1	---	---	---	---	V2.2
3	Keep the brake pedal constant! 保持制动踏板不动	---	Pump motor on 启动泵马达	---	---	---	---
4		---	Inlet valves front axle on 启动前轴进液阀	---	Inlet valves rear axle on 启动后轴进液阀	---	---
5		---	AVFL on	t5	AVFL off	t6	---
6		V3.1	Measurement pressure decrease FL 测量前左轮减压	t7	---	---	V3.2
7		---	EVFL off	t1.3	EVFL on	t.3.3	---
8		---	AVFR on	t5	AVFR off	t6	
9		V4.1	Measurement pressure increase FL & decrease FR 测量前左轮增压和前右轮减压	t7	---	---	V4.2
10		---	EVFR off	t1.3	EVFL on	t3.3	---
11		---	AVRL on	t5	AVRL off	t6	---
12		V5.1	Measurement pressure increase FR & decrease RL 测量前右轮增压和前左轮减压	t7	---	---	V5.2
13		---	EVRL off	t1.4	EVRL on	t3.4	---

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

14		---	AVRR on	t5	AVRR off	t6	---
15		V6.1	Measurement pressure increase RL & decrease RR 测量后左轮增压和后右轮减压	t7	---	---	V6.6
16		---	EVRR off	t1.4	EVRR on	t3.4	---
17		V7.1	Measurement pressure increase RR 测量后右轮增压	t7	---	---	V7.2
18	Release the brake pedal! 松开制动踏板	---	Inlet valves front axle off 断电前轴进液阀	---	Inlet valves rear axle off 断电后轴进液阀	---	---
19		V8.1	---	t8	Pump motor off 断电泵马达	---	V8.2

Remarks: 备注：

After the final test speed is reached the rolls are detached by the test rig at first. Then the freewheeling deceleration (ΔV1 = V1.1-V1.2) at all individual wheels are captured (start message 1). These values are subtracted from the afterwards measured wheel decelerations. As next step the operator is prompt to apply the brake pedal till the expected deceleration is reached and measurement for the base brake test (ΔV2 = V2.1-V2.2) is executed (start message 2). Afterwards the operator has to keep the brake pedal constant.
在达到最终测试速度后，试验台首先断开转毂以测量各车轮的自由转动减速度（ΔV1 = V1.1-V1.2）（启动信息 1），这些值将从随后测得的车轮减速度中减去。下一步，操作者将被提示踩下制动踏板直至达到减速度期望值并完成基础制动测试测量（ΔV2 = V2.1-V2.2）（启动信息 2）。随后操作者应保持制动踏板不动。

The tester switches on the pump motor (start message 3) and all inlet valves (start message 4). Afterwards the pressure at the wheel FL is released by switching on of the outlet valve for the time t5 (start message 5).
测试设备启动泵马达（启动信息 3）和所有进液阀（启动信息 4）。随后在 t5 时间内启动出液阀以释放前左轮中压力（启动信息 5）。

After the stabilisation time t6 is expired the ECU measures the decelerations (ΔV3 = V3.1-V3.2) of all wheels with the dummy message (start message 6) for the time t7 to ensure a correct executed pressure release function at the wheel FL.
在稳定时间 t6 结束后，ECU 在 t7 时间内测量所有车轮减速度（ΔV3 = V3.1-V3.2）（启动信息 6），以确保前左轮减压功能正确执行。

As next step the inlet valve FL is immediately opened for the time t1.3 to increase the pressure in the wheel. In parallel the pressure in the wheel FR is released by switching on of the outlet valve for the time t5 (start messages 7 & 8). After this test step respectively after the stabilisation time t3.3 (t6) is expired the decelerations (ΔV4 = V4.1-V4.2) of all wheels are measured again with execution of the dummy message (start message 9) for the time t7. This is required to ensure the pressure increase at the wheel FL and the pressure decrease at the wheel FR.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

下一步，前左轮进液阀将立即打开 t1.3 的时间以增加该轮压力，同时通过启动出液阀 t5 的时间来减少前右轮的压力（启动信息 7&8）。在该步骤完成且稳定时间 t3.3（t6）结束后，在 t7 的时间内再次测量所有车轮的减速度（ΔV4 = V4.1-V4.2）（启动信息 9），以确保前左轮增压和前右轮减压。

Afterwards the inlet valve FR is immediately opened for the time t1.3 to increase the pressure in the wheel. In parallel the pressure in the wheel RL is released by switching on of the outlet valve for the time t5 (start messages 10 & 11).
之后，前右轮进液阀将立即打开 t1.3 的时间以增加该轮压力，同时通过启动出液阀 t5 的时间来减少后左轮的压力（启动信息 10&11）。

This is ensured by measuring of the wheel decelerations (ΔV5 = V5.1-V5.2) with execution of the dummy message (start message 12) for the time t7 after the stabilisation time t3.3 (t6) is expired.
The test at rear axel is done in the same way (start messages 13 till 17; measurements of wheel decelerations ΔV6 till ΔV7).
这是通过在稳定时间 t3.3（t6）结束后的 t7 时间内，测量车轮减速度（ΔV5 = V5.1-V5.2）（启动信息 12）来实现的。后轴测试与前轴的方式相同（启动信息 13 至 17；测量车轮减速度 ΔV6 至 ΔV7）。

At the end of the ABS test the operator is prompt to release the brake pedal. Afterwards all inlet valves are immediately switched off (start message 18). The pump motor is switched off after the time t8 (start message 19). This is required to discharge the accumulator chambers. In parallel the decelerations of all wheels are measured again and then they are compared with the freewheeling forces which were measured at the beginning of the base brake test.
在 ABS 测试结束时，操作者被提示松开制动踏板。随后，所有进液阀将被立即断电（启动信息 18）。在时间 t8 后断电泵马达（启动信息 19），从而排空蓄液仓。同时，再次测量所有车轮的减速度，并将其与基础制动测试最初的自由转动减速度相比较。

Details to used diagnostic function see chapter 5.6.
关于采用的诊断功能具体信息请参考 5.6 小节。

<u>Parameter:</u> 参数

See chapter 4.3.3.4. 参考 4.3.3.4 小节。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.3.2.4 Proposal of the evaluation of the base brake and dynamic ABS test 基础制动测试和动态 ABS 测试的评估逻辑建议

a) Evaluation of the freewheeling before the base brake test (exemplary for the wheel FL):
在基础制动测试前评估自由转动减速度（以前左轮为例）：

If
ΔV1/wheel_FL ≤ ΔV_freewheeling_max

then ΔV1/wheel_FL = ΔV_freewheeling and execute test step b)

else stop process and output “Freewheeling at the wheel FL nok”

如果 ΔV1/wheel_FL ≤ ΔV_freewheeling_max，
则 ΔV1/wheel_FL = ΔV_freewheeling 并执行测试步骤 b)；
否则停止流程并输出“前左轮自由转动减速度不合格”。

b) Evaluation of the base brake test (exemplary for the front axle):
评估基础制动测试（以前轴为例）

If
(ΔV2/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_brakes
and
(ΔV2/wheel_FL – V_brakes_tolerance) ≤ ΔV2/wheel_FR ≤ (ΔV/wheel_FL + V_brakes_tolerance)

then execute test step c)

else stop process and output “Base brake test at the front axle nok”

如果 (ΔV2/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_brakes
且 (ΔV2/wheel_FL – V_brakes_tolerance) ≤ ΔV2/wheel_FR ≤ (ΔV/wheel_FL + V_brakes_tolerance)，
则执行测试步骤 c)；
否则停止流程并输出“前轴基础制动测试不合格”。

c) Evaluation ABS test „pressure decrease“ (exemplary for the wheel FL):
评估 ABS 测试“减压”（ 以前左轮为例）：

If
ΔV3.1/wheel_FL ≤ (ΔV_unpressurised + ΔV_freewheeling_FL)

then execute test step d)

else stop process and output “ABS pressure decrease at the wheel FL nok”

如果 ΔV3.1/wheel_FL ≤ (ΔV_unpressurised + ΔV_freewheeling_FL)，
则执行测试步骤 d)；
否则停止流程并输出“前左轮 ABS 减压不合格”。

d) Evaluation ABS test „pressure increase“ (exemplary for the wheel FL):

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

评估 ABS 测试“增压”（以前左轮为例）：

If
(ΔV4.1/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_pressure_increase_brakes

then execute test step e)

else stop process and output “ABS pressure increase at the wheel FL nok”

如果(ΔV4.1/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_pressure_increase_brakes，
则执行测试步骤 e)；
否则停止流程并输出“前左轮 ABS 增压不合格”。

e) Evaluation of the freewheeling after the ABS test (exemplary for the wheel FL):
在 ABS 测试后评估自由转动减速度（以前左轮为例）：

If
ΔV8/wheel_FL ≤ ΔV_freewheeling_max

then output “base brake test and dynamic ABS test ok”;

else output “ Freewheeling at the wheel FL nok ”

如果 ΔV8/wheel_FL ≤ ΔV_freewheeling_max，
则输出“基础制动测试和动态 ABS 测试合格”；
否则输出“前左轮自由转动减速度不合格”。

Example: 范例：

Evaluation variables*: 评估变量*：
ΔV_unpressurised = 0,5 km/h
ΔV_freewheeling_max = 0,8 km/h
ΔV_FA_brakes = 2 km/h
ΔV_FA_pressure_increase_brakes = 1,5 km/h
V_brakes_tolerance = 25%

*All mentioned variables for evaluation are valid for the measure time of 500ms. All values are examples and must be finally defined by the customer!
*所有上述评估变量仅适用于 500ms 的测量时间。所有值只是范例，最终还需由客户定义。

a) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step “freewheeling before the base brake test”:
评估测试步骤“基础制动测试前的自由转动减速度”中计算出的各车轮速度差异（km/h）：

ΔV1_FL	ΔV1_FR	ΔV1_RL	ΔV1_RR
0,5	0,5	0,3	0,3

b) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step “base brake test”:
评估测试步骤“基础制动测试”中计算出的各车轮速度差异（km/h）：

ΔV2_FL	ΔV2_FR	ΔV2_RL	ΔV2_RR
2,8	2,9	2,6	2,5

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

c) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "ABS test – pressure decrease FL":
评估测试步骤"ABS 测试-前左轮减压"中计算出的各车轮速度差异（km/h）：

ΔV3.1_FL	ΔV3.1_FR	ΔV3.1_RL	ΔV3.1_RR
0,9	1,0	0,8	0,8

d) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "ABS test – pressure increase FL":
评估测试步骤"ABS 测试-前左轮增压"中计算出的各车轮速度差异（km/h）：

ΔV4.1_FL	ΔV4.1_FR	ΔV4.1_RL	ΔV4.1_RR
2,3	0,7	0,4	0,8

e) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "freewheeling after ABS test":
评估测试步骤"ABS 测试后的自由转动减速度"中计算出的各车轮速度差异（km/h）：

ΔV8_FL	ΔV8_FR	ΔV8_RL	ΔV8_RR
0,6	0,5	0,4	0,3

<u>Judgement of each test step for the wheel FL respectively FA:</u>
前左轮和前轴的每个测试步骤判断值：

a) Freewheeling at the wheel FL before the base brake test: 0,5 ≤ 0,8 **= ok**
基础制动测试前的前左轮自由转动减速度：0,5 ≤ 0,8 **= ok**
b) Base brake test front axle: ((2,8-0,5) ≥ 2) && (2,8-0,7) ≤ 2,9 ≤ (2,8+0,7) **= ok**
前轴基础制动测试：((2,8-0,5) ≥ 2) && (2,8-0,7) ≤ 2,9 ≤ (2,8+0,7) **= ok**
c) ABS test „pressure decrease FL": 0,9 ≤ (0,5+0,5) **= ok**
ABS 测试"前左轮减压"：0,9 ≤ (0,5+0,5) **= ok**
d) ABS Test „pressure increase brakes FL": (2,3-0,5) ≥ 1,5 **= ok**
ABS 测试"前左轮增压"：(2,3-0,5) ≥ 1,5 **= ok**
e) Freewheeling at the wheel FL after ABS test: 0,6 ≤ 0,8 **= ok**
ABS 测试后的前左轮自由转动减速度：0,6 ≤ 0,8 **= ok**

4.3.3 Base brake and dynamic ESP® test (function test „brake pipe correlation, pressure increase and release") 基础制动和动态 ESP®测试（功能测试"制动管路关联性，增压减压"）

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Base brake and dynamic ESP® test (function test „brake pipe correlation, pressure increase and release“)
基础制动测试和动态 ESP®测试（功能测试“制动管路关联性，增压减压”）

Operator: 操作者	Test rig: 试验台	To be evaluated brake forces: 待评估制动力		Demanded brake forces: 要求的制动力	Measure sample: 测量样本
→ Don’t apply the brake pedal! 禁止踩制动踏板	Measuring of the freewheeling decelerations 测量自由转动的减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{FR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RL} \leq \Delta V_{freewheeling_max}$	ΔV1 (V1.1-V1.2)
→ Apply the brake pedal till the expected deceleration is reached! 踩下制动踏板直至达到期望减速度	Pressure increase at all wheels 所有车轮增压 Measuring of all decelerations 测量所有减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_brakes}$ $\Delta V_{FL} - V_{brakes_tolerance} \leq \Delta V_{FR} \leq \Delta_{FL} + V_{brakes_tolerance}$ $\Delta V_{RL} \geq \Delta V_{RA_brakes}$ $\Delta_{RL} - V_{brakes_tolerance} \leq \Delta V_{RR} \leq \Delta_{RL} + V_{brakes_tolerance}$	ΔV2 (V2.1-V2.2)
→ Release the brake pedal! 松开制动踏板	Measuring of the freewheeling decelerations 测量自由转动的减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{FR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RR} \leq \Delta V_{freewheeling_max}$	ΔV3 (V3.1-V3.1)
→ Don’t apply the brake pedal! 禁止踩制动踏板	Pressure increase FL 前左轮增压 Measuring of all decelerations 测量所有减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} > (\Delta V_{FA_pressure_increase} - \Delta V_{_freewheeling_FL})$ $\Delta V_{FR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FR})$ $\Delta V_{RL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RL})$ $\Delta V_{RR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RR})$	ΔV4 (V4.1-V4.2) ΔV4.n (V4.n-V4.n+1)
→ Don’t apply the brake pedal! 禁止踩制动踏板	Pressure decrease FL 前左轮减压	---	---	---	---
→ Don’t apply the brake		Wheel FL: Wheel FR:	Δ xx km/h	$\Delta V_{FL} \leq (\Delta V_{_unpressurised} +$	ΔV5 (V5.1-V5.2)

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

pedal! 禁止踩制动踏板	Pressure increase FR 前右轮增压 Measuring of all decelerations 测量所有减速度	Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{_freewheeling_FL}$) ΔV_{FR} > ($\Delta V_{FA_pressure_increase}$ - $\Delta V_{_freewheeling_FR}$) ΔV_{RL} ≤ ($\Delta V_{_unpressurised}$ + $\Delta V_{_freewheeling_RL}$) ΔV_{RR} ≤ ($\Delta V_{_unpressurised}$ + $\Delta V_{_freewheeling_RR}$)	ΔV5.n (V5.n-V5.n+1)
→ Don't apply the brake pedal! 禁止踩制动踏板	Pressure decrease FR 前右轮减压	---	---	---	---
→ Don't apply the brake pedal! 禁止踩制动踏板	Pressure increase RL 后左轮增压 Measuring of all decelerations 测量所有减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	ΔV_{FL} ≤ ($\Delta V_{_unpressurised}$ + $\Delta V_{_freewheeling_FL}$) ΔV_{FR} ≤ ($\Delta V_{_unpressurised}$ + $\Delta V_{_freewheeling_FR}$) ΔV_{RL} > ($\Delta V_{RA_pressure_increase}$ - $\Delta V_{_freewheeling_RL}$) ΔV_{RR} ≤ ($\Delta V_{_unpressurised}$ + $\Delta V_{_freewheeling_RR}$)	ΔV6 (V6.1-V6.1) V6.n (V6.n-V6.n+1)
→ Don't apply the brake pedal! 禁止踩制动踏板	Pressure decrease RL 后左轮减压	---	---	---	---
→ Don't apply the brake pedal! 禁止踩制动踏板	Pressure increase RR 后右轮增压 Measuring of all decelerations 测量所有减速度	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	ΔV_{FL} ≤ ($\Delta V_{_unpressurised}$ + $\Delta V_{_freewheeling_FL}$) ΔV_{FR} ≤ ($\Delta V_{_unpressurised}$ + $\Delta V_{_freewheeling_FR}$) ΔV_{RL} ≤ ($\Delta V_{_unpressurised}$ + $\Delta V_{_freewheeling_RL}$) ΔV_{RR} > ($\Delta V_{RA_pressure_increase}$ - $\Delta V_{_freewheeling_RR}$)	ΔV7 (V7.1-V7.2) ΔV7.n (V7.n-V7.n+1)
→ Don't apply the brake pedal! 禁止踩制动踏板	Pressure decrease RR 后右轮减压	---	---	---	---
→ Don't apply the brake pedal! 禁止踩	Measuring of	Wheel FL: Wheel FR:	Δ xx km/h Δ xx km/h	ΔV_{FL} ≤ ($\Delta V_{_unpressurised}$ + $\Delta V_{_freewheeling_FL}$)	ΔV8 (V8.1-V8.2)

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

制动踏板	the freewheeling decelerations 测量自由转动的减速度	Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h	$\Delta V_{FR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FR})$ $\Delta V_{RL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RL})$ $\Delta V_{RR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RR})$	

Remarks: 备注：

The base brake and ESP® tests are executed directly after the WSS tests. As first step during the base brake test the freewheeling decelerations are measured. After the final test speed is reached the rolls are detached at first then the freewheeling decelerations are measured. Afterwards the operator is asked to apply the brake pedal immediately. As soon as the expected decelerations are reached, the operator has to keep the brake pedal constant and shouldn't release it till the deceleration values of the base brake test are measured. Afterwards the operator has to relapse the brake pedal before the ESP® tests are started.
在 WSS 测试后立即进行基础制动测试和 ESP®测试。基础制动测试的第一步是测量自由转动的减速度。在达到最终测试速度后，首先断开转毂以测量自由转动减速度，而后要求操作者立即踩下制动踏板。当减速度达到期望值后，操作者必须保持制动踏板不动，直至基础制动测试的减速度值测量完为止。随后操作者应当在 ESP®测试开始前将制动踏板回位。

The ESP® test is executed immediately after the base brake test. The operator has to keep the brake pedal released. The tester activates the corresponding valves and the pump motor step by step for the ESP® test. In addition the tester measures the corresponding decelerations at each step and wheel.
ESP®测试将在基础制动测试后立即进行。操作者仍然必须保持松开制动踏板。测试设备将在 ESP®测试中依次启动相应的电磁阀和泵马达。另外，测试设备将在每一步骤中测量各车轮相应的减速度。

The operator might not apply the brake pedal the complete ESP® test!
在整个 ESP®测试中，操作者不得踩下制动踏板。

The respective wheel decelerations are calculated by the tester and compared with the fixed limits.
测试设备计算出各车轮减速度并将其与固定极限值进行对比。

To reduce the entire cycle time of the ESP® test, the test cycle respectively corresponding valve activation for each wheel can overlap.
为了降低 ESP®测试整个循环时间，测试循环和各车轮相应的阀动作可以相互重叠。

The base brake test has to be executed at all vehicles independent of the installed control system. The test is required to ensure the function and tightness of base brake.
在所有车辆上都必须进行基础制动测试，这与车辆安装的控制系统无关。该测试用于确保基础制动系统的功能和密封性正常。

The base brake and ESP® test have to ensure the following functions and assembly processes:
执行基础制动测试和 ESP®测试以确保下列功能和装配过程正常：

1) To ensure the function of the base brake system
   确保基础制动系统功能正常
2) To ensure the correlation of the brake pipes between the ESP® hydraulic ports and each wheel

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

确保 ESP®液压单元的各个轮缸回路螺纹孔通过制动管路关联到正确的相应车轮

3) To ensure the function of the EV and AV valves after brake filling process (e.g. to exclude a impairment of function caused by contamination, etc.)
在加注制动系统（例如：为了消除污染物对系统功能造成的损害）后，确保 EV 和 AV 阀功能正常

4) To ensure delivery rate of both pumps respectively the active ESP® pressure increase function at all wheels
确保两个泵单元的泵液速度以及 ESP®在所有车轮上的主动增压功能正常

5) To ensure the correct correlation between the wheels speed sensors and the wheels
确保轮速传感器与车轮关联性正确

ESP® test ok ⇒ **Continue**

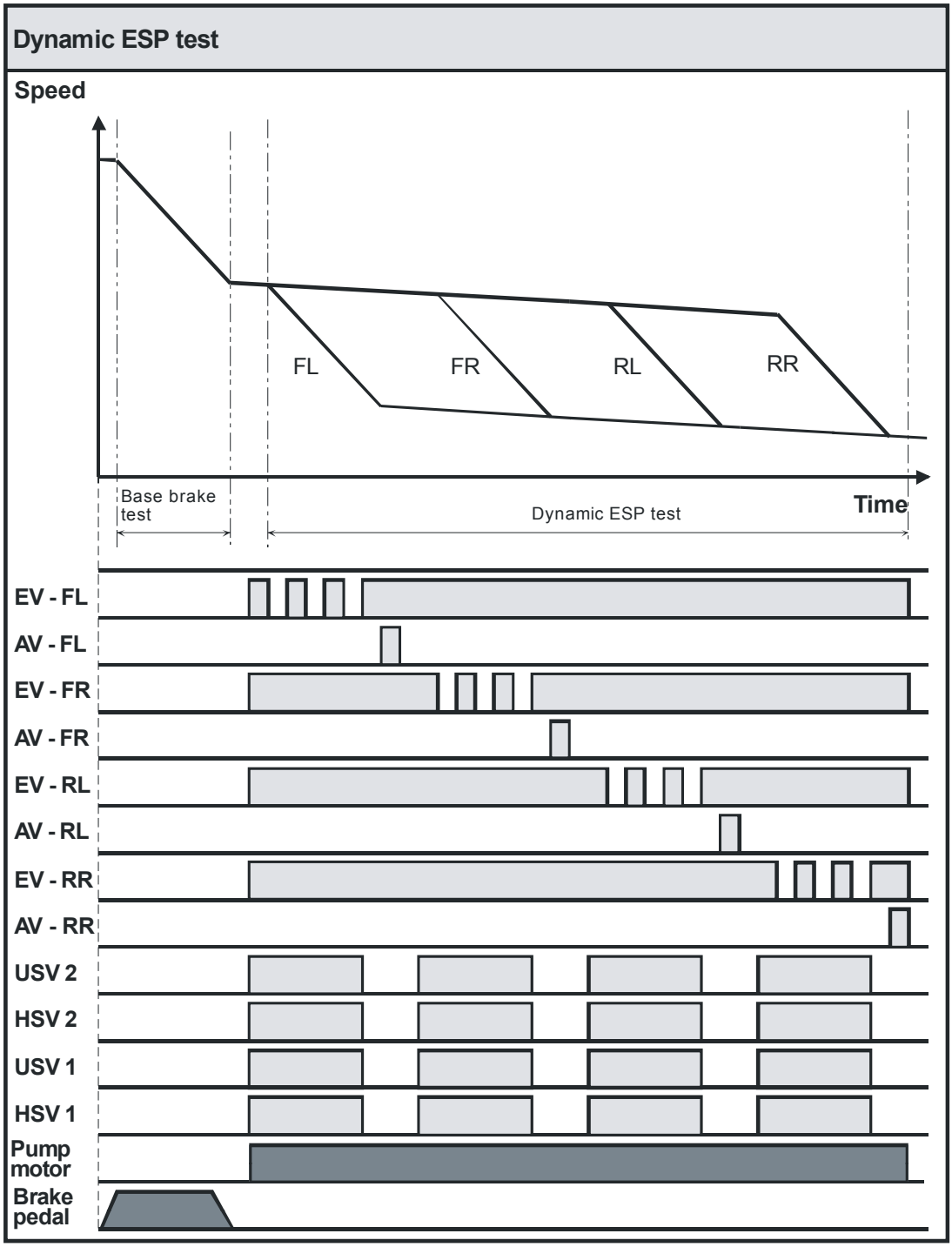
ESP®Test nok ⇒ **Failure message, abort, print out fault message and send vehicle to repair area**

ESP®测试合格 ⇒ **继续下线检测流程**
ESP®测试不合格 ⇒ **退出，打印错误信息并将车辆送至维修区**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.3.3.1 Diagram base brake and dynamic ESP® test (independent of the brake split distribution) 基础制动测试和动态 ESP®测试（与制动回路分布类型无关）



Remarks: 备注：

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Please note the maximal allowed valve activation times. More information about these times can be read in the chapter 6. 请注意阀启动最大允许时间。关于这些时间更多信息请参考第 6 章。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

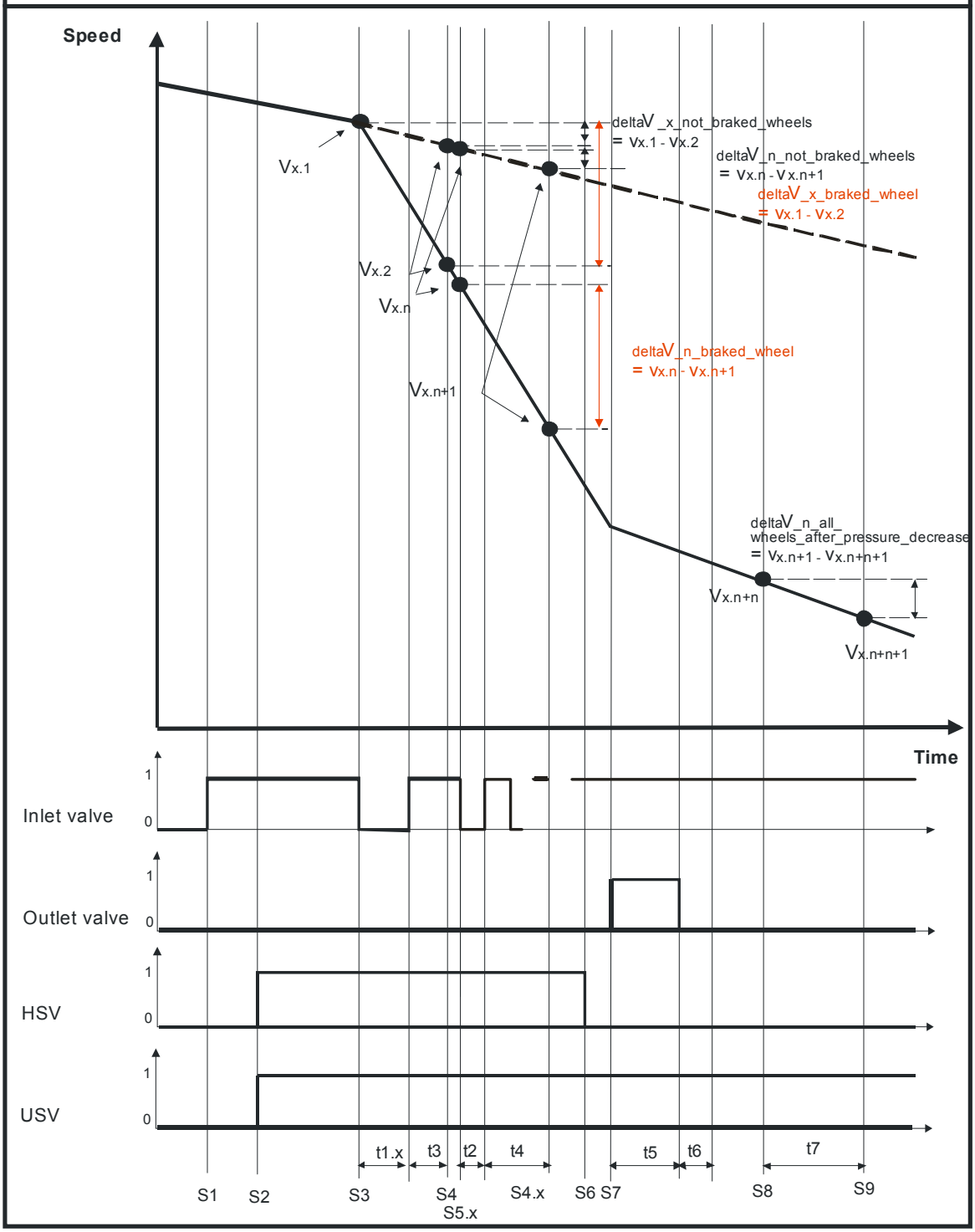
4.3.3.2 Diagram to the dynamic ESP® test exemplary at one wheel 单个车轮的动态 ESP®测试示意图

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

Dynamic ESP test

Test diagram (exemplary at one wheel):



Remarks: 备注：

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

- The diagnostic tester sends the request (S1) to close the inlet valve. Then it sends a second request (S2) to close the pilot valve (USV) and to open the priming (HSV) valve.
  诊断设备首先发送请求（S1）关闭进液阀，随后发送第二个请求（S2）关闭回路控制阀（USV）并打开高压开关阀（HSV）。

- As next step the tester sends a further request (S3). The ECU takes the first speed sample (Vx.1) and then it executes the first pressure increase pulse for the defined time (t1.x) to reduce the air gap between the brake disk and pads.
  下一步测试设备再度发送请求（S3）。ECU 采集第一个速度样本（Vx.1）并在规定时间（t1.x）内执行第一个增压脉冲，以降低制动盘和摩擦垫片之间的气隙。

- The ECU takes the second speed sample (Vx.2) after the defined stabilisation time (t3) is expired.
  在规定的稳定时间（t3）结束后，ECU 采集第二个速度样本（Vx.2）。

- The ECU takes these two speed samples, calculates the differences speed for the pressure increase cycle (Vx.1-Vx.2 = ΔV_braked_wheel/wheel_xx & ΔV_not_braked_wheels/wheel_xx) and outputs the result if the tester request it (S4).
  对于这两个速度样本，ECU 计算增压循环的速度差异（Vx.1-Vx.2 = ΔV_braked_wheel/wheel_xx & ΔV_not_braked_wheels/wheel_xx），并在测试设备请求时（S4）输出结果。

- Depending on the expected deceleration the tester sends a number of requests within a loop (S5.x) to increase the pressure (n pressure increase cycles with the time t2 and the stabilisation time t4). The ECU takes these individual speed samples, calculates the differences speed for the loop of pressure increase cycles (Vx.n-Vx.n+1) and outputs the result if the tester request it (S4.x). The loop is executed as long as the expected deceleration is reached.
  取决于减速度期望值大小，测试设备将在一个循环（S5.x）内发送若干个增压请求（在时间 t2 和稳定时间 t4 内的 n 个增压循环）。ECU 采集这些单独的速度样本，计算该增压循环内的速度差异（Vx.n-Vx.n+1）并在测试设备请求时（S4.x）输出结果。该循环将执行至达到减速度期望值为止。

- As next step the tester sends the request (S6) to close the priming valve (HSV). Afterwards the tester executes a pressure decrease for the defined time t5 and waits then the stabilisation time t6 (S7).
  在下一步中，测试设备将发送请求（S6）以关闭高压开关阀（HSV）。随后，测试设备在规定时间 t5 内减压，并等待 t6 的稳定时间（S7）。

- The ECU takes before and after the measure time (t7) further speed samples (Vx.n+n; Vx.n+n+1).
  在测量时间（t7）前后 ECU 将采集更多的速度样本（Vx.n+n; Vx.n+n+1）。

- The ECU uses these speed samples and calculates the differences speed for the pressure decrease cycle (Vx2.1-Vx.2.2 = ΔV_all_wheels_after_pressure_decrease/wheel_xx) and outputs the result if the tester request it (S9).
  对于这些速度样本，ECU 计算减压循环的速度差异（Vx2.1-Vx.2.2 = ΔV_all_wheels_after_pressure_decrease/wheel_xx），并在测试设备请求时（S9）输出结果。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

To guarantee the comparability between the individual calculated speed differences, the measure times between the first longer pressure increase cycle, the following shorter increase cycles (loop of pulses) and the pressure decrease cycle should always be identically --> (t1+t3) = (t2+t4) = (t5+t6) = t7.
为了确保计算的各个速度差异值之间具有可比性，时间较长的第一个增压循环、随后较短的增压循环（脉冲循环）和减压循环的测量时间必须始终保持相同--> (t1+t3) = (t2+t4) = (t5+t6) = t7。

To save test time, the pressure decrease function can be ensured by evaluation of the speed difference at the next tested wheel.
为了节省测试时间，可以通过评估下一个测试轮的速度差异来确保减压功能正常。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.3.3.3 Test sequence for the dynamic ESP® test (independent of the brake split distribution) 动态 ESP®测试序列（与制动回路分布类型无关）

Start message: 启动信息	Operator prompt: 操作者提示信息	Measure sample: 测量样本	Activation before the time 1: 时间 1 前动作	Time 1: 时间 1	Activation before the time 2: 时间 2 前动作	Time 2: 时间 2	Measure sample: 测量样本
1	Don't apply the brake pedal! 禁止踩下制动踏板	V1.1	---	t7	---	---	V1.2
2	Apply the brake pedal till the expected decelerations are reached! 踩下制动踏板直至达到减速度期望值	V2.1	---	t7	---	---	V2.2
3	Release the brake pedal! 松开制动踏板	V3.1	---	t7	---	---	V3.1
4	Don't apply the brake pedal! 保持制动踏板不动	---	Pump motor on 启动泵马达	---	---	---	---
5		---	Inlet valves front axle on 启动前轴进液阀	---	Inlet valves rear axle on 启动后轴进液阀	---	---
6		---	USV1 and USV2 on	---	---	---	---
7		---	HSV1 and HSV2 on	---	---	---	---
8		V4.1	EVFL off	t1.1	EVFL on & measurement pressure increase FL	t3.1	V4.2
---		Loop pressure increase (number x1) 循环增压（数量 x1）					
9.x		V4.n	EVFL off	t2	EVFL on & measurement pressure increase FL	t4	V4.n+1
10		---	HSV1 and HSV2 off	---	---	---	---
11		---	AVFL on	t5	AVFL off	t6	---
12		---	HSV1 and HSV2 an	---	---	---	---

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

13		V5.1	EVFR off	t1.1	EVFR on & measurement pressure release FL & increase FR	t3.1	V5.2
---		Loop pressure increase (number x1) 循环增压（数量 x1）					
14.x		V5.n	EVFR off	t2	EVFR on	t4	V5.n+1
15		---	HSV1 and HSV2 off	---	---	---	---
16		---	AVFR on	t5	AVFR off	t6	---
17		---	HSV1 and HSV2 an	---	---	---	---
18		V6.1	EVRL off	t1.2	EVRL on & measurement pressure in-	t3.2	V6.2
---		Loop pressure increase (number x1) 循环增压（数量 x1）					
19.x		V6.n	EVRL off	t2	EVRL on & measurement pressure in-crease RL	t4	V6.n+1
20		---	HSV1 and HSV2 off	---	---	---	---
21		---	AVRL on	t5	AVRL off	t6	---
22		---	HSV1 and HSV2 an	---	---	---	---
23		V7.1	EVRR off	t1.2	EVRR on & measurement pressure release RL & increase RR	t3.2	V7.2
---		Loop pressure increase (number x1) 循环增压（数量 x1）					
24.x		V7.n	EVRR off	t2	EVRR on & measurement increase RR	t4	V7.n+1
25		---	HSV1 and HSV2 off	---	---	---	---
26		---	AVRR on	t5	AVRR off	t6	---
27		V8.1	measurement pres-sure release RL	---	---	t7	V8.2
28		---	USV1 and USV2 off	---	---	---	---
29		---	Inlet valves front axle off 断电前轴进液阀	---	Inlet valves rear axle off 断电后轴进液阀	---	---

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

30		---	---	t8	Pump motor off 断电泵马达		

Remarks: 备注：

After the final test speed is reached the rolls are detached by the test rig at first. Then the freewheeling deceleration (ΔV1 = V1.1-V1.2) at all individual wheels are captured (start message 1). These values are subtracted from the afterwards measured wheel decelerations. As next step the operator is prompt to apply the brake pedal till the expected deceleration is reached and measurement for the base brake test (ΔV2 = V2.1-V2.2) is executed (start message 2). These values are used to judge the base brake function. Afterwards the operator has to release the brake pedal. This is ensured by measuring of the wheel decelerations (ΔV3=V3.1-V3.2) again (start message 3).
在达到最终测试速度后，试验台首先断开转毂以测量各车轮的自由转动减速度（ΔV1 = V1.1-V1.2）（启动信息 1），这些值将从随后测得的车轮减速度中减去。下一步，操作者将被提示踩下制动踏板直至达到减速度期望值并完成基础制动测试测量（ΔV2 = V2.1-V2.2）（启动信息 2）。这些值用于判断基础制动功能。随后操作者应松开制动踏板，为了确保这一点，应再次测量车轮减速度（ΔV3=V3.1-V3.2）（启动信息 3）。

Then the tester switches on the pump motor (start message 4), all inlet valves (start message 5), both USV (start message 6) and both HSV valves (start message 7).
随后测试设备启动泵马达（启动信息 4）、所有进液阀（启动信息 5）、所有 USV（启动信息 6）和所有 HSV（启动信息 7）。

To get a sufficient deceleration at the wheel FL, the tester runs a number of pressure increase pulses. The first pulse should be longer than the following. This is required to reduce the air gap between the brake pads and disk (start message 8 with the pressure increase time FA t1.1 & stabilisation time FA t3.1). Afterwards the tester runs a number of loops with a smaller pressure increase time (start message 9.1 till 9.n; pressure increase time - general t2 & stabilisation time – general t4).
为了保证前左轮达到充分减速度，测试设备将运行若干个增压脉冲。第一个增压脉冲的时间应当长于之后的脉冲，这是为了降低摩擦垫片和制动盘之间的气隙（启动信息 8，前轴增压时间 t1.1&前轴稳定时间 t3.1）。随后，测试设备运行若干个增压时间相对较短的循环（启动信息 9.1 至 9.n，增压时间 - 常规 t2&稳定时间 – 常规 t4）。

The loop is executed as long as the expected deceleration (ΔV_FA_pressure_increase) or the maximal allowed number (x1) of pressure increase pulses is reached.
执行该循环直至达到减速度期望值（ΔV_FA_pressure_increase）或达到增压脉冲最大允许数量（x1）。

The actual speed is measured by the ECU before each pressure increase pulse is executed and after the stabilisation time is expired.
在每个增压脉冲执行前且稳定时间结束后，ECU 测量实际速度。

The ECU uses these speed samples and calculates the speed differences for the pressure increase cycle (ΔV4=V4.n-V4.n). After the stop criteria are reached, the tester switches off both HSV valves (start message 10) at first, then it releases the pressure at the wheel FL by switching on of the outlet valve (start message 11, time t5).
通过这些速度样本，ECU 可以计算出增压循环的速度差异值（ΔV4=V4.n-V4.n）。在达到停止标准后，测试设备将首先断电所有 HSV 阀（启动信息 10），其次通过启动出液阀（启动信息 11，时间 t5）释放前左轮中的压力。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

To be able to execute the same procedure at the wheel FR both HSV valves have to be switched on again (start message 8). Afterwards the pressure is first increased then released at the wheel FR (start messages 9-12).
当在前右轮执行相同步骤时，所有 HSV 阀必须再次启动（启动信息 8）。之后，对前右轮先增压再减压（启动信息 9-12）。

The pressure release function is guaranteed by evaluation of the deceleration values after first pressure increase cycle at the wheel FR. This is ensured by evaluation of the wheel decelerations (ΔV5 = V5.1-V5.2).
通过评估第一次增压循环后前右轮的减速度值，可以确保减压功能正常。这是通过评估车轮减速度（ΔV5 = V5.1-V5.2）来实现的。

The test at the wheel FR and rear axle is done in the same way (start messages 12 till 27; measurements of wheel decelerations ΔV5 till ΔV8).
后轴测试方法与前轴相同（启动信息 12 至 27；测量车轮减速度 ΔV5 至 ΔV8）。

At the end of the ESP® all valves, which are still switch on, are switched off (start messages 28&29). The pump motor is switched off after the time t8 (start message 30).
在 ESP®测试结束时，断电所有已启动的阀（启动信息 28&29）。然后在 t8 的时间后断电泵马达（启动信息 30）。

Details to used diagnostic function see chapter 5.6 or chapter 5.7.
关于采用的诊断功能具体信息请参考 5.6 或 5.7 小节。

Important remark: 重要备注：

At ESP® systems pilot valves without pressure limitation are used. To protect these systems against over pressure, the pressure is limited by controlling of the pilot valve current. This current corresponds to the maximal system pressure (fluctuation by the allowed tolerances possible). More details can be read in the chapter 6.
ESP®系统中使用的回路控制阀没有限压。为了避免系统压力过大，将通过调节回路控制阀的电流来限制压力。该电流 与最大系统压力相对应（可能在允许公差范围内波动）。更多信息请参考第 6 章。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.3.3.4 Parameter (base brake, dynamic ABS and ESP® test) 参数（基础制动测试、动态 ABS 测试和动态 ESP®测试）

Kind of parameter:	Parameter name:		Unit:	Physical value:
Time 时间	1. pressure increase pulse FA (ESP® test) 前轴增压脉冲（ESP®测试）	t1.1	Milliseconds (ms)	120
	1. pressure increase pulse RA (ESP® test) 后轴增压脉冲（ESP®测试）	t1.2		80
	general pressure increase pulse FA (ABS test) 前轴常规增压脉冲（ABS 测试）	t1.3		60
	general pressure increase pulse RA (ABS test) 后轴常规增压脉冲（ABS 测试）	t1.4		40
	pressure increase loop pulse (general) 增压循环脉冲（常规）	t2		20
	stabilisation time FA (1. pressure increase pulse – ESP®) 前轴稳定时间（增压脉冲 – ESP®）	t3.1		380
	stabilisation time RA (1. pressure increase pulse – ESP®) 后轴稳定时间（增压脉冲 – ESP®）	t3.2		420
	stabilisation time FA (pressure increase pulse – ABS) 前轴稳定时间（增压脉冲 – ABS）	t3.3		440
	stabilisation time RA (pressure increase pulse – ABS) 后轴稳定时间（增压脉冲 – ABS）	t3.4		460
	stabilisation time general – after the following pressure increase pulses (loop) 常规稳定时间 – 在接下来的增压脉冲（循环）之后	t4		480
	pressure decrease pulse – general 减压脉冲 – 常规	t5		300
	stabilisation time (general) after the pressure decrease pulse 减压脉冲后的稳定时间（常规）	t6		200
	Measurement time – general 测量时间 – 常规	t7		500
	Post run time of the pump motor 泵马达滞后运行时间	t8		2000
Number 数量	Number of loops for the pressure increase 增压循环数	x1	---	5
Evaluation parameter 评估参数	ΔV_FA_pressure_increase ΔV 前轴增压		Kilometre per hour (km/h)	tbd
	ΔV_RA_pressure_increase ΔV 后轴增压			tbd
	ΔV_unpressurised ΔV 无压力			tbd
	ΔV_freewheeling_max ΔV 自由转动最大值			tbd
	ΔV_FA_brakes ΔV 前轴制动			tbd
	ΔV_FA_pressure_increase_brakes ΔV 前轴增压制动			tbd
	ΔV_RA_brakes ΔV 后轴制动			tbd
	ΔV_RA_pressure_increase_brakes ΔV 后轴增压制动			tbd
	V_brakes_tolerance ΔV 制动公差		Percent (%)	25

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Remarks: 备注：

The mentioned times are proposed times such as the activation times for the pressure increase and release cycles, the stabilisation and measure times. All these times depend on the vehicle (e.g. size of brake, orifices in the hydraulic unit, etc.) respectively the test rig, therefore these times must be confirmed respectively modified during trials at the customer plant.
上述时间值为建议值，例如增压减压循环的启动时间、稳定时间和测量时间等。所有这些时间值都取决于车辆（如制动系统尺寸、液压单元螺纹孔径等）和试验台，因此必须在客户工厂中通过试验确认或调整。

Also the mentioned evaluation parameter have to be defined by a statistic evaluation with the first tested vehicles .The definition of all these data is up the customer responsibility.
同时，上述的评估参数必须在最初测试车辆上通过统计学评估来定义。所有数据的定义取决于客户。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.3.3.5 Proposal of the evaluation of the base brake and dynamic ESP® test 基础制动测试和动态 ESP®测试的评估逻辑建议

a) Evaluation of the freewheeling before the base brake test (exemplary for the wheel FL):
在基础制动测试前评估自由转动减速度（以前左轮为例）：

If
ΔV1/wheel_FL ≤ ΔV_freewheeling_max

then ΔV1/wheel_FL = ΔV_freewheeling and execute test step b)

else stop process and output “Freewheeling at the wheel FL nok”

如果 ΔV1/wheel_FL ≤ ΔV_freewheeling_max，
则 ΔV1/wheel_FL = ΔV_freewheeling 并执行测试步骤 b)；
否则停止流程并输出“前左轮自由转动减速度不合格”。

b) Evaluation of the base brake test (exemplary for the front axle):
评估基础制动测试（以前轴为例）

If
(ΔV2/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_brakes
and
(ΔV2/wheel_FL – V_brakes_tolerance) ≤ ΔV2/wheel_FR ≤ (ΔV/wheel_FL + V_brakes_tolerance)

then execute test step c)

else stop process and output “Base brake test at the front axle nok”

如果(ΔV2/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_brakes
且(ΔV2/wheel_FL – V_brakes_tolerance) ≤ ΔV2/wheel_FR ≤ (ΔV/wheel_FL + V_brakes_tolerance)，
则执行测试步骤 c)；
否则停止流程并输出“前轴基础制动测试不合格”。

c) Evaluation of the freewheeling after the base brake test (exemplary for the wheel FL):
在基础制动测试后评估自由转动减速度（以前左轮为例）：

If
ΔV3/wheel_FL ≤ ΔV_freewheeling_max

then execute test step d)

else stop process and output “Freewheeling at the wheel FL nok”

如果 ΔV3/wheel_FL ≤ ΔV_freewheeling_max，
则执行测试步骤 d)；
否则停止流程并输出“前左轮自由转动减速度不合格”。

d)) Evaluation of the ESP® test „pressure increase“ (exemplary for the wheel FL):

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

评估 ESP®测试“增压”（以前左轮为例）：

If

(ΔV4.n/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_pressure_increase

then execute test step e)

else stop process and output “ESP® pressure increase at the wheel FL nok ”

如果(ΔV4.n/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_pressure_increase，
则执行测试步骤 e)；
否则停止流程并输出“前左轮 ESP®增压不合格”。

e) Evaluation of the ESP® test „pressure decrease“ (exemplary for the wheel FL):
评估 ESP®测试“减压”（以前左轮为例）：

If

ΔV5.1/wheel_FL ≤ (ΔV_unpressurised + ΔV_freewheeling_FL)

then output “Base brake and dynamic ESP® test ok”;

else output “ESP® pressure decrease at the wheel FL nok”

如果 ΔV5.1/wheel_FL ≤ (ΔV_unpressurised + ΔV_freewheeling_FL)，
则输出“基础制动系统和动态 ESP®测试合格”；
否则输出“前左轮 ESP®减压不合格”。

Example: 范例：

Evaluation variables*: 评估变量*：
ΔV_FA_pressure_increase = 2,0 km/h
ΔV_unpressurised = 0,5 km/h
ΔV_freewheeling_max = 0,8 km/h
ΔV_FA_brakes = 2 km/h
V_brakes_tolerance = 25%

*All mentioned variables for evaluation are valid for the measure time of 500ms. All values are examples and must be finally defined by the customer!
*所有上述评估变量仅适用于 500ms 的测量时间。所有值只是范例，最终还需由客户定义。

a) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step “freewheeling before the base brake test”:
评估测试步骤“基础制动测试前的自由转动减速度”中计算出的各车轮速度差异（km/h）：

ΔV1_FL	ΔV1_FR	ΔV1_RL	ΔV1_RR
0,5	0,5	0,3	0,3

b) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step “base brake test”:
评估测试步骤“基础制动测试”中计算出的各车轮速度差异（km/h）：

ΔV2_FL	ΔV2_FR	ΔV2_RL	ΔV2_RR
2,8	2,9	2,6	2,5

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

c) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "freewheeling after the base brake test":
评估测试步骤"基础制动测试后的自由转动减速度"中计算出的各车轮速度差异（km/h）：

ΔV3_FL	ΔV3_FR	ΔV3_RL	ΔV3_RR
0,6	0,5	0,4	0,3

d) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "ESP® test – pressure increase FL":
评估测试步骤"ESP®测试-前左轮增压"中计算出的各车轮速度差异（km/h）：

ΔV4.n_FL	ΔV4.n_FR	ΔV4.n_RL	ΔV4.n_RR
2,8	0,7	0,4	0,8

e) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "ESP® test – pressure decrease FL":
评估测试步骤"ESP®测试-前左轮减压"中计算出的各车轮速度差异（km/h）：

ΔV5.1_FL	ΔV5.1_FR	ΔV5.1_RL	ΔV5.1_RR
0,9	1,0	0,8	0,8

<u>Judgement of each test step for the wheel FL respectively FA:</u>
前左轮和前轴的每个测试步骤判断值：

a) Freewheeling before the base brake test: 0,5 ≤ 0,8 **= ok**
基础制动测试前的前左轮自由转动减速度：0,5 ≤ 0,8 **= ok**
b) Base brake test front axle: ((2,8-0,5) ≥ 2) && (2,8-0,7) ≤ 2,9 ≤ (2,8+0,7) **= ok**
前轴基础制动测试：((2,8-0,5) ≥ 2) && (2,8-0,7) ≤ 2,9 ≤ (2,8+0,7) **= ok**
c) Freewheeling after the base brake test: 0,6 ≤ 0,8 **= ok**
基础制动测试后的前左轮自由转动减速度：0,6 ≤ 0,8 **= ok**
d) ESP® test „pressure increase FL": (2,8-0,5) ≥ 2,0 **= ok**
ESP®测试"前左轮增压"：(2,8-0,5) ≥ 2,0 **= ok**
e) ESP® test „pressure decrease FL": 0,9 ≤ (0,5+0,5) **= ok**
ESP®测试"前左轮减压"：0,9 ≤ (0,5+0,5) **= ok**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

4.4 Finishing actions & standstill tests 结束动作和静态测试

4.4.1 Evaluation of the fault code memory 评估故障码内存

Evaluation of the fault code memory 评估故障码内存	
Tester: 测试设备	Remarks: 备注
Reading and evaluating of the fault code memory. 读取并评估故障码内存。	More details can also be read in the chapter 4.1.10. 更多信息请参考 4.1.10 小节。

Remarks: 备注

At the end of the ABS/ ESP® end of line (EOL) tests the fault code memory is finally checked. In case of fault codes are stored, the vehicle has to send to the rework area and after the root cause of is removed the ABS/ ESP® EOL test has to repeat again.
在 ABS / ESP®下线检测结束时，将对故障码内存做最后检查。如果其中储存有故障码，车辆必须送至维修区。在故障根本原因排除后，必须重新执行 ABS / ESP®下线检测。

The evaluation of the fault code memory has following purpose:
评估故障码内存目的如下：

1) To ensure that the ABS / ESP® system is free of faults after the EOL tests
   确保 ABS / ESP®系统在下线检测结束后无故障

ABS/ESP® system free of faults ⇒ Continue

ABS/ESP® system not free of faults ⇒ Abort, print out fault message and send vehicle to repair area

ABS/ESP® 系统无故障 ⇒ 继续下线检测流程
ABS/ESP® 系统有故障 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.4.2 Warning lamp test (change to the default diagnostic mode) 告警灯测试（切换为默认诊断模式）

Warning lamp test (change to the default diagnostic mode) 告警灯测试（切换为默认诊断模式）	
Operator / Tester: 操作者/测试设备	Remarks: 备注
Check and confirm if all ABS / ESP® warning and signal lamps are on! 检查并确认是否 ABS / ESP®所有告警灯和信号灯都点亮	All ABS/ ESP® signal and warning lamps are activated in the customer specific diagnostic mode! 在客户指定诊断模式下 ABS / ESP®所有告警灯和信号灯将点亮！
Change to the default diagnostic mode. 切换为默认诊断模式	
Check and confirm if all ABS / ESP® warning and signal lamps are off after the time $t_{signal_warning_lamps_off}$ is expired! 在 $t_{signal_warning_lamps_off}$ 的时间之后检查并确认是否 ABS / ESP®所有告警灯和信号灯都关闭	ABS/ ESP® executes a software reset. After the time $t_{signal_warning_lamps_off}$ is expired respectively the system is free of faults all signal and warning lamps are switched off. ABS / ESP®重启软件。在 $t_{signal_warning_lamps_off}$ 的时间之后且系统无故障时，所有信号灯和告警灯将关闭。
Ignition/engine off. 点火开关/发动机关闭	
Evaluation of the entire test data & print of a test protocol 评估所有测试数据并打印测试协议	

Remarks: 备注

As long as the ECU is within the customer specific diagnostic mode all control functions are deactivated and all ABS / ESP® warning respectively signal lamps are illuminated. This special mode (test mode) can be left by changing into the customer default mode (no hardware or software reset required).
当 ECU 处于客户指定诊断模式时，所有控制功能将禁用，同时 ABS / ESP®所有告警灯和信号灯将点亮。如果想脱离这种特殊模式（测试模式），可以通过切换入客户默认模式（不需要硬件或软件重启）来实现。

Parameter: 参数

Kind of parameter:	Parameter name:	Unit:	Physical value:
Time	$t_{signal_warning_lamps_off}$	Milliseconds	≤ 5000

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

After the EOL tests are finished the tester has to print out the complete test results and a short judgement :ok / nok
在下线检测结束后，测试设备应当打印出完整测试结果以及一个简短判断：合格/不合格

The test of the ABS/ ESP® signal and warning lamps has following purpose:
ABS / ESP®信号灯和告警灯测试目的如下：

1) To ensure the function of all signal and warning lamps
   确认所有信号灯和告警灯功能正常

Signal and warning lamp test ok ⇒ **End of the EOL tests**

Signal and warning lamp test nok ⇒ **Abort, print out fault message and send vehicle to repair area**

信号和告警灯测试合格 ⇒ 下线检测结束
信号和告警灯测试不合格 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

4.5 EOL driving test (only ESP® systems - optional) 下线检测行驶测试（仅限 ESP®系统 – 可选）

The EOL driving test is a optional test (customer specific). This test can be executed as additional test to the previous described tests. This test consists of a standstill and a dynamic driving tests.

下线检测行驶测试为可选测试（客户指定），可以作为上一个测试的附加部分来执行。该测试包括静态和动态行驶测试两部分。

EOL driving test 下线检测行驶测试	
Standstill tests: 静态测试	- Pressure sensor test 压力传感器测试 - Yaw rate and lateral signal offset test 横摆角速度和侧向信号偏移测试 - Storing of the lateral sensor signal offset 侧向传感器信号偏移储存
EOL driving tests: 下线检测行驶测试	- Sensitivity test of the yaw rate sensor signal 横摆角速度传感器信号灵敏度测试 - Sensitivity test of the lateral sensor signal 侧向传感器信号灵敏度测试 - Steering angle sensor signal test 方向盘转角传感器信号测试

Remarks: 备注：

The EOL driving test is activated via diagnostics. Afterwards a defined driving manoeuvre has to be executed. The driver identifies the state of the test by checking of the signal and warning lamps. The individual activation of these lamps shows which test has to be executed as next (e.g. standstill test finished; ECU waits for execution of the dynamic driving test). In addition these lamps show if the tests are finished successfully or if a sensor fault is present.
下线检测行驶测试是由诊断功能启动的，随后将执行规定的行驶试验。驾驶员通过检查信号灯和告警灯来确认测试状态，这些灯通过单个点亮可以显示下个应该执行的测试是什么（例如静态测试结束；ECU 等待执行动态驾驶测试）。此外，这些灯还可以显示测试是否顺利结束或有传感器故障存在。

The EOL driving tests are designed in way, that each individual test can be selected via diagnostics and to do the tests via a very simple manoeuvre.
下线检测行驶测试是这样设计的：通过诊断功能可以选择每个单独的测试，并且测试是通过一个很简单的行驶试验来进行的。

Details to the EOL driving test can be read in the customer specific documentations.
关于下线检测行驶测试更多信息请参考客户规范文件。

EOL driving test ok ⇒ End of the EOL driving test (successfully finished)
EOL driving test nok ⇒ Abort, print out fault message and send vehicle to repair area, repeat EOL driving test

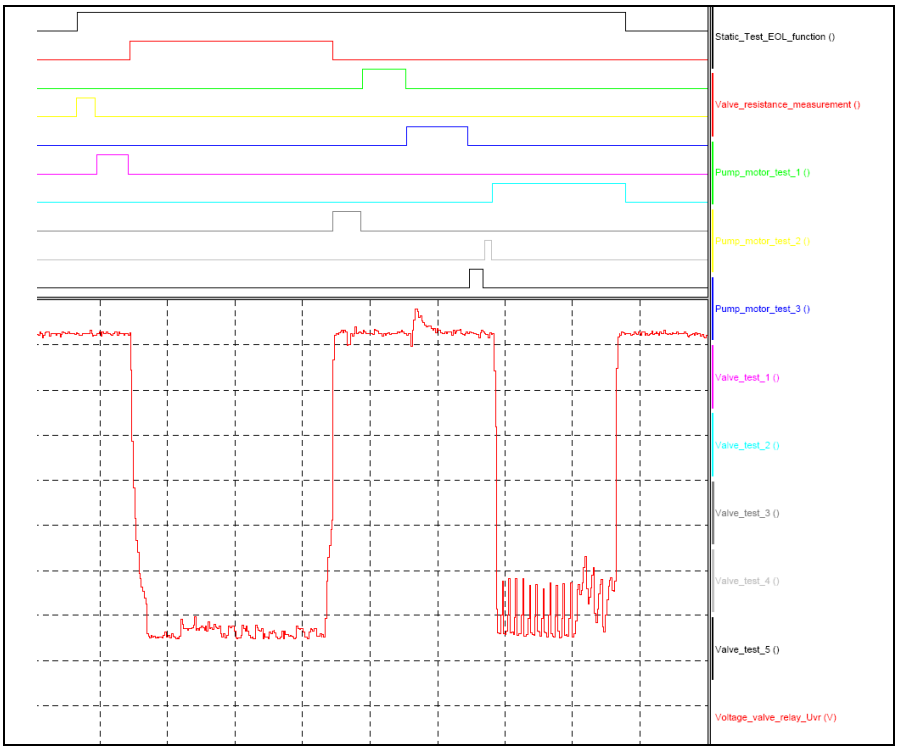
下线检测行驶测试合格 ⇒ 下线检测行驶测试结束（成功完成）
下线检测行驶测试不合格 ⇒ 退出，打印错误信息并将车辆送至维修区

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

5 Bosch specific EOL diagnostic routines at generation 9 诊断程序说明

5.1 Diagnostic routine „Static Test“



5.1.1 Start parameter „Static Test“

Start parameter „Static Test“

Remarks:

This routine has no specific start parameter 详见项目诊断功能描述.

5.1.2 Evaluation parameter „Static Test“

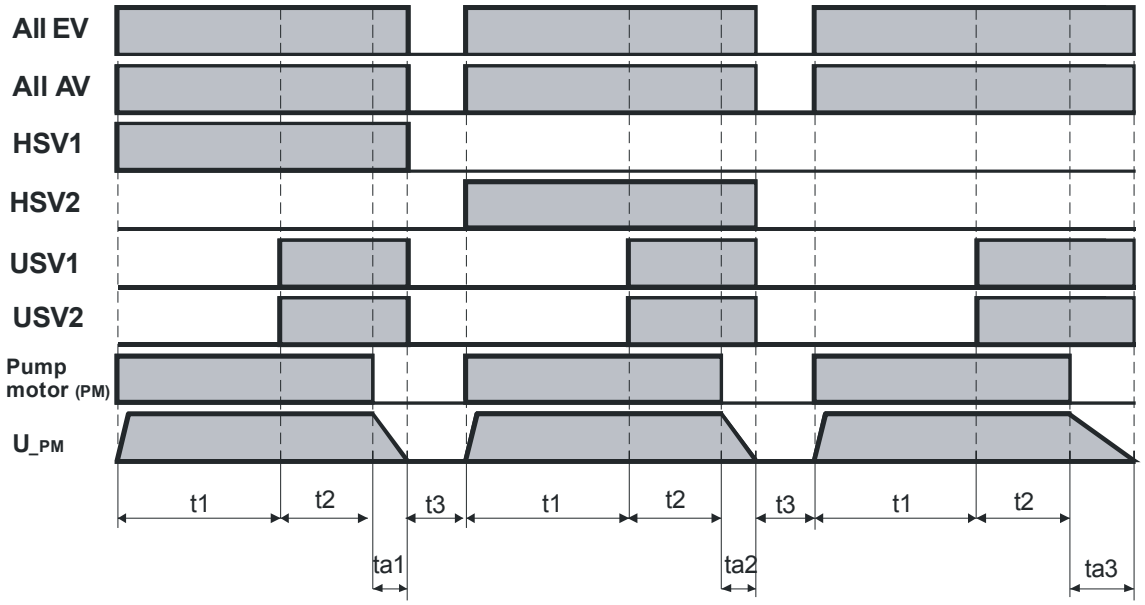
Evaluation parameter „Static Test“

Remarks:

This routine has no specific evaluation parameter 详见项目诊断功能描述.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

5.2 Diagnostic routine „Pump motor slow-down time test“



5.2.1 Start parameter „Pump motor slow-down time test“

Start parameter „Pump motor slow-down time test“
Delay time t_1 (time till the USV1 & USV2 is switched on)
Delay time t_2 (time till the PM is switched off)

Remarks:
The length (byte or word) of the start parameter and resolution is customer specific 详见项目诊断功能描述.

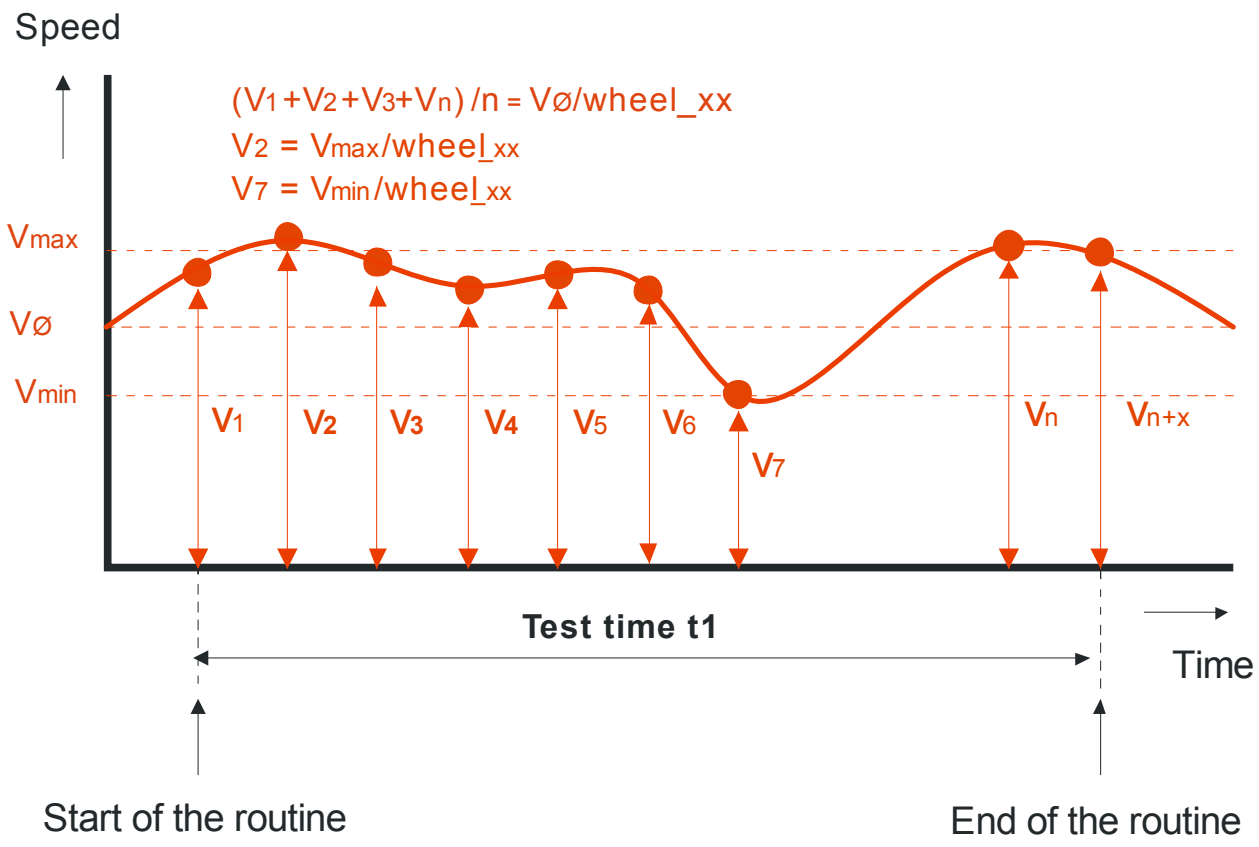
5.2.2 Evaluation parameter „Pump motor slow-down time test“

Evaluation parameter „Pump motor slow-down time test“
Post run time t_{a1}
Post run time t_{a2}
Post run time t_{a3}

Remarks:
The length (byte or word) of the evaluation parameter and resolution is customer specific 详见项目诊断功能描述.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

5.3 Diagnostic routine „Wheel Speed Sensor Test“



5.3.1 Start parameter „Wheel Speed Sensor Test“

Start parameter „Wheel Speed Sensor Test“
Test time t_1

Remarks:

The length (byte or word) of the start parameter and resolution is customer specific 详见项目诊断功能描述.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

5.3.2 Evaluation parameter „Wheel Speed Sensor Test“

Evaluation parameter „Wheel Speed Sensor Test“
V_max/wheel_FL
V_min/wheel_FL
V_max/wheel_FR
V_min/wheel_FR
V_max/wheel_RL
V_min/wheel_RL
V_max/wheel_RR
V_min/wheel_RR
*V_Ø/wheel_FL
*V_Ø/wheel_FR
*V_Ø/wheel_RL
*V_Ø/wheel_RR

Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific 详见项目诊断功能描述.

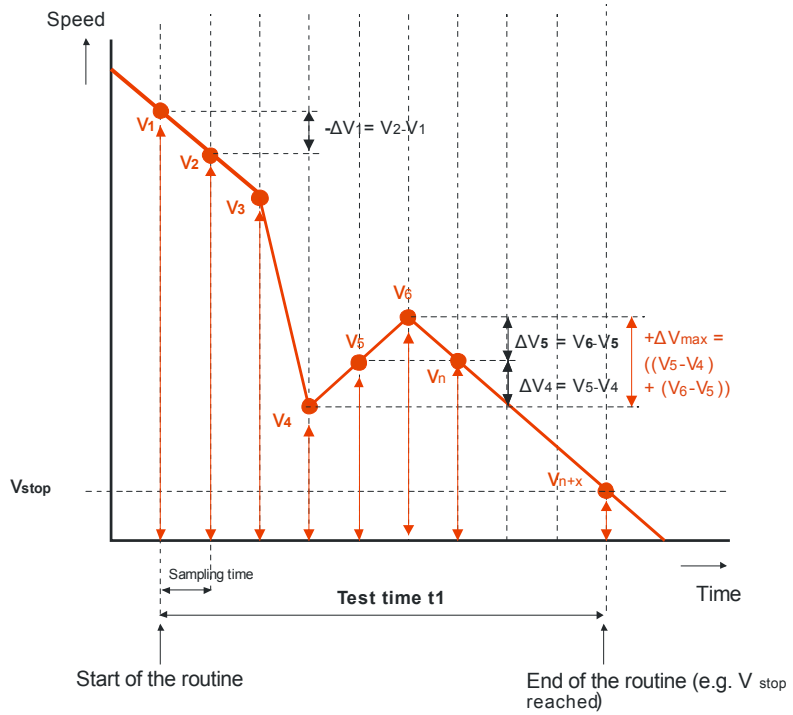
* The output of the average speed values can be switched on or off depending on the customer request （具体客户需求）.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

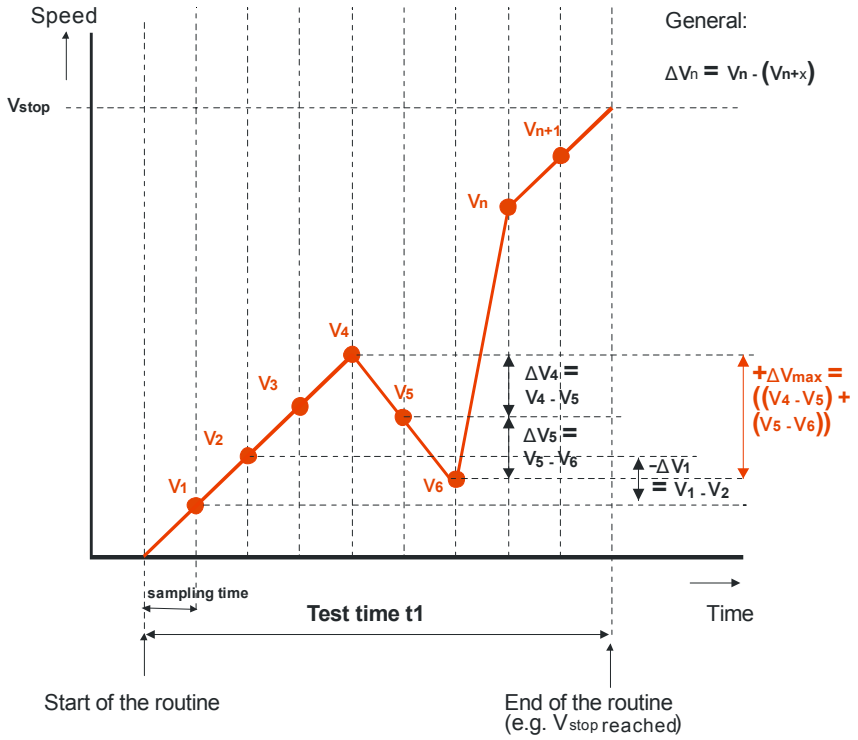
对本文有任何疑问，以英文为准。

5.4 Diagnostic routine „Wheel Speed Sensor Test – accelerated/decelerated vehicle“

Test phase: during deceleration:



Test phase: during acceleration:



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

5.4.1 Start parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“

Start parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“
Test time t_1
V_{Stop}
Test phase (during acceleration or deceleration)

Remarks:

The test time t_1 for the wheel speed sensor signal quality test has to be adapted on the roller speed. This is required to ensure that one complete rotation of the wheel is tested. The alternate parameter stop speed V_{Stop} has to be defined in the same way. Which of both parameters are reached at first (expired test time t_1 respectively stop speed V_{Stop} reached) stops the test routine. After that the tester can pick up and evaluate the test results. V_{stop} was defined that the routine stops if the last wheel exceeds (acceleration phase) or is below (deceleration phase) the selected stop speed.

The routine evaluation algorithm can be changed by the parameter “test phase”, so that the test can be executed during acceleration or deceleration of the vehicle.

The length (byte or word) of the start parameter and resolution is customer specific 详见项目诊断功能描述.

5.4.2 Evaluation parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“

Evaluation parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“
ΔV_{max}/Wheel_FL
ΔV_{max}/Wheel_FR
ΔV_{max}/Wheel_RL
ΔV_{max}/Wheel_RR

Remarks:

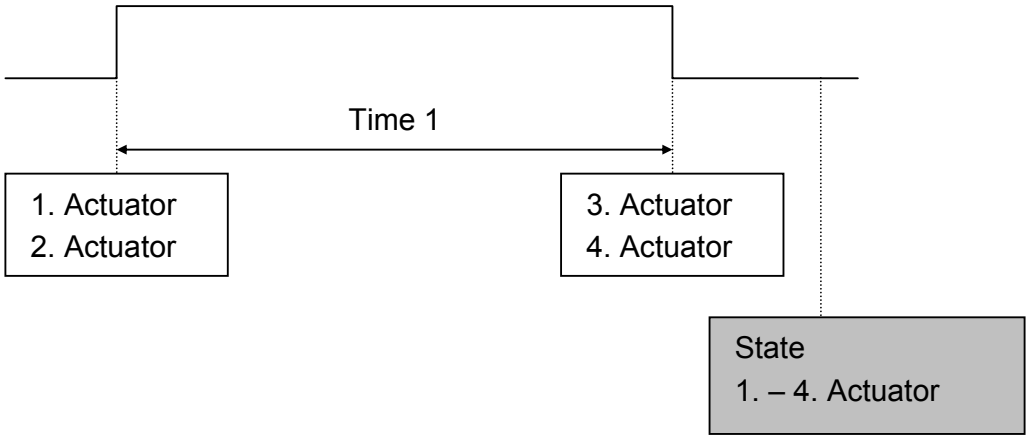
The length (byte or word) of the evaluation parameter and resolution is customer specific.

The routine provides the sum of all in a row following positive difference speed values of each wheel (e.g. ΔV_{max}/wheel_FL, etc.) after the test time t_1 or the selected stop speed V_{stop} is reached.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

5.5 Diagnostic routine „Actuator Test“



5.5.1 Start parameter „Actuator Test“

Start parameter „Actuator Test“
Actuation value 1. actuator
1. actuator
Actuation value 2. actuator
2. actuator
Time 1
Actuation value 3. actuator
3. actuator
Actuation value 4. actuator
4. actuator

Remarks:

The length (byte or word) of the start parameter and resolution is customer specific 详见项目诊断功能描述.

The number and definition of the required actuators respectively the definition of the actuation value is customer specific.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

5.5.2 Evaluation parameter „Actuator Test“

Evaluation parameter „Actuator Test”
State (on/off) 1. actuator
State (on/off) 2. actuator
State (on/off) 3. actuator
State (on/off) 4. actuator

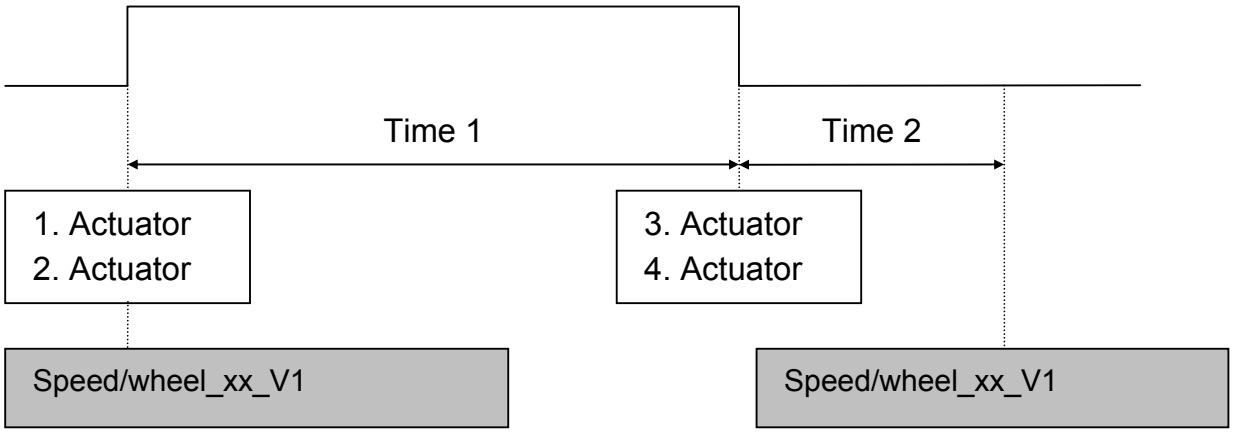
Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific 详见项目诊断功能描述.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

5.6 Diagnostic routine „Dynamic Test“



5.6.1 Start parameter „Dynamic Test “

Start parameter „Dynamic Test“
Actuation value 1. actuator
1. actuator
Actuation value 2. actuator
2. actuator
Time 1
Actuation value 3. actuator
3. actuator
Actuation value 4. actuator
4. actuator
Time 2

Remarks:

The length (byte or word) of the start parameter and resolution is customer specific 详见项目诊断功能描述.

The number and definition of the required actuators respectively the definition of the actuation value is customer specific.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

5.6.2 Evaluation parameter „Dynamic Test“

Evaluation parameter „ Dynamic Test “
Speed_Δ/Wheel_FL (= Speed/wheel_FL_V1 - Speed/wheel_FL_V2)
Speed_Δ/Wheel_FR (= Speed/wheel_FR_V1 - Speed/wheel_FR_V2)
Speed_Δ/Wheel_RL (= Speed/wheel_RL_V1 - Speed/wheel_RL_V2)
Speed_Δ/Wheel_RR (= Speed/wheel_RR_V1 - Speed/wheel_RR_V2)

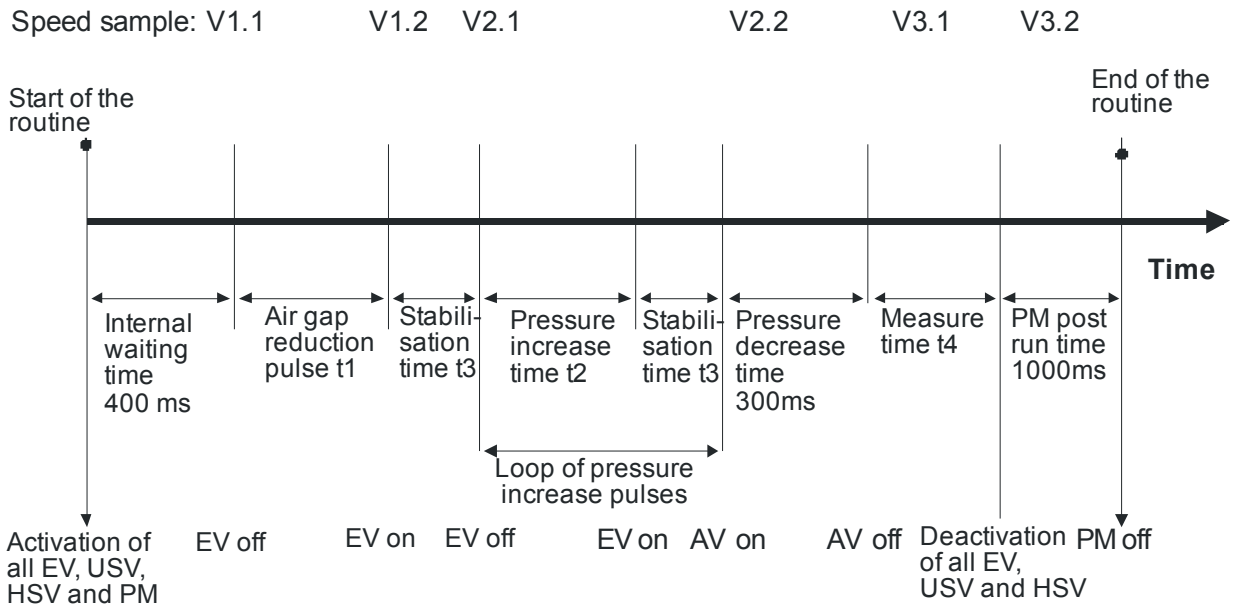
Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific 详见项目诊断功能描述.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

5.7 Diagnostic routine „Extended Dynamic Test“ – only for ESP® systems available



5.7.1 Start parameter „Extended Dynamic Test”

Start parameter „Extended Dynamic Test“
Selected Wheel (FL, FR, RL or RR)
EV actuation time t1 (air gap reduction pulse – to reduce the air ga tween disk and pad)
EV actuation time t2 (loop to increase the pressure)
Number of the max. allowed pressure increase pulses (actuation time t2)
Stabilisation time t3
Demanded speed difference ΔV
Measure time t4 to calculate the ΔV after the pressure release pulse

Remarks:

The length (byte or word) of the start parameter and resolution is customer specific 详见项目诊断功能描述.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

5.7.2 Evaluation parameter „Extended Dynamic Test“

Evaluation parameter „Extended Dynamic Test“
Demanded speed difference ΔV reached? (yes, no, failure)
Number of required pressure increase pulses
ΔV after air gap reduction pulse
ΔV after the 1. pressure increase pulse of the pressure increase loop
ΔV after the n. pressure increase pulse of the pressure increase loop
ΔV after the last pressure increase pulse of the pressure increase loop
ΔV after the pressure decrease pulse

Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific 详见项目诊断功能描述.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

6 Pump motor and valve protection measures respectively notes to the percentaged activation 阀保护措施

	ABS	ESP®
Valve coil:	Diagnostic mode	
EV 7,3 Ω	If U_{VR} > 9V, than lowering of the duty cycle to U_{eff} = 9V. If the valve activation value is > 0%; than the ECU starts the up/down counter and counts with a ratio up/down = 2/1. The total time is limited to 120s. For each EV valve a separate counter is used.	
AV 7,3 Ω	If the valve activation value is > 0%; than the ECU starts the up/down counter and counts with a ratio up/down = 2/1. The total time is limited to 30s. For each AV valve a separate counter is used.	
HSV 3,3 Ω	---	20ms with 100% duty cycle, than lowering of the duty cycle to U_{eff} = 3,5V and cyclic all 200ms for 20ms with 100%
USV 7,3 Ω	---	Max. duty cycle is limited to max. I = 0,73 A (~ 110 ± 20 bar at active pressure increase)
USV 4,3 Ω		tbd
PM	---	If the USV1 & HSV1 or/and USV2 & HSV2 and the PM are switched on, than the ECU starts the PM up/down counter and counts with a ratio up/down = 2/1. After the PM counter has reached its limitation of 30s the PM activation is reset.

Remarks:

The described valve and PM protection measures are always switched on. They cannot be disabled. These measures are required to prevent predamages at the valve coils caused by overheating. In addition the current limitation of the USV coils protect the system against predamages by overpressure. In case of the allowed activation times are exceeded, the valve activations are immediately reset.
阀组与电机保护功能是不可关闭的，当温度过高或压力过大时保护系统开始工作。

Following activation values can be selected by the tester:

Valve coil:	Kind of output stage activation / valve:	Kind of allowed activation via tester:
EV	duty cycling of the voltage / regulator valve	percentaged in 1% steps between 0-100%
AV	duty cycling of the voltage / switching valve	digital: 0 % = off; 100% = on
USV	current controlled / regulator valve	percentaged in 1% steps between 0-100%
HSV	duty cycling of the voltage / switching valve	digital: 0 % = off; 100% = on

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

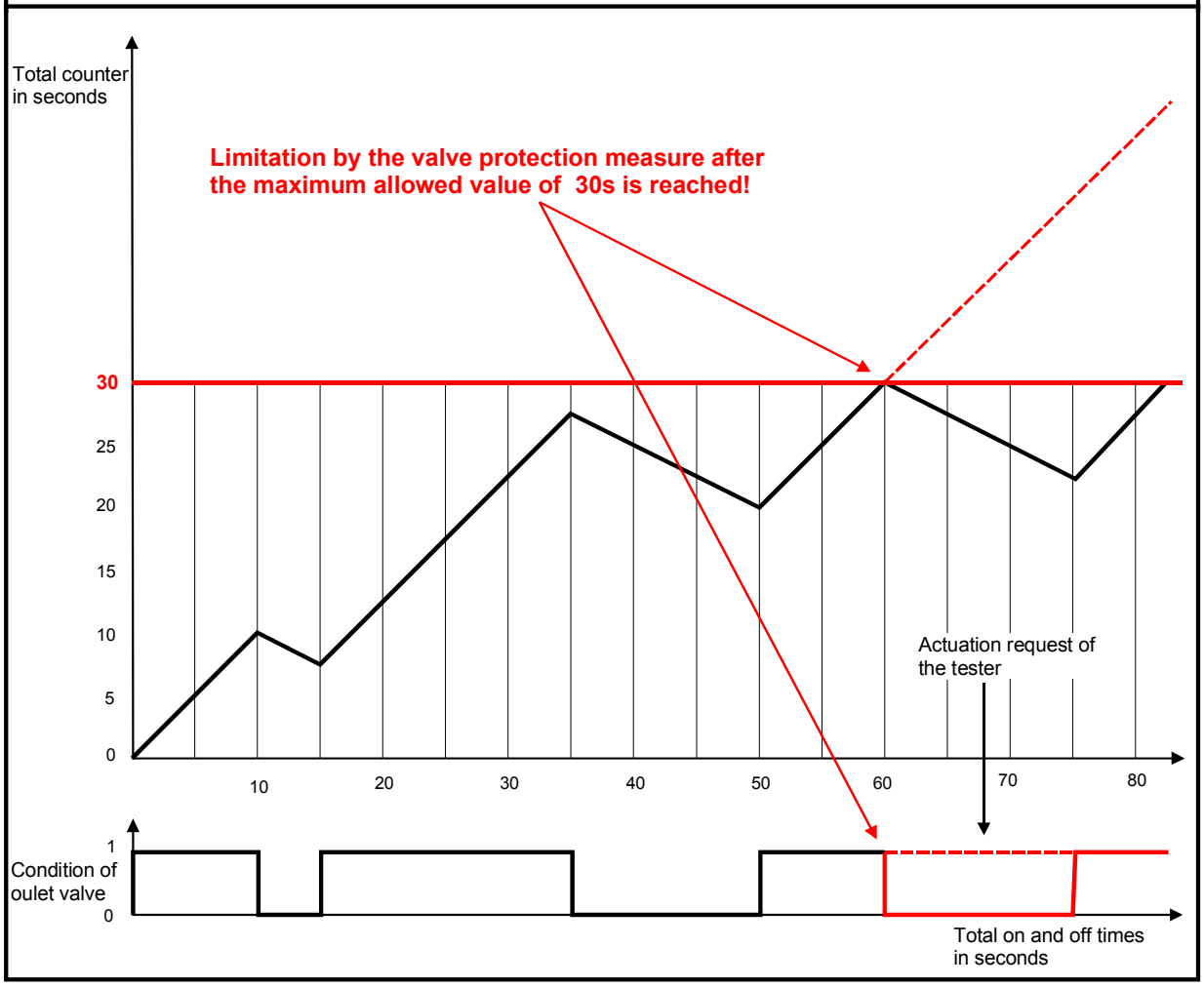
Remarks:

The AV valves are switching valves. To guarantee a correct function (opening or closing of the valve), the valves have been switched on only with 100% and off with 0% duty cycle.
The HSV valves have always been switched on with 100% duty cycle to guarantee, that they have been completely opened also in case of the brake pedal is applied. After that the duty cycle can be reduced to a lower level. This HSV switching on and reduction profile is realised with the valve protection measures therefore the tester does not need to reduce the duty cycle after the HSV valves were switched on with 100%.
For the voltage controlled EV and current controlled USV valves following notes should be noticed:
Depending on the selected percentaged activation value the brake pressure can be hold respectively controlled. A 100% activation value corresponds the maximal brake pressure which is allowed by the valve protection measure. This is especially valid for the USV valves.

Graphical presentation of a up/down counter (AV coil):

Gen 9: Valve protection measure: Up/down counter with a ratio of = 2/1

Example: Outlet valve with a maximum limitation to 30 seconds



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。

Remarks.

The forced limitation by the valve protection measure is active as long as valve coils and the ECU are sufficient protected against predamages caused by overheating. Only after an appropriate cooling time the valves can be activated again.
To guarantee a sufficient cooling time also after restart of the system (e.g. ignition cycle) the provided coil resistance values are evaluated by the valve protection measure. The resistance the coils are measured after each system power up and then cyclic all 20 seconds. These values are used to initialize the up and down counter value for the EV and AV coils.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

7 Test rig information 测试设备信息

1. Dynamometer design 功率计设计:
The ABS / ESP® tests are designed to be performed either on a 2-axle brake dynamometer (low speed) or a roller test bench with 4 independent rolls (high speed).
To allow individual wheel test cycles the rolls of each wheel should be independently driven by the test rig drives.
本测试可在两轴转毂（低速）或 4 个独立转毂（高速）上完成。为了实现单个车轮的测试，每个转毂需要有单独的驱动。

2. Diagnostic Information 诊断信息:
The diagnostic computer should be as close as possible near the vehicle to avoid communication problems. The timing (time gaps between messages and bytes within messages) should be set close to the minimum possible limits in order to reduce the test time.
To be able to execute high speed tests the communication has to be started below the customer specific diagnostic speed limit. In addition this speed limitation function has to be switched off with the corresponding customer specific diagnostic function.
诊断电脑和车辆应尽可能的接近，以确保通讯的良好。为了减少测试时间，各信息之间的时间间隔应接近最小值。高速测试时应在提速前完成建立通讯。根据具体客户需求，限速功能需关闭。

3. Process bytes 过程字节:
The final definition of the process byte data must be synchronized between the EOL facility and the evac/fill resepectively repair bleed equipment supplier or the customer.
最终的过程字节需由客户或设备供应商（下线检测与抽真空加注）确定。

4. Editor menus / vehicle types 编辑菜单/车辆型号:
ECU Identification codes may change during production due to updates of software revisions.
It is important to set up the software program to modify identification numbers easily by using an editor menu.
The timing of the dynamic valve test must be found during test trials.
Timing of valve actuation cycles, such as actuation times and stabilization times should be included in an editor menu.
Upper and lower limit values for evaluation should also be included in an editor menu for simple adaptation or modifications.
Due to the various vehicle types and configurations, different actuation times for the valves cycles of the dynamic ABS / ESP® tests might be required. These data should be also available in separate parameter file within the test equipment software and also easily be modified by using an editor menu.
The selection of the vehicle type should be done by either a bar code reading or an operator selection when driving into the test rig.
在生产过程中，由于软件的升级，可能需要修改 ECU 识别码，编辑菜单中应设置对识别码的修改。
阀的动态测试的时间需经过试验确定，并在编辑菜单中有设置。待评估数据的上/下限值应在编辑菜单中有设置，以便修改。
根据不同的变量代码，测试的数据会有所不同。这些变量代码与数据需写入测试设备中以便操作或修改。当进入测试设备时可通过条形码扫描或由操作者选择来完成车辆型号的匹配。

5. Test protocol 测试协议:
In case of an ok test result, a print out (”test passed”) is made.
In case of a nok test result, a failure protocol must be printed out.
This protocol must include all test data of the test step that failed in order to allow a better failure analysis.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

An entire test protocol of all test results must be available on request.
如果检测通过，须打印“测试通过”的标签。如果测试未通过，须打印该车辆在下线检测过程中所有测试数据以便分析。可以提供所有批量测试的结果。

6. Vehicle conditions 车辆状态:
The battery must be charged properly in order to allow a sufficient power supply to the system.
The engine should be running in order to apply a sufficient power supply.
电瓶需满载以确保测试顺利进行。为确保电瓶供电，发动机可以启动。

7. Electrical circuit diagrams 电路图:
It is the user’s responsibility to make sure they are using the latest available electrical circuit diagram!
一定要使用最新的电路图，以避免不必要的麻烦。

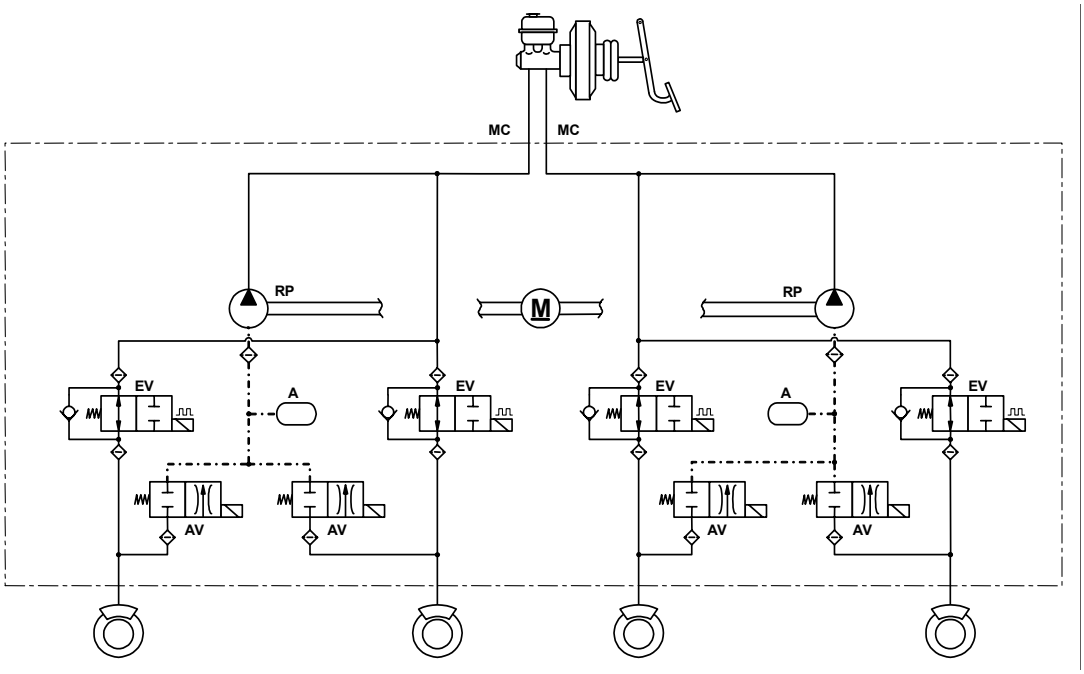
© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

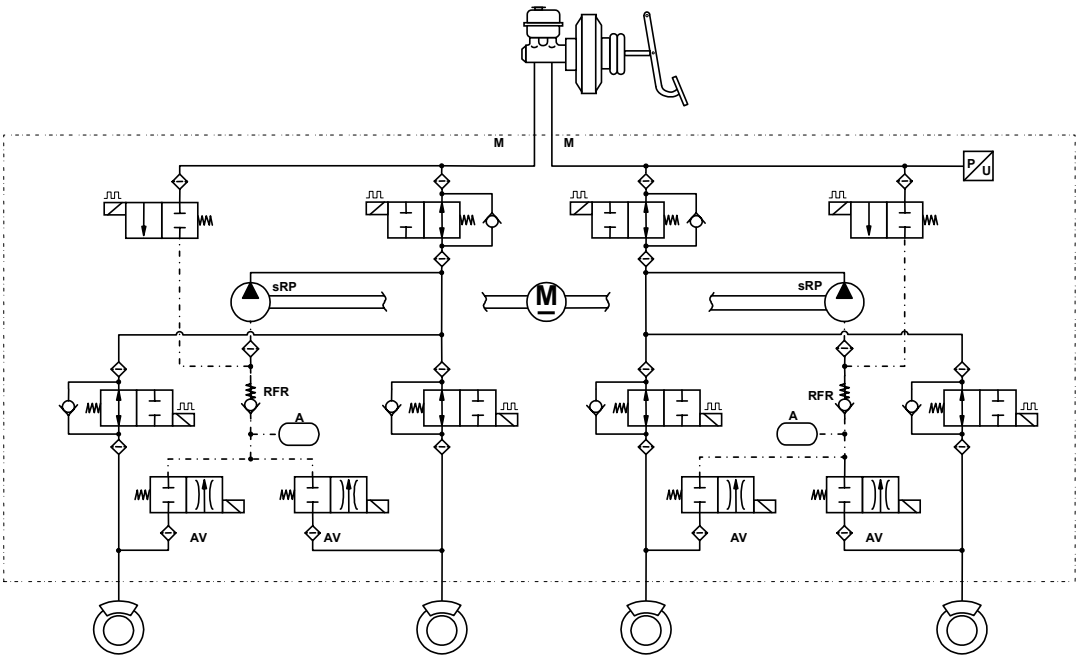
Attachments 附件

7.1 Hydraulic circuit diagrams 液压图

7.1.1 ABS 9 – independent of the brake split distribution



7.1.2 ESP® 9- independent of the brake split distribution

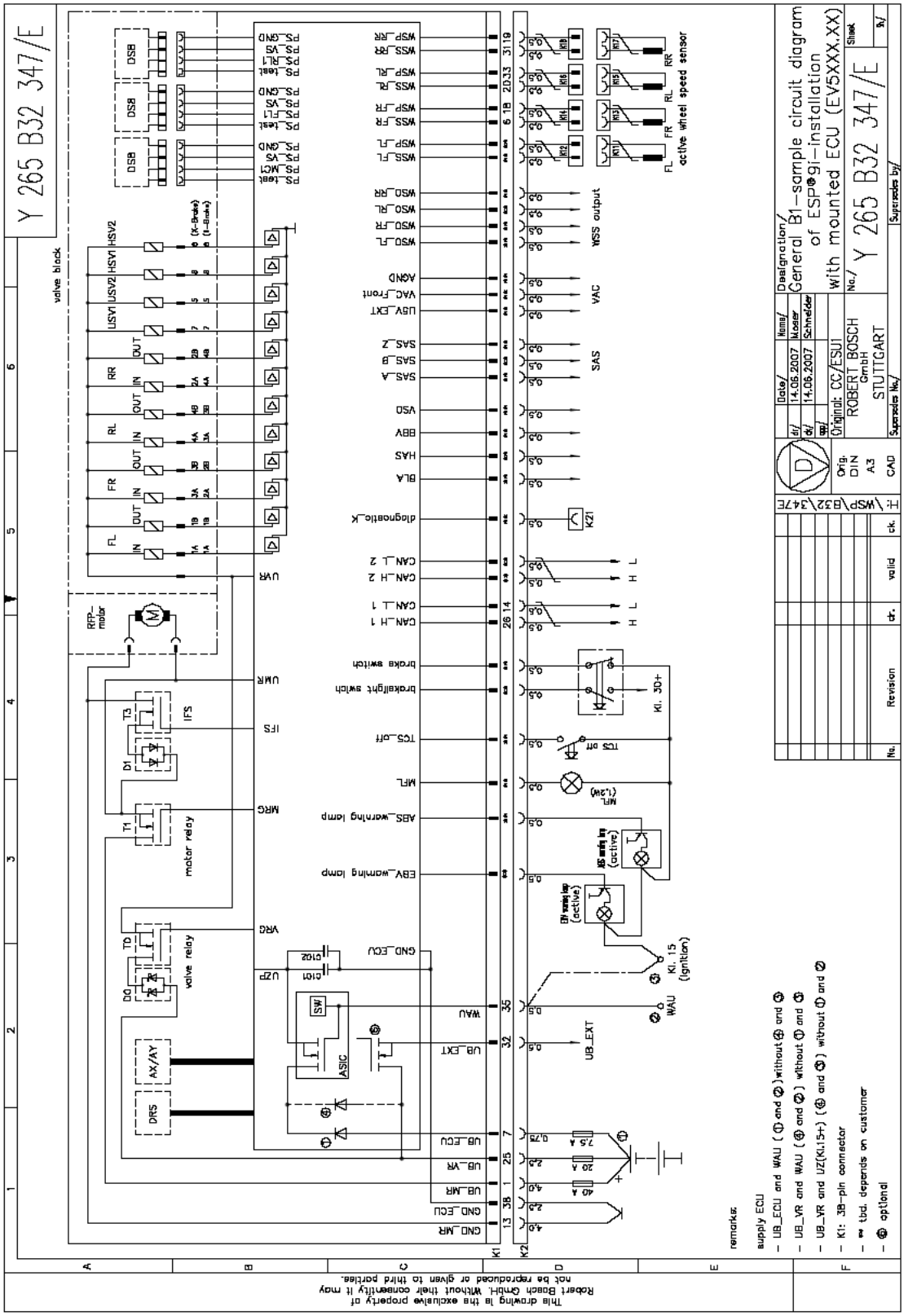


The dashed line within the ABS respectively ESP® Hydraulic shows the area of the secondary and priming circuits 虚线为第二回路

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

对本文有任何疑问，以英文为准。

7.2 Electrical circuit diagram ESP® 9i 电路图



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.
对本文有任何疑问，以英文为准。